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THE NATIONAL ENERGY COUNCIL

INDONESIA ENERGY OUTLOOK 2023

**BUREAU OF ENERGY POLICY AND ASSEMBLY FACILITATION
SECRETARIAT GENERAL OF
THE NATIONAL ENERGY COUNCIL
2023**

REMARK BY CHAIRMAN OF THE NATIONAL ENERGY COUNCIL



I would like to appreciate the effort taken by the Secretariate General of the National Energy Council for publishing the book of Indonesia Energy Outlook 2023. This book does not only give the overview of the current energy condition, but also shows Indonesia energy projection in the future as well as the overview of energy condition by region that would be very beneficial as an input in the making of the policy and planning in Indonesia's energy sector.

The economic growth and population growth strongly influences the supply-demand energy. Furthermore, the government's policies on the utilization of Solar Power Plant, the acceleration of electric vehicle, and the B-35 biofuel implementation should be used as a consideration in evaluating and formulating medium and long-term strategy.

In this opportunity, allow me to convey my gratitude to Members of the National Energy Council, Secretariate General of the National Energy Council, especially for the Editorial Board for the hard work. I hope the book of Indonesia Energy Outlook 2023 would be beneficial for all the stakeholders and energy policy makers.

Jakarta, December 2023

Minister of Energy and Mineral Resources
as the Chairman of the National Energy Council

Arifin Tasrif

FOREWORD BY MEMBER OF THE NATIONAL ENERGY COUNCIL



Let us praise to God the Almighty for His blessing that the National Energy Council has finished the book of Indonesia Energy Outlook 2023.

Indonesia Energy Outlook portrays the energy in Indonesia including the current energy condition and energy projection in the future.

Indonesia Energy Outlook 2023 is jointly written by Members of the National Energy Council from Stakeholders and Editorial Board at the Secretariate General of the National Energy Council with the assistance from the experts. The formulation of Indonesia Energy Outlook also involves units in the Ministry of Energy and Mineral Resources namely the Data and Information Technology Center (Pusdatin), Directorate General of Electricity, Directorate General of Oil and Gas, Directorate General of New Renewable Energy and Energy Conservation, as well as Directorate General of Mineral and Coal especially on the collection of current data and planning data.

In Energy Outlook 2023, the energy supply and demand projection are carried out for the next 10 years in 2 (two) scenarios namely Hymne which depicts the scenario of Business as Usual (BaU) with the deployment of the existing policy and Mars which is assumed to deploy policies towards developed country by 2045 and Net Zero Emission (NZE) by 2060. Next, the projection from the two scenarios will be compared with the projection that has been set in National Energy Plan (RUEN).

In addition to this 2023 publication, Indonesia Energy Outlook also presents the result of projection based on 7 regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku and Papua.

Jakarta, December 2023
Member of the National Energy Council
from Industrial Stakeholder

Herman Darnel Ibrahim

FOREWORD BY SECRETARY GENERAL OF THE NATIONAL ENERGY COUNCIL



Indonesia Energy Outlook 2023 is an annual publication by the Secretariate General of the National Energy Council since 2014.

The energy supply and demand projection are conducted based on LEAP (Low Emissions Analysis Platform) modeling using 2022 as the baseline year taken Pusdatin MEMR and Indonesia Statistic Agency (BPS) as well as data from related units.

In the effort to improve the quality of energy supply and demand projection data, Secretariate General of the National Energy Council would enhance the relationship with related parties so that the Indonesia Energy Outlook can become a reliable and trusted reference.

Jakarta, December 2023

Secretary General of the National Energy Council

Djoko Siswanto

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- Members of the National Energy Council from stakeholders;
- Directorate General of Oil and Gas, Directorate General of Electricity, Directorate General of Mineral and Coal, Directorate General of New Renewable Energy and Energy Conservation, Pusdatin MEMR and Geological Agency;
- Danish Energy Agency;
- Energy experts for their contribution in formulating Indonesia Energy Outlook 2023.

DISCLOSURE

Indonesia Energy Outlook 2023 is an analysis on the long-term national energy demand and supply projection (2023-2033) with particular assumptions which are developed for the purpose of future energy scenario planning. The assumption and projection are based on energy technology development including fossil and renewable energy based on current data and condition. The data in Indonesia Energy Outlook is taken from official publication, as well as temporary data or data that are continuously updated by the sources.

EXECUTIVE SUMMARY

IEO is an annual publication about energy demand and supply projection for the next 10 years (2023-2033) based on social, economy, and future technology development assumptions by using year 2022 as the baseline. The developed scenario differs each year. In 2023, the two scenarios used are Hymne scenario which depicts Business as Usual (BaU) by implementing existing policy and Mars scenario which deploys the assumptions of 2045 Advance Indonesia and Net Zero Emission (NZE) 2060. The result of projection in the two scenarios will be compared with the result of projection in RUEN. To have a further overview of energy distribution by region, this 2023 edition will divide the energy supply demand projection based on 7 regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku and Papua.

In the modeling, energy consumption is data in each province which is added up following the division by region. Data of energy consumption by province is derived from units in MEMR and energy companies. Oil fuel and LPG consumption/sales data by province is derived from Directorate General of Oil and Gas. Main data for electricity sales is based on PLN data with the additional identification data of non PLN consumption and electricity in smelter industry from Directorate General of Electricity. Data on coal consumption by province refers to data from BPS Industrial Statistic. Meanwhile, data on gas consumption is from sales data in Handbook Energy and Economy Statistics of Indonesia (HEESI) with the data on consumption by industrial sector derived from BPS Industrial Statistic. Data on household consumption by province is from Indonesia Residential End Use Survey as the result of CLASP by Directorate General of New Renewable Energy and Energy Conservation.

The energy supply and demand analysis are conducted based on calculation of LEAP (Low Emissions Analysis Platform) model. LEAP is a modeling application in energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population and household or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

As modeling input, the data used is derived from HEESI published by Pusdatin, especially related to basic consumption data by sector and by type of energy in 2022. Furthermore, to project energy consumption for the next 10 years, population growth and economic growth projection data are used as driven. Meanwhile, PLN's RUPTL (Business Plan for Providing Electricity) 2021-2030 is used as input for power plant development.

Hymne scenario uses energy assumptions that refer to current condition and future projection based on historical data of the last several years such as additional city gas network, electric stove, electric vehicle, biofuel implementation, and others. The power plant development refers to RUPTL 2021-2030 with the assumption that the project completion is delayed for 2 years. Meanwhile, Mars scenario uses the economic and population growth assumption that are similar with Hymne scenario but the energy utilization uses the assumption that refers to 2045 Advanced Indonesia and NZE 2060, including in household sector in which the use of city gas and electric stove is projected to be higher than in Hymne scenario. In transportation sector, the use of EV and biofuel is projected to be higher than in Hymne scenario and the use of hydrogen would be started in 2032. In power plant, there will be additional capacity from NRE power plant especially solar PV, wind power plant, and co-firing power plant that is higher than in Hymne scenario and also nuclear power plant of 100 MW.

The energy supply and demand condition in the last 10 years shows that the final energy consumption is still dominated by transportation sector. However, in 2022 there is a shift of energy consumption to industrial sector due to the increasing coal demand in industry. Thus, industrial sector has the highest share with 45% followed by transportation with 37%. Meanwhile, energy consumption is around 13% in household, 4.2% in commercial sector, and 1% in other sectors (agriculture, mining and construction).

Based on the type of energy, final energy consumption for all types of energy experienced positive growth with the largest increase in coal. Meanwhile, gas and oil fuel experience negative growth of -2% and -0.5% respectively. Based on the type of energy, final energy consumption in 2022 is dominated by coal (26%), oil fuel (23%) and biodiesel (19%). Biodiesel consumption is a mixture of diesel oil and Fatty Acid Methyl Ester (FAME) or known as B-30, so that total oil fuel in final energy consumption is around 36% and electricity consumption is only 16%.

The total primary energy supply in 2022 reaches 246 million TOE. Coal utilization still dominates with around 42%, followed by oil with 31%, gas with 14% and NRE with 12.3% of the total primary energy supply. Thus, NRE target in RUEN will be difficult to achieve in the next 3 years.

On the energy transformation side, there are activities of oil refineries, LPG refineries, LNG refineries and power plants. The current installed oil refinery capacity is only 1.17 million bpd (barrels per day). The crude oil demand in 2022 for oil refinery input of 322 million barrels comes from domestic oil production (68%) and imports (32%). After processing, the oil refinery will produce refined products in the form of gasoline, diesel oil, fuel oil, kerosene, aviation fuel, avgas and other refinery products (non-fuel) such as LPG, lubricant, naphtha and others. Oil fuel refinery production in 2022 is 261 million barrels and non-fuel production is 51 million barrels.

Meanwhile, LPG refinery capacity reaches 4.7 million tons per year, but only 3.9 million tons per year operates in 2022. Domestic LPG production can only supply around 20% of domestic LPG demand. Meanwhile, the LNG refinery capacity reaches 31.2 million tons per year with total LNG production in 2022 amounting to 14,993.2 thousand metric tons. Most of the LNG is used for export purposes and only around 23% is used domestically, especially to meet power plant demand.

By the end of 2022, Indonesia has total power plant capacity of 83.8 GW, consisting of 79.8 GW of on-grid power plant and 3.95 GW of off-grid power plant. This figure shows that there has been an increase in power plant of almost 1.7 times in the last 10 years. The power plant is still dominated by coal which makes up half of the total national capacity, followed by gas energy at around 25%. Meanwhile, New and Renewable Energy (NRE) based power plant reaches only 15%, or around 6 GW in the last 10 years. The use of NRE in power plant is dominated by hydropower (58%), geothermal (20%), and biomass (18%). Meanwhile, the use of solar energy, both on-grid and off-grid, reaches only 225 MW.

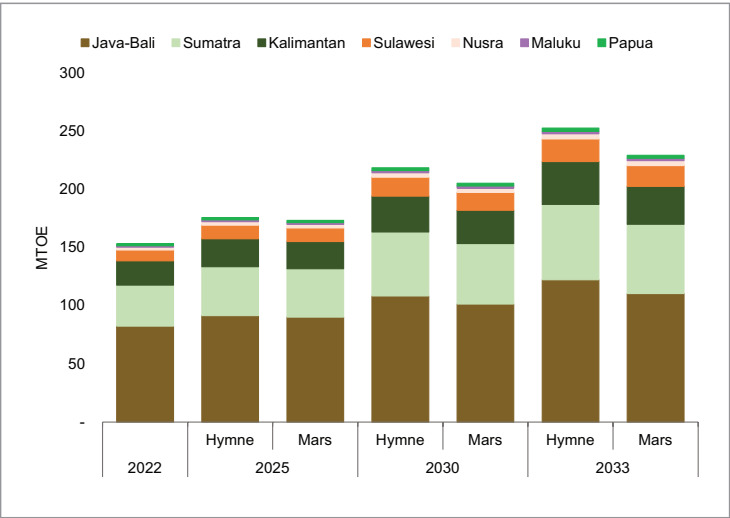
Total CO₂ emissions in 2022 is around 696.7 million tonnes of CO₂ or experiences an average increase of 4.1% per year. Based on fuel burning activities in 2022, the largest CO₂ emissions come from power plant activities (42.6%), followed by industry (29.6%), transportation (22.3%) and household and commercial sectors (0.6% and 0.3% respectively).

In the next ten years, final energy demand in Hymne scenario is projected to grow by an average of 4.6%. Meanwhile, final energy demand in the Mars scenario grows lower

at an average of 3.5% per year in line with energy efficiency in all sectors. Based on projection results, oil fuel will dominate final energy consumption in 2022 for the next 10 years. Thus, in 2033 oil fuel consumption reaches 113 million TOE (Hymne) and 96 million TOE (Mars). However, the average growth in oil fuel consumption in both scenarios is below the average growth in total final energy consumption, namely 4% (Hymne) and 2.5% (Mars). This condition is influenced by oil fuel to electricity substitution, the use of biodiesel B40 and ethanol E5, as well as the use of 0.04 million TOE hydrogen in the transportation sector starting in 2031.

The largest growth in final energy consumption based on energy type is NRE (biodiesel and bioethanol). Each increases by an average of 7.4%/year (Hymne) and 7.5%/year (Mars) with an increase in the FAME mixture in diesel oil by 40% in 2025 and ethanol in gasoline by 10% in 2025.

In 2033, energy consumption per region will still be dominated by Java-Bali region, but the average growth in energy consumption in Java-Bali region is the smallest compared to other regions. Meanwhile, the largest growth in energy consumption is in Sulawesi region, reaching 6.9% in Hymne and 5.8% in Mars which is influenced by the growth of the mineral processing and refining industry, especially in Southeast Sulawesi, South Sulawesi, Central Sulawesi and West Sulawesi.



Final Energy Consumption by Region

Total electricity demand (including electricity consumption for several smelter industries) in 2033 is projected to increase from 313 TWh in 2022 to 479 TWh (Hymne) and 488 TWh (Mars). The highest growth in electricity consumption in the next 10 years is in the transportation sector, amounting to 47% (Hymne) and 60% (Mars). Thus, the consumption will increase from 0.14 TWh in 2022 to 9.5 TWh (Hymn) and 24 TWh (Mars) in 2033. The issuance of Presidential Regulation Number 55 of 2019 concerning the Acceleration of the Battery-Based Electric Vehicle (KBLBB) Program for Road Transportation which is followed by several derivative regulations including Presidential Instruction Number 7 of 2022 concerning the Use of Battery Based Electric Vehicles as Operational Service Vehicles and/or Individual Service Vehicles for Central Government and Regional Government Agencies and the Minister of Industry Number 6 of 2023 which regulates subsidies of Rp. 7 million for the purchase of two-wheeled electric vehicles are expected to support the oil fuel to electric vehicles substitution program.

With government policies that support the development of EVs, it is projected that there will be 0.9 million units (Hymne) and 1.9 million units (Mars) of cars and 1.9 million units (Hymne) and 9.8 million (Mars) motorcycles in 2033.

Based on the region, the highest electricity demand until 2033 is in Java-Bali region and the smallest demand is in Papua. In the next 10 years, electricity demand in Papua will increase by around 4.8% (Hymne) and 8.8% (Mars), but its share will still be at around 0.7% of the total national electricity demand.

Electricity production in the next 10 years will still be dominated by fossil energy, especially coal. In 2022, electricity production from coal-fired power plant is around 71% of total production. However in the next 10 years it is projected that the share will fall to 63% (Hymne) and 38% (Mars) in 2033. A significant decline in the share of coal producing electricity in Mars scenario is influenced by the co-firing program at several coal fired power plants. The generating capacity of coal fired power plant implementing co-firing in 2022 is 14 GW and will increase to around 17 GW in 2033. In Hymne scenario it is assumed that co-firing in coal fired power plant will reach 5% starting in 2030, while in Mars scenario co-firing in coal fired power plant is targeted to reach 15% starting in 2030.

Electricity production from NRE power plant will experience significant growth in the next ten years, namely 9% (Hymne) and 15% (Mars) respectively. Specifically in Mars scenario, power plant development is directed at optimizing NRE power plants, especially solar

power plant which has enormous potential in Indonesia as well as biomass-based power plant (including co-firing) and other bioenergy-based power plants such as biogas and waste. In addition, there is a program to substitute diesel power plant to NRE or gas power plant. As the result, the share of electricity production from diesel power plant in 2033 will decline to around 1%. The diesel power plant still exists, but it is prioritized for the outermost and remote areas.

Electricity production from NRE power plant will increase from 49 TWh in 2022 to 124 TWh (Hymne) and 228 TWh (Mars). Similar to current conditions, electricity production from NRE in 2033 in Hymne scenario will be dominated by hydropower (38%) and geothermal (33%). Meanwhile, electricity production in Mars scenario will be dominated by solar power plant and Bioenergy power plant (which includes Biomass, biogas and waste power plant) at around 28% and 32% respectively.

The largest growth in NRE electricity production comes from solar energy, namely 61%. Thus the production increases from 0.4 TWh in 2022 to 68 TWh in 2033. This condition is supported by the continued decline in the cost of installing solar power plant in accordance with data from IRENA (International Renewable Energy) which shows that the cost of solar power plant continues to decline from USD 0.444/kWh in 2010 to USD 0.049/kWh in 2022.

In 2032, nuclear power plants will be introduced in Mars scenario with electricity production of around 701 GWh to support the development of low-carbon electricity.

Total primary energy supply in 2033 will increase approximately 4.2% and 3.6% respectively to 379 million TOE (Hymne) and 353 million TOE (Mars). In Hymne scenario, coal primary energy supply grows by an average of 3.4% per year, but in the Mars scenario it grows negatively by -0.4% per year due to the declining use of coal in power plants in Mars scenario with the co-firing program in several coal fired power plants. On the other hand, primary energy supply from NRE shows the highest at around 8% (Hymne) and 12% (Mars) which is influenced by the increase in NRE power plant, especially solar power plant and biofuel utilization program. In 2033, NRE supply in Hymne will reach 70 million TOE, or around 18.5% of the total primary energy supply. Meanwhile, in Mars scenario, NRE supply will be around 102 million TOE, or 28.9% of the total primary energy supply.

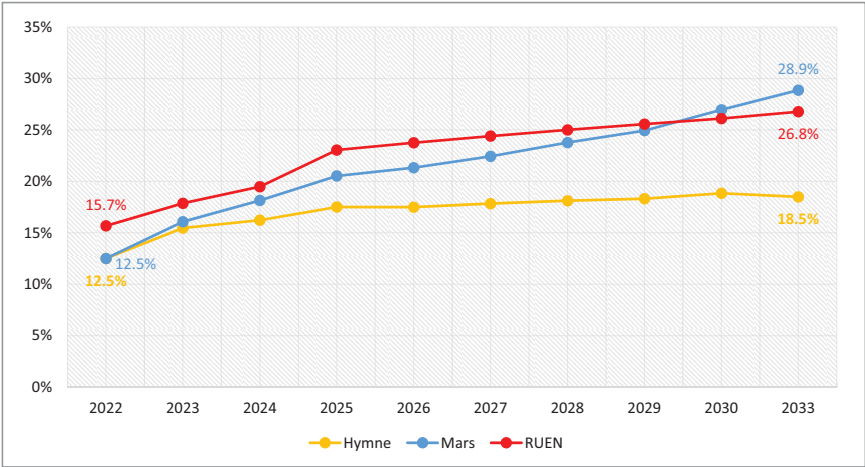
Meanwhile in 2033, electricity demand per capita will be around 1,662 kWh/capita in Mars scenario and only 1,630 kWh/capita in Hymne scenario or below the target of electricity per capita in National Energy Policy (KEN) of 2,500 kWh/capita in 2025.

In 2033, primary energy supply per capita will reach 1.20 TOE/capita in Hymne scenario and 1.12 TOE/capita in Mars scenario. The projection results are still far below the target in the KEN, namely 1.4 TOE/capita in 2025.

Total emissions in 2033 are projected to increase to 1,080 million tonnes CO₂eq (Hymne) and 840 million tonnes CO₂eq (Mars). The emission from both scenarios are still lower than the emission target in the energy sector NDC of 1,355 million tons of CO₂eq in 2030.

In 2033, power plants will still be the largest emitter throughout the projected year because of the dominant use of fossil energy, especially coal, of around 47% (Hymne) and 44% (Mars). The second largest emitter is the transportation sector due to the dominant use of oil fuel in vehicles. The share of emissions from the transportation sector in 2033 will reach 25% (Hymne) and 29% (Mars).

The result of the modeling, for both the Hymne and Mars scenarios, shows that only coal is expected to reach the KEN target in 2025. Meanwhile, NRE is still far below the target. The NRE mix in 2025 is expected to reach 17.5% in Hymne and 20.5% in Mars. This is quite reasonable considering that the realization of the NRE mix in 2022 will only reach 12.3%. Meanwhile, the RUEN estimates the need to reach 15.7% in 2022 to achieve the KEN target.



NRE Mix Projection 2022 – 2033

The NRE mix is projected to continue to increase until 2033 to 18.5% in Hymne and 28.9% in Mars Scenario. In 2030, the NRE mix in Mars scenario will reach 27% or exceed the 26% NRE mix in the RUEN.

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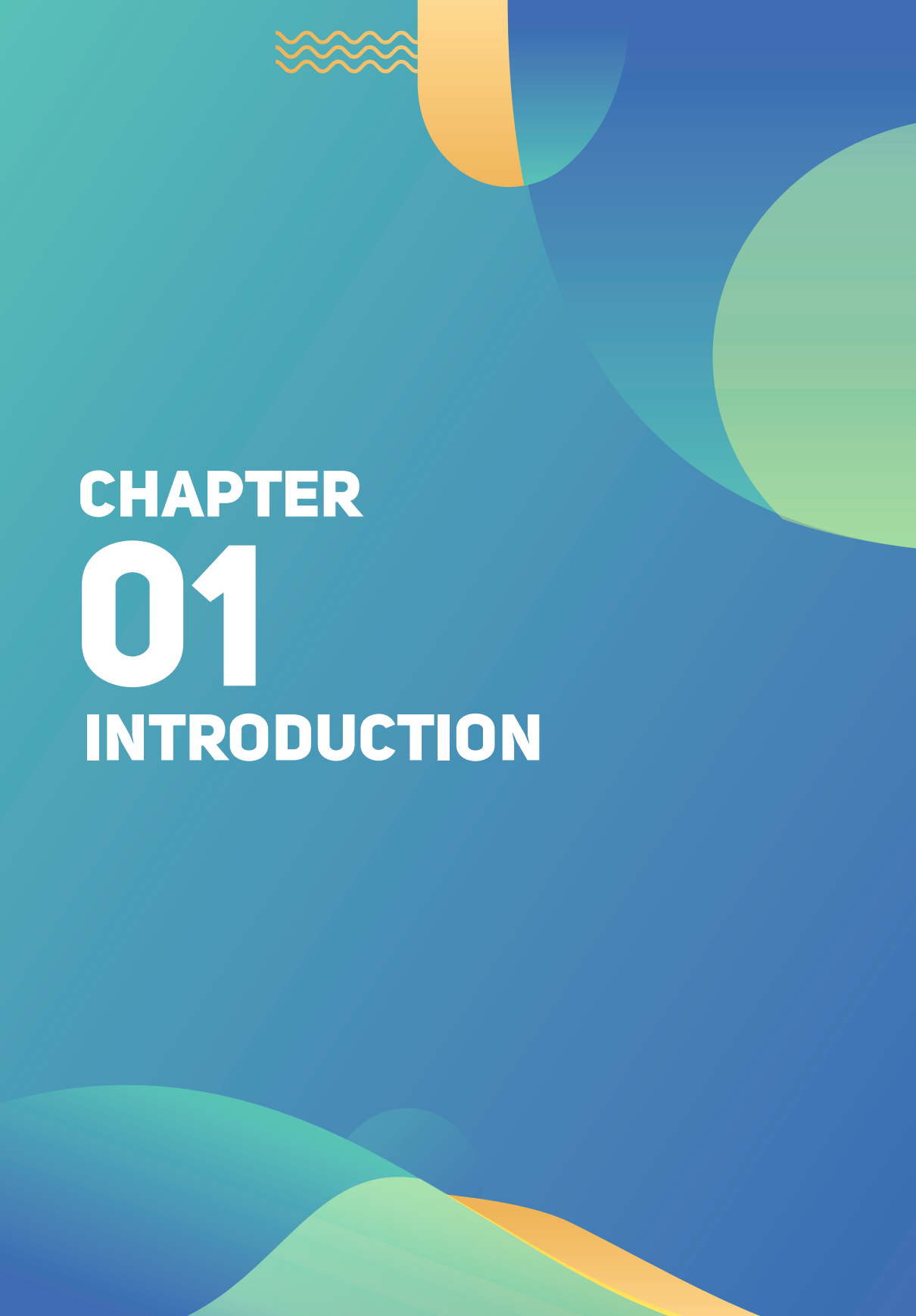
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CHAPTER **01** **INTRODUCTION**



01

INTRODUCTION

1.1 BACKGROUND

The book of IEO which is published annually is a result of an analysis that gives an overview of the national energy condition especially on energy supply and demand projection for the next 10 years (2023-2033).

The 2023-2033 projection uses two scenarios namely Hymne scenario which uses current assumptions or Business as Usual (BaU) and Mars scenario which uses new policy assumptions towards developed country by 2045 and Net Zero Emission (NZE) by 2060.

The main data source in IEO 2023 is Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2022 publication by Pusdatin MEMR, RUPTL PLN 2021-2030, Statistics Indonesia 2022-BPS and Industrial Statistics of BPS.

1.2 METHODOLOGY

1.2.1 Modeling Analysis Framework

Modeling analysis is divided into three stages namely final energy demand analysis, energy transformation, and primary energy supply. Final energy demand analysis is done

by using the assumptions of GDP growth, population growth, policy, strategic planning and roadmap on current energy development. The primary energy supply analysis also takes into account the use of various energy sources and energy resources potential including the current policies and energy technology development. Meanwhile, energy transformation analysis is done by considering RUPTL of PLN (National Electricity Company). The modeling analysis framework is shown in Figure 1.1.

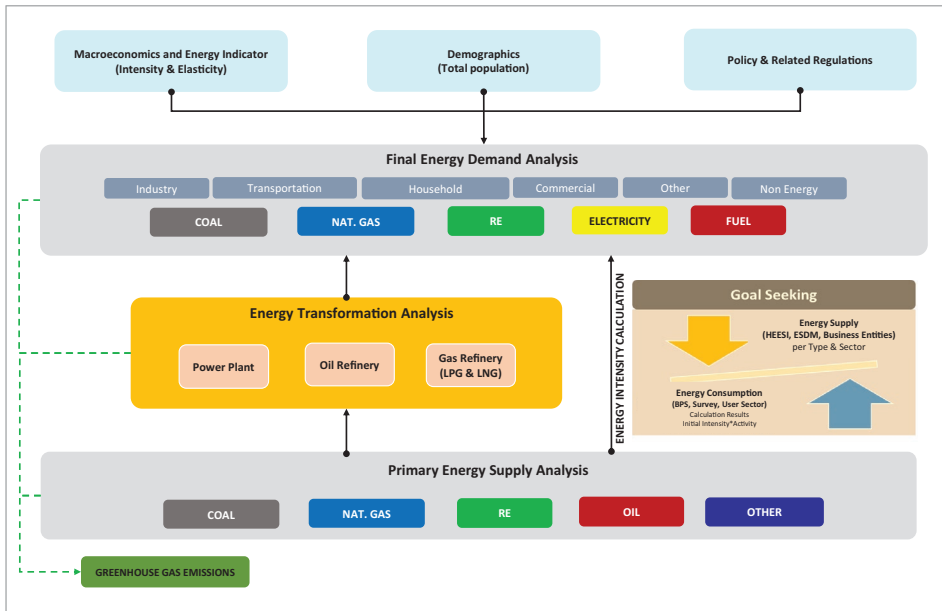


Figure 1.1 Modeling Analysis Framework

The energy supply and demand analysis are conducted based on calculation of LEAP model as a simulation model in energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population and household or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

Based on the modeling analysis framework in Figure 1.1, the parameters in final energy demand projection are social economic data (population and economic growth), historical data on energy consumption to identify energy intensity, and energy consumption pattern as the impact of people's lifestyle that is influenced by the projection of GDP increase, or a more efficient technology.

In 2022, the energy supply demand projection by region will also be carried out to obtain the overview of energy demand and supply in each region. The projection is done in seven regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusra, Maluku and Papua. The consideration in determining the seven regions is based on the geographical condition of several provinces located close to each other.

In the modeling, energy consumption uses the data on each province. It is then calculated based on per region. The data of energy consumption per province is obtained from units in MEMR and energy business entities. Data on oil fuel and LPG consumption/sales per province is specifically derived from Directorate General of Oil and Gas. Meanwhile, the main data of electricity sales is derived from PLN sales data with additional identification of non-PLN consumption data and electricity in smelter industry from Directorate General of Electricity. Data on coal consumption per province is obtained from Directorate General of Mineral and Coal. However, the sales of electricity by industrial sub sector refers to data from Industrial Statistics BPS. Specifically for data on gas consumption is based on sales data at HEESI, while consumption by industrial sub sector uses Industrial Statistics BPS data. The data on household consumption by province uses data from Indonesia Residential End Use Survey as the result of the joint study by CLASP and Directorate General of New Renewable Energy and Energy Conservation.

1.2.2 Energy Projection Scenario

1.2.2.1 Hymne Scenario

National economic growth used in the projection is derived from BPS, Institute for Research on Economics and Society – Indonesia University (LPEM UI), Ministry of National Development Planning (Bappenas). This assumption is also used to project energy demand in the ongoing revision of KEN. The energy growth in 2022 refers to Law No. 6 of 2021 on State Budget of Fiscal Year 2022. Thus, Indonesia economic growth in 2022 is predicted to reach 5.2%.

The projection of economic growth by province uses the data of Gross Regional Domestic Product growth projection of 34 provinces in 2021-2035 from Bappenas. The economic growth by region is determined by summing up and aggregating the data per province. Meanwhile, assumption of population is derived from BPS publication (Population Census 2020, March 2023). Similar to economic growth, population growth by region also uses aggregate method from each province.

The energy assumptions in Hymne scenario refers to current condition and future projection based on historical data in the last several years, such as additional city gas network, electric stove, electric vehicle, biofuel implementation, and others. The power plant development refers to RUPTL 2021-2030 with the assumption that the project completion is delayed for 2 years.

1.2.2.2 Mars Scenario

In Mars scenario, the economic and population growth assumption is similar to Hymne scenario, but energy utilization use assumptions that lead to developed country in 2045 and NZE in 2060, including:

- a. Household sector: the use of city gas and electric stove is projected to be higher than Hymne scenario.
- b. Transportation sector: the use of Electric Vehicle (EV) and biofuel is projected to be higher than Hymne scenario and the use of hydrogen would be started after 2030.
- c. Industrial sector: the substitution of diesel to electric is carried out in stages, and industrial conservation (which are not found in Hymne scenario).
- d. Power plant: additional capacity from NRE power plant especially solar PV, wind power plant, and steam power plant with co-firing is higher than in Hymne scenario, and the nuclear power plant will be started in 2032.

The details of assumptions used in the two scenarios above can be seen in Table 1.1.

Table 1.1 Scenario Assumption

INDICATOR	Unit	Hymne	Mars
DEMOGRAPHICS AND ECONOMICS			
Population Growth			
- Sumatra	%	1.32 (2023-2025), 1.09 (2026-2030), 0.93 (2031-2033)	
- Java and Bali		0.88 (2023-2025), 0.65 (2026-2030), 0.54 (2031-2033)	
- Kalimantan		1.72 (2023-2025), 1.82 (2026-2030), 1.32 (2031-2033)	
- Sulawesi		1.24 (2023-2025), 1.03 (2026-2030), 0.90 (2031-2033)	
- Nusra		1.55 (2023-2025), 1.39 (2026-2030), 1.27 (2031-2033)	
- Maluku		1.33 (2023-2025), 1.20 (2026-2030), 1.06 (2031-2033)	
- Papua		1.51 (2023-2025), 1.17 (2026-2030), 0.99 (2031-2033)	
Economic Growth (Average)			
- Sumatra	%	5.9 (2022-2030), 6.9 (2031-2032)	
- Java and Bali		5.3 (2022-2030), 6.0 (2031-2032)	
- Kalimantan		6.4 (2022-2030), 7.2 (2031-2032)	
- Sulawesi		6.9 (2022-2030), 7.7 (2031-2032)	
- Nusra		5.8 (2022-2030), 6.6 (2031-2032)	
- Maluku		9.6 (2022-2030), 9.1 (2031-2032)	
- Papua		7.5 (2022-2030), 8.3 (2031-2032)	
DEMAND			
Electric Vehicle			
Electric Cars 2033			
- Java	%	5	10
- Outside Java	%	5	10
Electric Motorcycle 2033			
- Java	%	2	10
- Outside Java	%	2	10
Electric Bus 2033			
- Java	%	0.1	1
- Outside Java	%	0.1	1
Electric Truck 2033			
- Java	%	-	0.1
- Outside Java	%	-	0.1
Hydrogen Vehicle			
Cars	%	-	1
Truck	%	-	1
ELECTRICITY 2033			
Nuclear	Capacity	RUPTL 2021-2030	100 MW
Co-Firing Coal	%	5% of the Co-Firing Coal PP Capacity	15% of the Co-Firing Coal PP Capacity

1.3 Writing Framework

The writing framework of the Indonesia Energy Outlook 2023 is as follows:

CHAPTER I INTRODUCTION

This chapter includes brief explanation on the purpose and objective of the Indonesia Energy Outlook 2023, along with methodology, scenario, and assumption in modeling.

CHAPTER II ENERGY CONDITION 2013 - 2022

This chapter describes the overall Indonesia energy condition in the last 10 years which include: energy potential and reserves, primary energy supply, final energy consumption, energy infrastructure, energy price, and emission from energy activity.

CHAPTER III ENERGY OUTLOOK 2023-2033

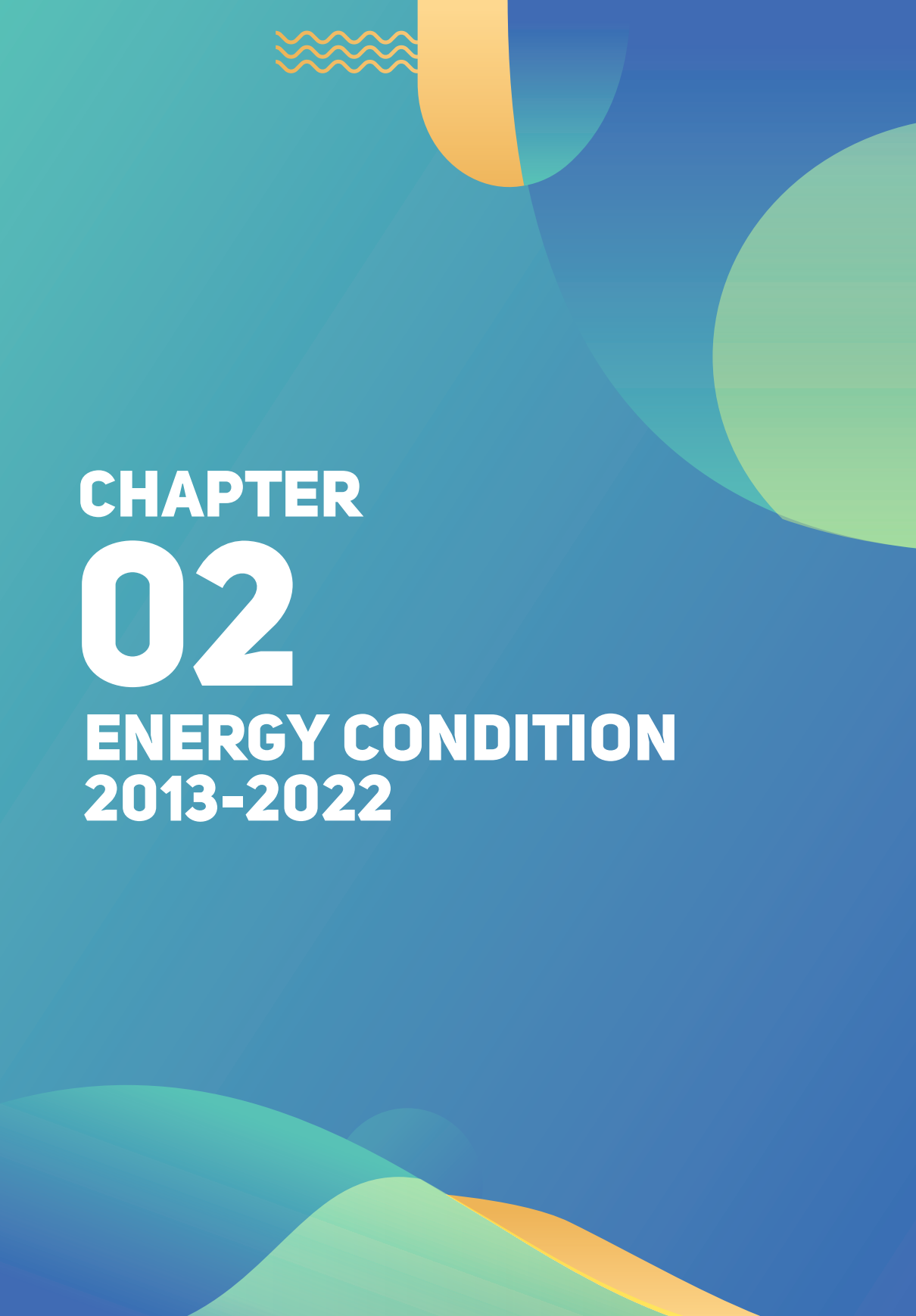
This chapter provides Indonesia energy condition projection for the next 10 years along with the analysis which include: final energy consumption, electricity, primary energy supply, and energy indicator. This chapter also discusses the relation between current achievement and primary energy mix target based on Presidential Regulation on RUEN.

CHAPTER IV NATIONAL ENERGY SECURITY AND PROGRESS IN ENERGY TRANSITION

This chapter presents the progress of the implementation of several national policies that are mainly related to energy security and energy transition in Indonesia as the focus of the National Energy Council.

CHAPTER V CLOSING

This chapter presents summary of the analysis on energy supply and demand projection in the next 10 years.



CHAPTER

02

**ENERGY CONDITION
2013-2022**

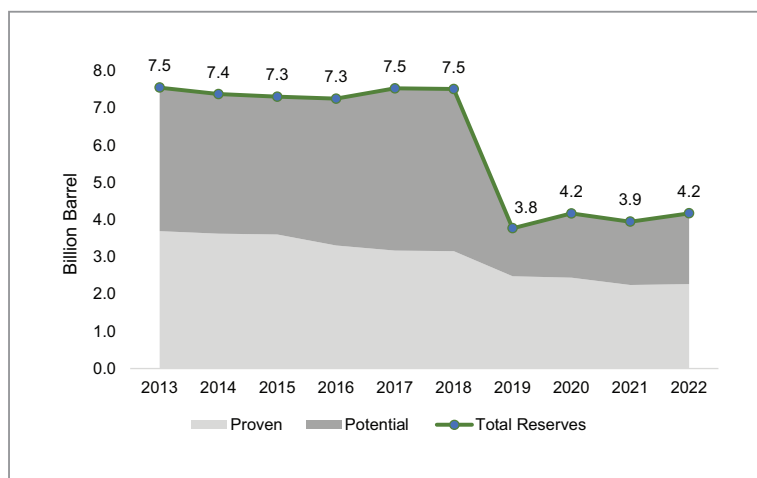
02

ENERGY CONDITION 2013-2022

2.1 RESOURCES AND RESERVES

2.1.1 Oil

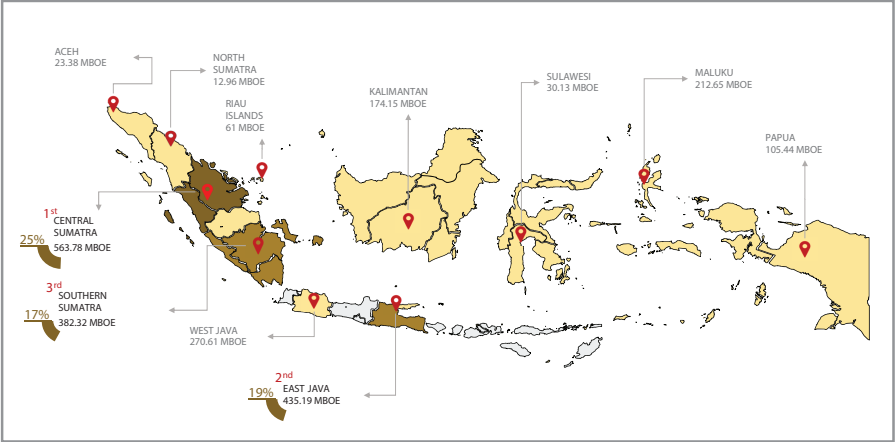
Total oil reserves in 2022 reaches 4.2 billion barrels, but only 2.3 billion barrels are proven reserves while the rest is potential reserves. In 2019, the reserves are calculated with new method. Consequently, the trend of reserves in the last 10 years cannot be compared. The complete description of the oil reserves trend is shown in Figure 2.1.



Source: HEESI, 2022

Figure 2.1 Oil Reserves 2013-2022

Based on its distribution, the biggest proven oil reserves in 2022 is found in central Sumatra with 563.78 million barrels, followed by reserves found in East Java with 435.19 million barrels and in southern Sumatra with 382.32 million barrels. The distribution of proven oil reserves per region can be seen in Figure 2.2

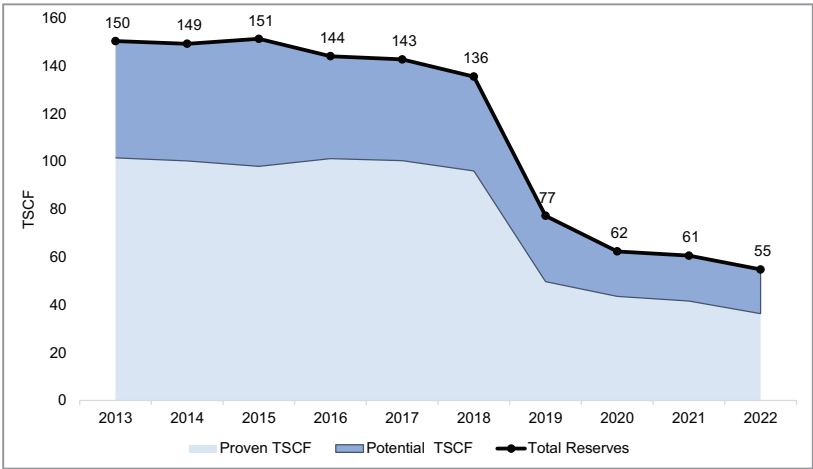


Source: MEMR, 2022 processed by Secretary General of National Energy Council (Sec Gen of NEC)

Figure 2.2 Map of Indonesia Proven Oil Reserves Distribution 2022

2.1.2 Gas

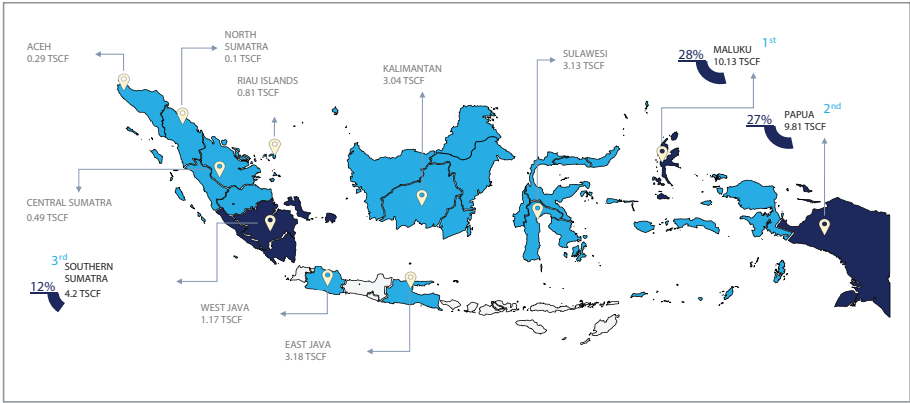
Indonesia’s total gas reserves in 2022 are around 55 TSCF, but the proven reserves are only 36 TSCF. Similar to oil reserves, calculation of gas reserves also changes since 2019. Thus, it is difficult to compare the trend of reserves in the last 10 years. The gas reserves trend in 2013-2022 can be seen in Figure 2.3.



Source: HEESI, 2022

Figure 2.3 Gas Reserves 2013 - 2022

The biggest proven gas reserves in 2022 is found in Maluku (Masela field) with total potential of around 10 TSCF, followed by reserves found in Papua of around 9.8 TSCF and in southern Sumatra of around 4.2 TSCF as shown in Figure 2.4.

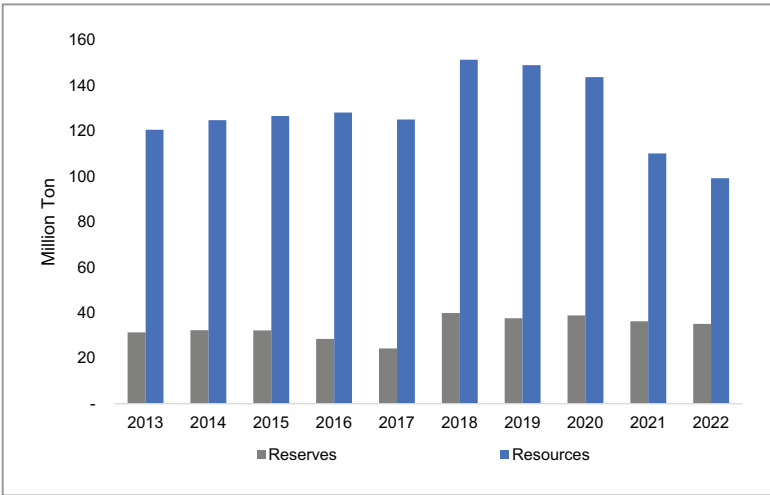


Source: KESDM, 2022 processed by Sec Gen of NEC

Figure 2.4 Map of Indonesia Proven Gas Reserves Distribution 2022

2.1.3 Coal

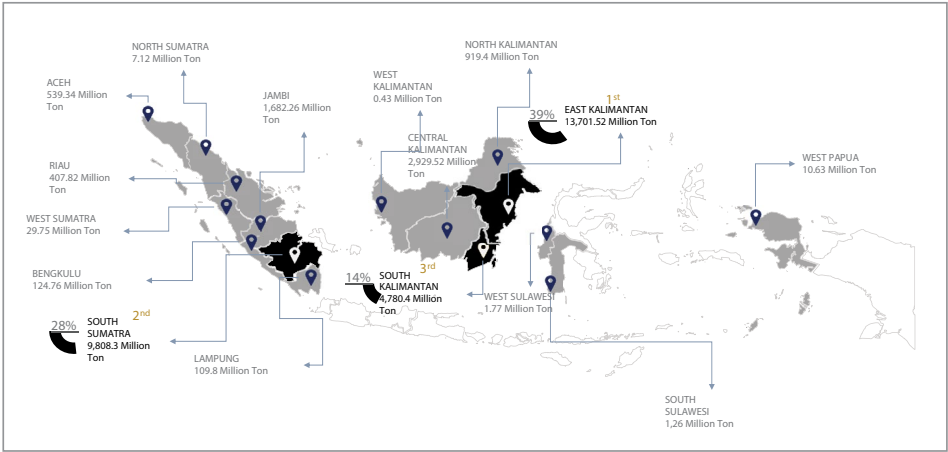
Total coal resources in 2022 are 99 billion tons, but the proven reserves in 2022 are only around 35 billion tons. Coal reserves continues to increase since 2013 from only 31 billion tons. The details of coal resources and reserves trend in the last 10 years can be seen in Figure 2.5.



Source: HEESI, 2022

Figure 2.5 Coal Resources and Reserves 2013-2023

The biggest coal reserves in Indonesia are found in East Kalimantan and South Sumatra with 13.7 million tons and 9.8 million tons each. The complete distribution of coal reserves can be seen in Figure 2.6.



Source: MEMR, 2022 processed by Sec Gen of NEC

Figure 2.6 Map of Indonesia Proven Coal Reserves Distribution 2022

2.1.4 New Renewable Energy

Indonesia has significant renewable energy and new energy potential. In 2021, the renewable energy potential data has been updated as seen in Table 2.1 below.

Table 2.1 Renewable Energy Power Plant Potential and Utilization 2022

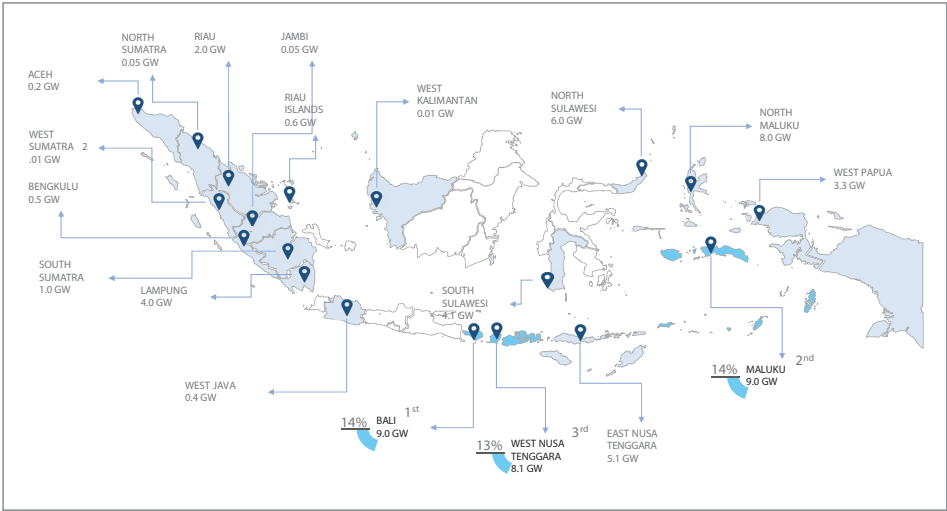
Type of Renewable Energy	Total Potential (GW)	Utilization (GW)	% Utilization
Ocean	63	-	-
Geothermal	23	2.4	10.3%
Bioenergy	57	3.1	5.4%
Wind	155	0.2	0.1%
Hydro	95	6.7	7.0%
Solar	3,294	0.3	0.01%
Total	3,687	12.6	0.3%

Source: DG NREEC, 2023

The total renewable energy potential for power plant is 3,687 GW, but until 2022 only 0.3% or 12.6 GW is utilized. The low NRE utilization for electricity generation is due to the relatively high cost of NRE-based power plant. Thus, it is difficult to compete with fossil power plant especially coal. In addition, investment in renewable energy requires high capital due to the lack of support from domestic industry in NRE component and the difficulty to obtain funding or loan with low interest.

2.1.4.1 Ocean Energy

Based on data from Directorate General of New Renewable Energy and Energy Conservation, the potential ocean energy in Indonesia is 63 GW. The biggest potential is in Bali and Maluku with 9.0 GW each and West Nusa Tenggara with 8.1 GW as shown in Figure 2.7.



Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.7 Map of Indonesia Ocean Wave Potential Distribution

2.1.4.2 Geothermal

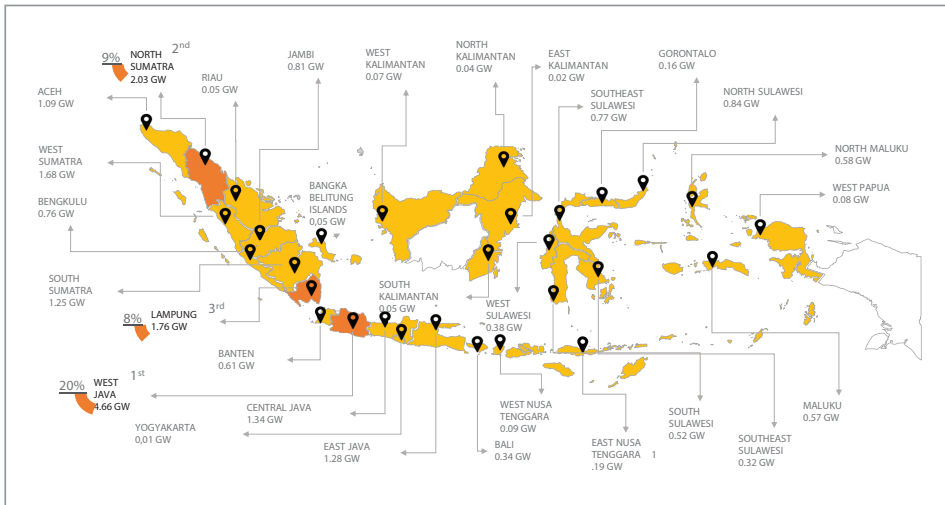
One of the energies that can be used as base load for power supply is geothermal. Total geothermal resources in Indonesia are the second biggest resources in the world with 23 GW. The trend of Indonesia geothermal resources and reserves in 2013-2022 can be seen in Table 2.2.

Table 2.2 Indonesia Geothermal Potential 2013-2022

Year	Resources (MW)					Total (MW)
	Speculative	Hypothetical	Reserves (MW)			
			Possible	Probable	Proven	
2013	7,377	4,973	13,449	823	2,288	28,910
2014	7,163	5,121	14,081	823	2,288	29,476
2015	7,055	4,943	14,435	823	2,288	29,544
2016	6,596	4,477	12,046	2,493	2,967	28,579
2017	6,617	4,456	11,975	2,493	2,967	28,508
2018	6,407	3,852	10,099	2,016	3,013	25,387
2019	5,952	3,387	9,696	1,876	3,055	23,966
2020	5,981	3,363	9,547	1,770	3,055	23,716
2021	5,981	3,363	9,547	1,770	3,105	23,766
2022	5,775	3,444	8,968	1,664	3,210	23,060

Source: HEESI, 2022

The biggest geothermal reserves are found in West Java (20%), North Sumatra (9%) and Lampung (8%). The distribution of geothermal reserves in details can be seen in Figure 2.8. From the total geothermal resources of 23 GW, only around 10% or 2.3 GW is utilized.

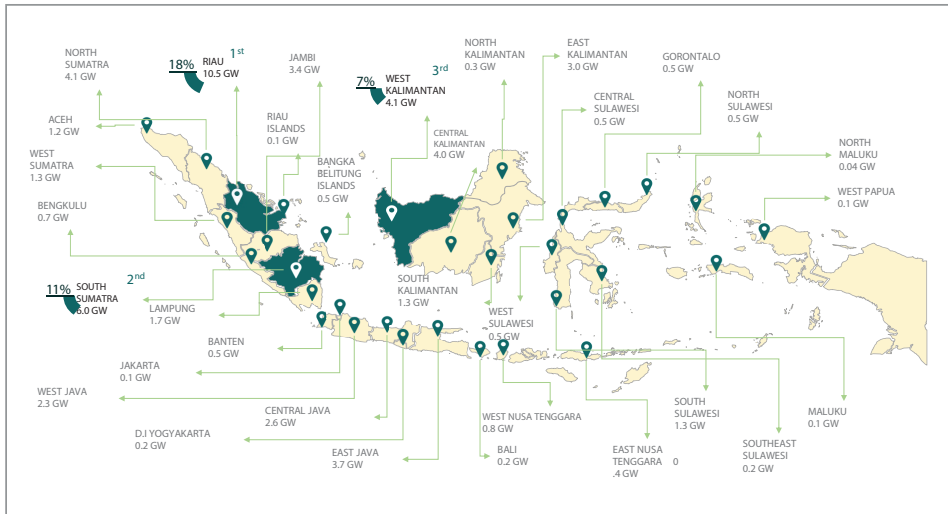


Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.8 Map of Indonesia Geothermal Potential and Reserves Distribution

2.1.4.3 Bioenergy

Indonesia bioenergy potential based on the latest data reaches 57 GWe consisting of 52.1 GWe biomass, 1.3 GWe POME, 1.4 GWe waste and 2.3 GWe biogas. Based on Figure 2.9, the biggest biomass potential is in Riau Province with 10.4 GWe from palm oil plantation. Meanwhile, the biggest waste and biogas potential is in West Java Province with 0.25 GWe and 0.46 GWe. The major potential of waste is equivalent to people's activity in urban areas and the large population in West Java compared to the other provinces.

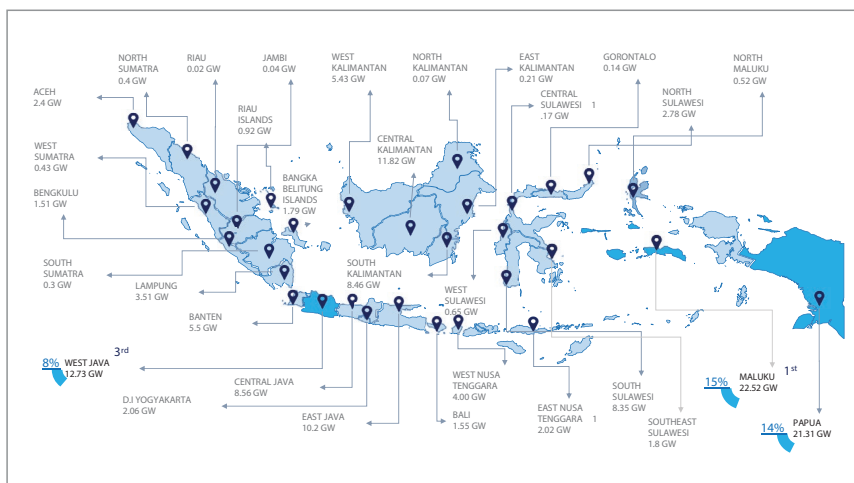


Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.9 Map of Indonesia Bioenergy Potential Distribution

2.1.4.4 Wind Energy

As an archipelago country with long coastal line, Indonesia has huge wind energy potential. Based on data by Geoportal ESDM, a number of areas in Indonesia has potential wind energy with the speed of 4 m/s – 6 m/s. In 2021, there is an additional wind energy potential from 60.6 GW to 154.9 GW. This increase occurs due to the additional offshore wind energy potential as seen in Figure 2.10 below. The biggest wind energy potential is in eastern Indonesia namely in Maluku and Papua.

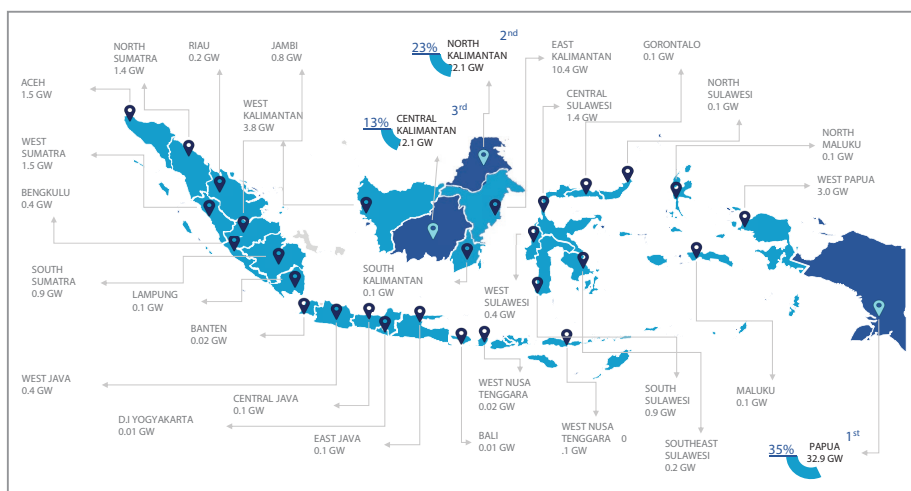


Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.10 Map of Indonesia Wind Energy Potential Distribution

2.1.4.5 Hydro Energy

In 2021, the updated hydro energy potential data shows an increase from 75 GW into 95 GW. It is derived from the update of hydro potential calculation method with water discharge data based on rainfall run off calculations using the water flow method developed by Research and Development Center for Water Resources (under cooperation between MEMR and Ministry of Public Works and Housing) and Hydro Power Plant potential update in Ministry of Public Works and Housing (until 2027). The biggest hydro potential is in Papua Province (34.6%) and North Kalimantan Province (23.3%) as described in Figure 2.11. below.

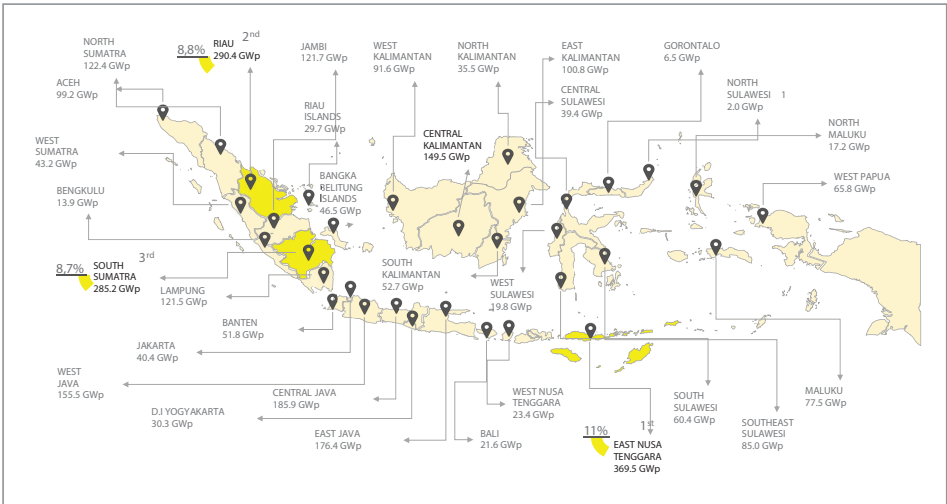


Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.11 Map of Indonesia Hydro Energy Potential Distribution

2.1.4.6 Solar Energy

As a tropical country with year-round sunshine in various regions, Indonesia has great potential for the development of Solar Power Plant. Based on updated solar energy data in 2021, the solar energy potential reaches 3,294.4 GW as shown in Figure 2.12. This update focuses on the solar potential calculation method based on radiation intensity classification starting from 3.75 kWh/m²/day and filtered by land cover maps (settlements, open land and savanna) excluding protected areas. In addition, the potential for floating solar power plants is also taken into account. Based on data per province, the largest solar potential is in the provinces of East Nusa Tenggara (NTT), Riau and South Sumatra.



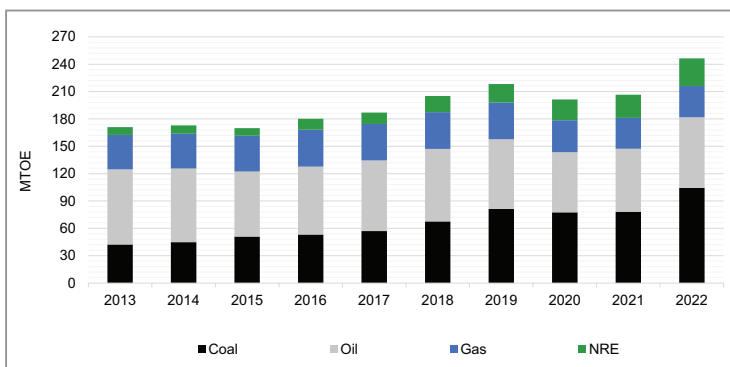
Source: DG NREEC, 2022 processed by Sec Gen of NEC

Figure 2.12 Map of Indonesia Solar Energy Potential Distribution

2.2 PRIMARY ENERGY

2.2.1 Primary Energy by Energy Type

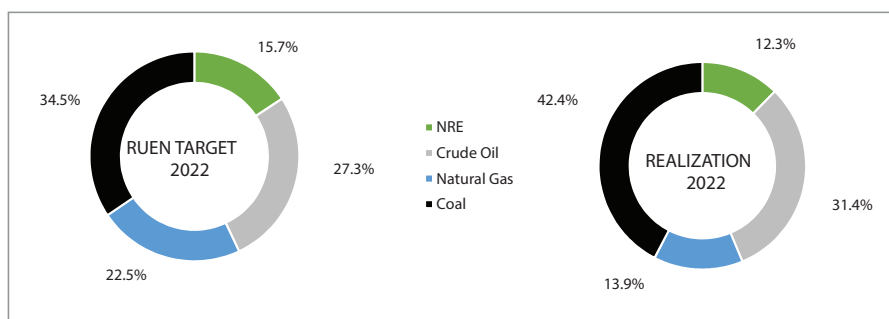
Total primary energy supply in 2022 reaches into 246 million TOE. It increases 19% from the previous year as shown in Figure 2.13 below. Based on its energy type, coal is still dominating with 42%, followed by oil with 31% and gas with 14%.



Source: HEESI, 2022

Figure 2.13 Primary Energy Supply 2013 – 2022

In 2022, NRE supply reaches 30 million TOE or 12.3% from the total primary energy supply. However, NRE target achievement is still below RUEN projection as shown in Figure 2.14 below. There is still an opportunity to increase NRE energy mix within 3 years ahead despite of the difficulty to meet the target of 23% NRE by 2025.



Source: RUEN, 2017 and HEESI, 2022

Figure 2.14 Primary Energy Mix 2022

2.2.2 Primary Energy by Region

In 2022, 58% of the total national primary energy or around 129 million TOE is centered in Java Bali. In Java-Bali, coal is still the main commodity with 53% share while the new NRE utilization is still 5.6%.

In Sumatra, primary energy supply reaches 20% of the total national. The energy use in Sumatra is dominated by oil with 37% share in energy mix, but the use of NRE in Sumatra has the biggest share of 27% compared to other regions. Meanwhile, the use of NRE reaches only 24% in Kalimantan, 9% in Sulawesi, 11% in Nusra, 4% in Maluku and 10% in Papua. The details of primary energy supply in 2022 in each region is shown in Figure 2.15 below.

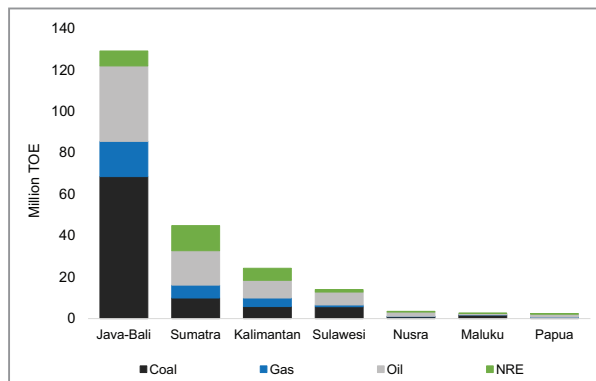


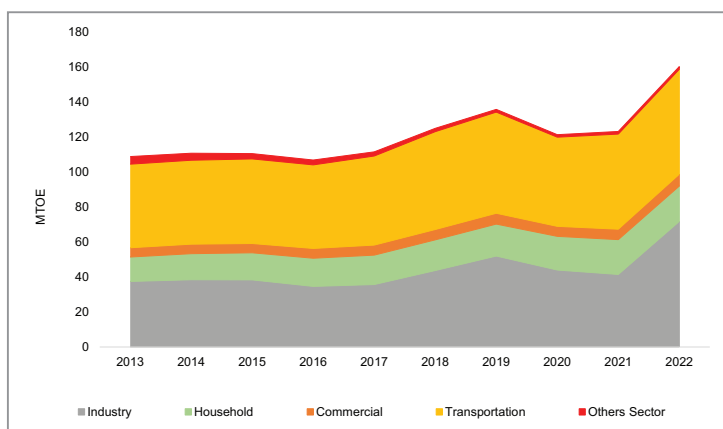
Figure 2.15 Primary Energy Supply by Region 2022

2.3 FINAL ENERGY CONSUMPTION

2.3.1 Final Energy Consumption by Sector

The final energy consumption in 2022 continues to increase after a drastic drop in 2020 due to Covid-19 pandemic. In 2022, final energy consumption reaches 160.3 million TOE or it grows around 4.4% per year during the period of 2013-2022.

In 2022, the biggest final energy consumption is industrial sector with 45%, followed by transportation with 37%. The increasing energy consumption in industry is influenced by coal consumption increase, including the growing smelter industry. Meanwhile, energy consumption is around 13% in household, 4.2% in commercial sector, and 1% in other sectors (agriculture, mining and construction). The complete description of the biggest final energy consumption per sector is shown in Figure 2.16.

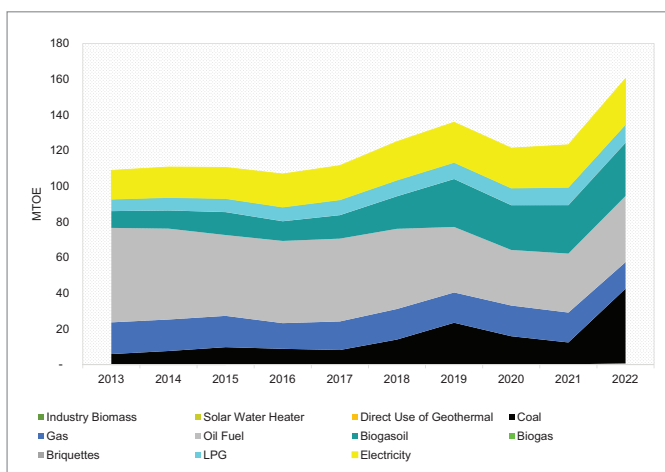


Source: HEESI, 2022

Figure 2.16 Final Energy Consumption by Sector 2013 – 2022

2.3.2 Final Energy Consumption by Energy Type

In the last 10 years, final energy consumption for all types of energy experiences a positive growth with the biggest increase in coal. Meanwhile, gas and oil fuel experience negative growth of -2% and -0.5% each. Based on type of energy, final energy consumption in 2022 is dominated by coal of around 41.9 million TOE (26%), followed by oil fuel of 37 million TOE (23%) and biodiesel of 30 million TOE (19%). Biodiesel is a mix between diesel and FAME. It is known as B-30. In total, oil fuel share in final energy consumption is around 58 million ton or 36%. Meanwhile, electricity consumption is only 16% from the total final energy consumption or 26 million TOE. The trend of final energy consumption per energy type can be seen in Figure 2.17 below.



Source: HEESI, 2022

Figure 2.17 Final Energy Consumption by Energy Type 2013 – 2022

2.3.3 Final Energy Consumption by Region

Based on region, final energy consumption in 2022 is still centered in Java-Bali with 54% share from Indonesia total final energy consumption. The share in other regions is 23% in Sumatra, 14% in Kalimantan, 6% in Sulawesi, 2% in Nusra, 1.1% in Papua and 0.9% in Maluku. This number is in line with the distribution of Indonesia population in which 58% are populated in Java-Bali and 22% are populated in Sumatra, while the rest are spread in Kalimantan, Nusra, Maluku and Papua.

In Java-Bali, Kalimantan, and Maluku, the energy use is dominated by the industry. Meanwhile in Sumatra, Sulawesi, Nusra and Papua, the use of energy is dominated by transportation sector. The details of final energy consumption per region is shown in Figure 2.18.

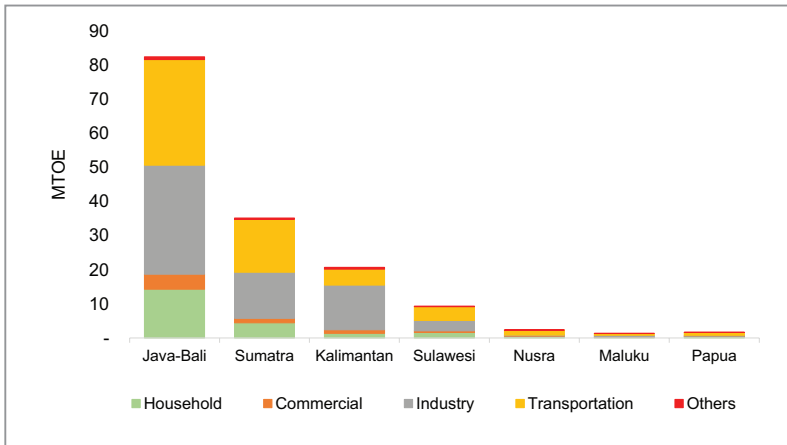


Figure 2.18 Regional Final Energy Consumption in 2022 by Sector

From the type of energy, all regions still rely on oil, especially in the form of oil fuel to meet the demand in transportation. The gap occurs in electricity commodity which is mainly used in Java-Bali, approaching 69% from the total national. It shows that electricity network infrastructure outside Java-Bali is still low. The details are shown in Figure 2.19.

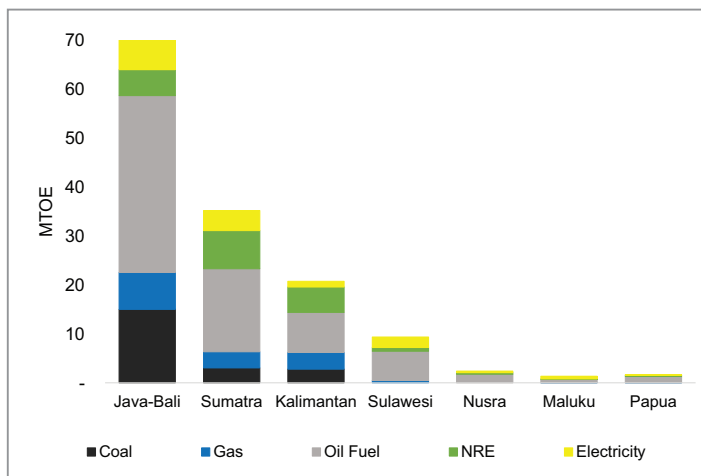


Figure 2.19 Regional Final Energy Consumption in 2022 by Energy Type

2.4 ENERGY INFRASTRUCTURE

2.4.1 Oil

Currently the installed oil refinery capacity is 1.17 million bph (barrel per day). The details of oil refinery and the capacity is shown in Table 2.3 below.

Table 2.3 Indonesia Oil Refinery Capacity 2022

Oil Refinery	Capacity (Thousand BPH)
Dumai	177.0
Musi	127.3
Cilacap	348.0
Balikpapan	260.0
Balongan	150.0
Cepu	3.8
Kasim	10.0
Tuban (TPPI)	100.0
Total Capacity	1,176.1
Total Operating Capacity	1,176.1

Source: HEESI, 2022

In the last 10 years, oil capacity in Indonesia does not change a lot. The last additional capacity to the refinery occurred in 2015 with the development of Tri Wahana Universal (TWU) refinery Unit II of 12 million bph in Bojonegoro, East Java. However, the refinery stops operating in early 2018 due to the limited feed to the refinery.

In 2022, crude oil for oil refinery feed is around 322 million barrels which is derived from domestic and import. Besides crude oil, oil refinery also needs gas and intermedia for the feed. After being processed, oil refinery will produce refinery products such as gasoline, diesel oil, fuel oil, kerosene, avtur, avgas and other non fuel products (LPG, lubricant, naphtha, and others). Oil fuel refinery production in 2022 is around 261 million barrels and non fuel production is around 51 million barrels.

2.4.2 Gas

Gas infrastructure is divided into LPG refinery, LPG Depot, LPG Filling Station and LNG Refinery.

2.4.2.1 LPG Refinery

Today Indonesia has LPG refinery with the capacity of 4.7 million tons. In 2022, only 3.9 million tons of the capacity is operating. The details of LPG refinery and the capacity is shown in Table 2.4 below.

Table 2.4 Indonesia LPG Refinery Capacity 2022

LPG Refinery	Capacity (MTPA)
LPG Refinery (Oil Refinery)	1,331.0
PT Pertamina (Dumai)	68.0
PT Pertamina (Plaju)	131.0
PT Pertamina (Cilacap)	318.0
PT Pertamina (Balikpapan)	91.0
PT Pertamina (Balongan)	548.0
PT TPPI	175.0
Upstream LPG Refinery	2,342.0
PT Badak NGL	1,000.0
PT Chevron*	90.0
PT Petrogas	14.0
PT Petrochina	600.0
PT Conoco Philips*	525.0
PT Saka Indonesia	113.0
Downstream LPG Refinery	1,067.3
PT Pertamina (P.Brandan)*	44.0
PT Maruta Bumi Prima*	17.0
PT Medco LPG Kaji*	73.0
PT Pertamina (Mundu)	37.0
PT Titis Sampurna	73.0
PT Sumber Daya Kelola (Tugu Barat)*	7.0
PT Bina Bangun Wibawa Mukti	55.0
PT Surya Esa Perkasa	82.0
PT Yudhistira Haka Perkasa*	44.0
PT Wahana Insannugraha	37.0
PT Media Karya Sentosa phase I*	58.0
PT Media Karya Sentosa phase II	84.0
PT Yudistira Energi	58.0
PT Gasuma Federal Indonesia	26.0
PT Pertasamtan Gas	259.0
PT Sumber Daya Kelola (Losarang)*	3.8
PT Arsynergy Resources	109.5
Total Capacity	4,740.3
Total Operating Capacity	3,878.5

Source: Oil and Gas Statistics Semester I 2022

Note: (*) Stop operating

2.4.2.2 DEPO LPG

Based on Directorate General of Oil and Gas MEMR, LPG depot in majority is owned by PT Pertamina (Persero) of around 90% (as of 1 May 2020) from total national LPG depot capacity amounting to 482,062 metric ton. The details of LPG depot for each business entity can be seen in Table 2.5 below.

Table 2.5 LPG Depot Capacity

Business Entity	Province	Capacity (MT)
PT. Pertamina (Persero)	all Indonesia	433,712
PT. Maspion Energy Mitratama	East Java	10,000
PT. Optima Sinergi Comvestama (OPSICO)	East Java	10,000
PT. Agrabudi Gas Utama	South Kalimantan	1,600
PT. Bosowa Duta Energasindo	South Sulawesi	10,000
PT. Misi Mulia Petronusa	East Java	10,000
PT. Pertamina Patra Niaga	West Nusa Tenggara	3,000
PT. Elnusa Petrofin	North Sulawesi	2,000
PT. Patra Trading	Riau	250
PT. Gemilang Asia Sejahtera	West Kalimantan	1,500
TOTAL		482,062

Source: DG of Oil and Gas, MEMR

2.4.2.3 LPG Filling Station

LPG Filling Station is a special tool to assign Pertamina (Persero) to distribute LPG (Liquified Petroleum Gas) to the people. The LPG Filling Station which receives approval and license based on the laws and regulations and is required by Pertamina to run the business and utilize LPG includes its activity from receiving LPG from the supply point appointed by Pertamina to storing and filling LPG to Non PSO (Public Service Obligation) LPG Bulk or other packaging based on Pertamina's requirements.

The LPG Filling Stations are mostly located in Java with 70 stations, Sumatra with 12 stations, Sulawesi with 2 stations and Kalimantan with 1 station. The distribution of LPG filling station per province can be seen in Table 2.6 below.

Table 2.6 Gas Filling Station Capacity per Region

Region	Capacity (MT)
Sumatra	720
Java-Bali	4,560
Kalimantan	2,360
Sulawesi	400
Others	810
Total	8,850

Source: DG of Oil and Gas, MEMR

2.4.2.4 LNG Refinery

Currently, Indonesia has LNG refinery capacity up to 31.2 million ton per year. The details of LNG refinery and its capacity is shown in Table 2.7 below.

Table 2.7 Indonesia LNG Refinery Capacity 2022

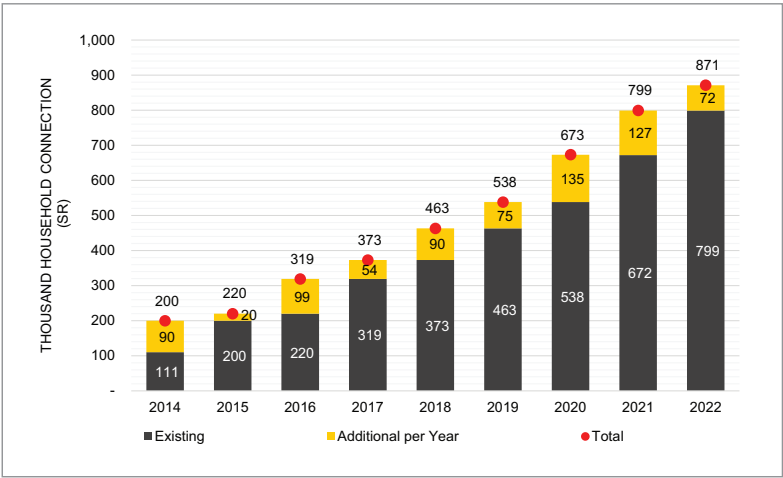
LNG Refinery	Capacity (MMTPA)
PT Badak	21.6
PT BP	7.6
PT Donggi Senoro	2
Total Capacity	31.2

Source: Oil and Gas Statistics Semester I 2021

Total LNG production in 2022 from the above three fields is 14,993.2 thousand metric ton. Most LNG is exported and only 23% is for domestic use especially to meet electricity demand.

2.4.2.5 City Gas

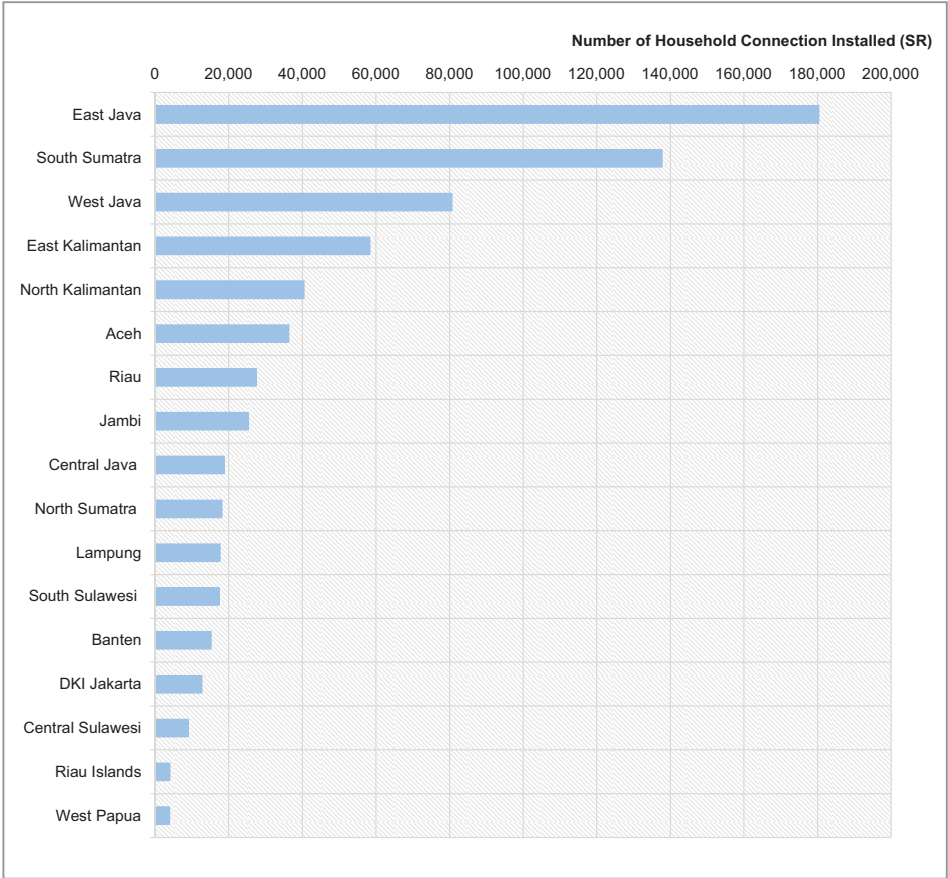
Since 2009 to 2022, city gas network for household has been built and funded by State Budget and Non-State Budget. In total, 871 thousand households have been connected with city gas. The development of city gas for household is shown in Figure 2.20 below.



Source: Working Performance MEMR 2022 & 2023 Target

Figure 2.20 Cumulative City Gas Development 2014-2022

The city gas infrastructure development is currently enjoyed by 17 provinces in all areas in Indonesia. The city gas is mostly built in East Java (26%) and South Sumatra (20%) based on the availability of gas resources. The distribution of city gas per province is shown in Figure 2.21 below.



Source: Oil and Gas Statistics Semester I 2022 and Presentation
 Material by DG Oil and Gas

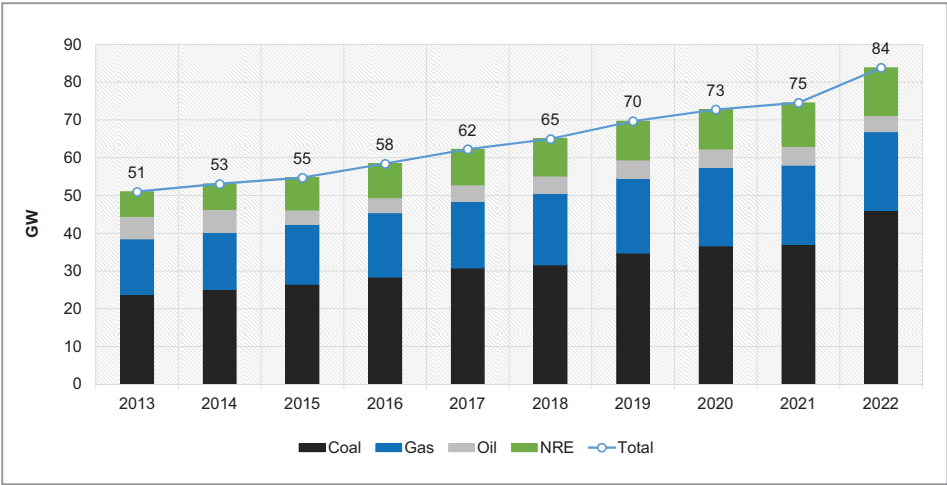
Figure 2.21 City Gas Network Infrastructure for Household Built by the Government per Province

2.4.3 Electricity

2.4.3.1 Power Plant

Until the end of 2022, Indonesia’s total power plant capacity is 83.8 GW, consisting of 79.8 MW on-grid power plant and 3.95 GW off-grid power plant. This number shows that additional power plant has doubled 1.7 times in the last 10 years as shown in Figure 2.22.

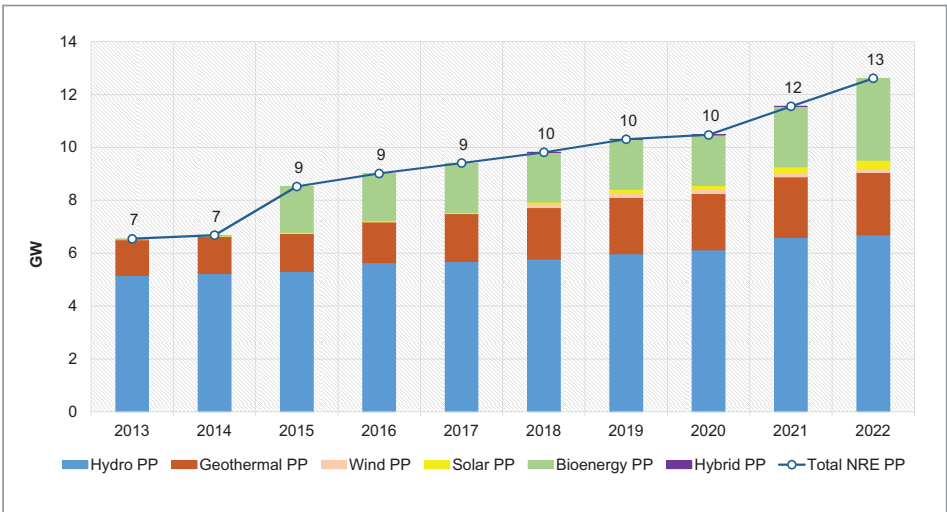
Coal Power plant still dominates half of the total national capacity, followed by gas power plant with 25%. Meanwhile, NRE power plant reaches only 15% or increases 6 GW in the last 10 years.



Source: HEESI, 2022

Figure 2.22 Power Plant Installed Capacity per Energy Type 2013-2022

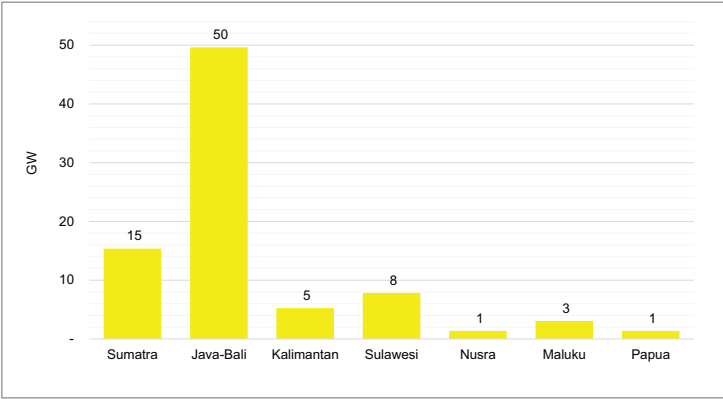
NRE utilization in power supply is dominated by hydro (58%), geothermal (20%), and biomass (18%) as shown in Figure 2.23. Meanwhile, the use of solar energy, both on-grid and off-grid reaches only 225 MW.



Source: HEESI, 2022

Figure 2.23 NRE Power Plant Installed Capacity 2013-2022

Power plant infrastructure is spread in all areas in Indonesia, but 62% of the power capacity is built in Java-Bali. Meanwhile, the capacity of power plant reaches 19% in Sumatra, 7% in Kalimantan, 10% in Sulawesi, 2% in Nusra, 4% in Maluku, and 2% in Papua (Figure 2.24). From this statistic, we can see that the power plant infrastructure is not spread evenly.

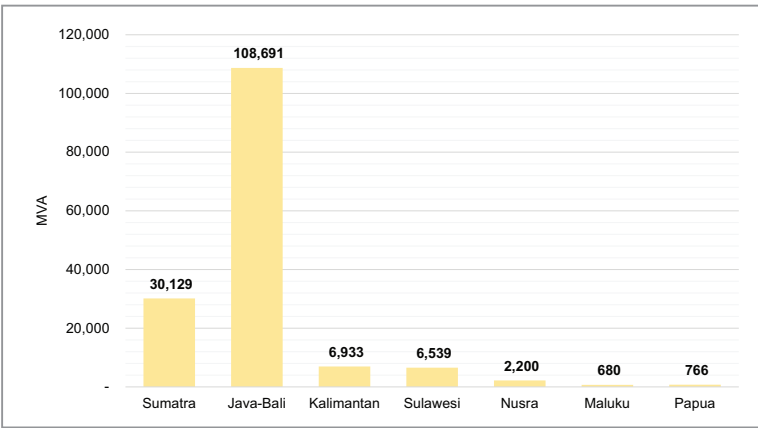


Source: DG of Electricity, processed by Sec Gen of NEC

Figure 2.24 Power Plant Capacity per Region 2022

2.4.3.2 Substation

Total substation capacity in 2021 is around 155,938 MVA and distribution substation capacity is 64,340.88 MVA based on Electricity Statistics, Directorate General of Electricity MEMR. From the total substation capacity, there are 2,269 national substations. The distribution of substation can be seen in Figure 2.25.



Source: Electricity Statistics, 2022

Figure 2.25 Substation per Region 2021

2.4.3.3 Electric Charging Station

Until 2022, there are 439 Public Electric Vehicle Charging Stations (SPKLU) and 975 Public Electric Vehicle Battery Swap Stations (SPBKLU) spread in 328 locations such as Fuel Filling Station and Gas Filling Station, shopping center, office area, parking area and hotels. The location of SPKLU and SPBKLU is shown in Table 2.8 below.

Table 2.8 Distribution of SPKLU and SPBKLU 2022

No.	Location	SPKLU (Unit)	SPBKLU (Unit)
1	Sumatra	45	25
2	Banten	25	227
3	DKI Jakarta	118	359
4	West Java	50	189
5	Central Java & DIY	30	35
6	East Java, Bali & Nusra	136	126
7	Sulawesi, Kalimantan & Maluku	35	14
Total		439	975
		1,414	

Source: Asosiasi Pengusaha Pengisian Kendaraan Listrik Indonesia (APPKLI), 2022

Currently, the issues in developing electric vehicle are high price of electric vehicle especially electric cars, the long queue for purchasing electric cars, and also the public's doubt on the availability of supporting infrastructure such as SPKLU and SPBKLU. In this regard, Minister of Energy and Mineral Resources Regulation Number 1 of 2023 concerning Electric Charging Infrastructure for Battery-Based Electric Vehicles (KBLBB) has been issued to regulate types of technology, integration of applications in providing electric charging infrastructure for battery-based electric vehicles, and the application of electricity tariffs for battery-based electric vehicles.

2.4.4 Biodiesel Refinery

The national biodiesel refinery has the total production volume of around 10.3 million kiloliters (KL) in 2021. This number increases 19% from the volume in 2020 of around 8.6 million KL. The biodiesel refinery is spread in Sumatra, Java, Kalimantan dan Sulawesi. The data of biodiesel refinery location in 2021 is shown in Table 2.9 below.

Table 2.9 Biodiesel Refinery

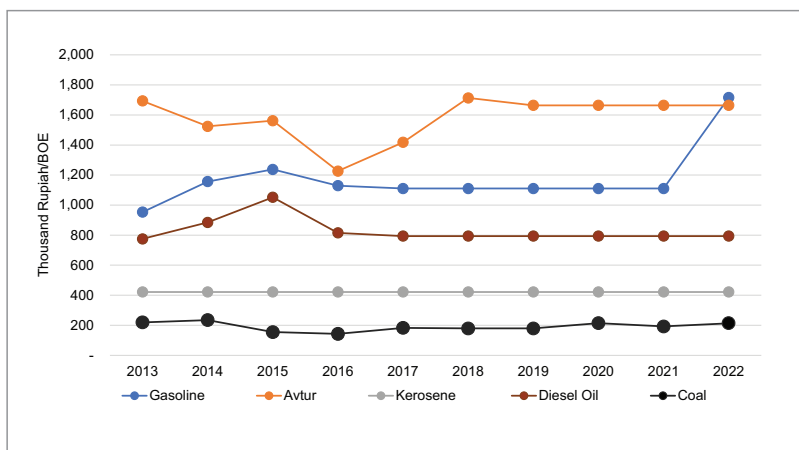
No	Biofuel Business Entity	Total Production Volume (Kiloliter)	Location
1	PT Wilmar Bioenergi Indonesia	1,505,025	Dumai, Riau
2	PT Wilmar Nabati Indonesia	1,955,813	Gresik, East Java
3	PT Multi Nabati Sulawesi	328,360	Bitung, North Sulawesi
4	PT Sinar Alam Permai	26,364	South Sumatra
5	PT Ciliandra Perkasa	280,256	Dumai, Riau
6	PT Musim Mas	829,400	Batam, Kepri dan Deli Serdang, North Sumatra
7	PT Intibenua Perkasatama	228,967	Dumai, Riau
8	PT Sukajadi Sawit Mekar	357,474	Kotawaringin Timur, Central Kalimantan
9	PT LDC Indonesia	375,633	Bandar Lampung, Lampung
10	PT Darmex Biofuels	211,028	Bekasi, West Java
11	PT Dabi Biofuels	24,035	Dumai, Riau
12	PT Bayas Biofuels	283,658	Indragiri Hilir, Riau
13	PT Tunas Baru Lampung	360,406	Bandar Lampung, Lampung
14	PT SMART Tbk	365,356	Kotabaru, South Kalimantan
15	PT Sinarmas Bio Energy	394,126	Bekasi, West Java
16	PT Kutai Refinery Nusantara	376,598	Balikpapan, East Kalimantan
17	PT Cemerlang Energi Perkasa	505,022	Dumai, Riau
18	PT Permata Hijau Palm Oleo	390,696	Medan, North Sumatra
19	PT Pelita Agung Agrindustri	487,632	Bengkalis & Dumai, Riau
20	PT Batara Elok Semesta Terpadu	338,794	Gresik, East Java
21	PT Energi Unggul Persada	424,661	Bontang, East Kalimantan
22	PT Sintong Abadi	1,105	Asahan, North Sumatra
23	PT Multimas Nabati Asahan	91,534	Batubara, North Sumatra
24	PT Jhonli Agro Raya	63,299	Tanah Bumbu, South Kalimantan
25	PT Sari Dumai Oleo	52,710	Dumai, Riau
Total Production Volume		10,257,952	

Source: DG NREEC, MEMR, 2021

2.5 ENERGY PRICE

In principle, the trend of energy demand is influenced by several factors. One of them is energy price. To support the objective of becoming a developed country through industrial development and other energy-using sectors, Indonesia needs energy with good quality at affordable prices. Since the enactment of Minister of Energy and Mineral Resources Decision Number: 37.K/HK.02/MEM.M/2022 in March 2022 concerning Types of Special Oil Fuel for Assignment, there has been a change in the type of special oil fuel

for assignment of minimum RON (Research Octane Number) 88 type to special oil fuel for assignment of type RON 90. This has resulted in a significant increase in the price of special oil fuel (gasoline) for assignment on the market. However, if energy prices are compared in the same unit (thousand rupiah/BOE), coal remains the cheapest energy source as shown in Figure 2.26 below.



Source: HEESI 2022, processed by Sec Gen of NEC

Figure 2.26 Energy Price 2013-2022

2.5.1 Electricity Price

Based on the PLN statistical report 2022, the average selling price of electricity (electricity tariff) per kWh in 2022 is IDR 1,137. It increases from the previous year of IDR 1,083. The tariff is still below the basic cost of providing electricity, which is IDR. 1,473 per kWh. Electricity prices in the household sector in the category of 450 VA power customers and some 900 VA power households (low-income households) are still subsidized. While for other sectors, economic prices are used. The determination of electricity tariffs by the government is regulated in Minister of Energy and Mineral Resources Regulation Number 3 of 2020 concerning the Fourth Amendment to Minister of Energy and Mineral Resources Regulation Number 28 of 2016 concerning Electricity Tariffs provided by PT PLN (Persero). This Minister of Energy and Mineral Resources Regulation also stipulates that PT PLN (Persero) is obliged to announce electricity tariff adjustments no later than one month before the new tariff adjustments are implemented. This is intended to increase transparency in the implementation of domestic electricity tariffs. The development of electricity tariffs per sector in the last ten years can be seen in Table 2.10.

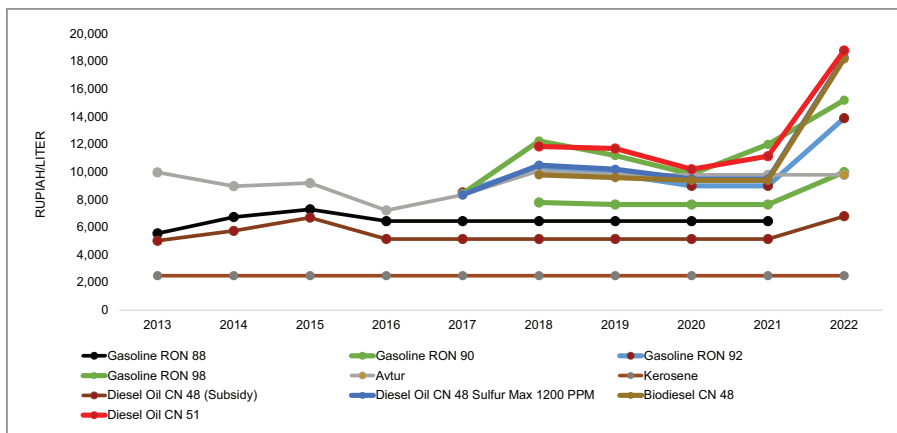
Table 2.10 Average Electricity Tariff per Customer Group (Rupiah/kWh)

Year	Household	Industry	Bussiness	Social	Government	Public Street Lighting	Average
2013	692.06	796.35	1116.58	756.88	1092.22	910.97	818.41
2014	758.16	977.77	1265.87	809.98	1256.2	1096.95	939.74
2015	837	1,142.70	1,284.20	812.4	1,324.50	1,459.10	1,034.50
2016	843.7	1,051.80	1,201.20	816	1,234.70	1,415.30	991.4
2017	1,056.00	1,088.80	1,245.60	821.3	1,278.50	1,461.50	1,105.10
2018	1,102.40	1,085.30	1,244.00	823.2	1,280.10	1,461.60	1,123.00
2019	1,098.80	1,100.70	1,258.30	830.7	1,291.50	1,465.10	1,129.60
2020	991.9	1,090.90	1,239.30	804.1	1,299.20	1,459.90	1,071.40
2021	1,024.01	1,086.22	1,234.68	806.37	1,292.85	1,447.06	1,083.30
2022	1,128.41	1,080.66	1,255.56	833.56	1,445.59	1,577.94	1,137.26

Source: PLN Statistics, 2022

2.5.2 Oil Fuel Price

Oil fuel prices are still subsidized, namely diesel oil for goods transportation and kerosene for households. These two types of oil fuel are a certain type of oil fuel. Meanwhile, subsidies for other oil fuels have been removed since 2014 and the prices follow economic prices. In March 2022, the government designated gasoline with an octane rating of 90 as a special type of oil fuel for assignment. It thereby replaces the previous type of premium gasoline with an octane rating of 88. The price of gasoline with an octane rating of 90 after being designated as special oil fuel for assignment will be IDR. 10,000 per liter. It increases from the previous price of Rp. 7,650 per liter, which is also influenced by the increase in world oil prices. The average trend of oil fuel prices in the last ten years is shown in Figure 2.27.

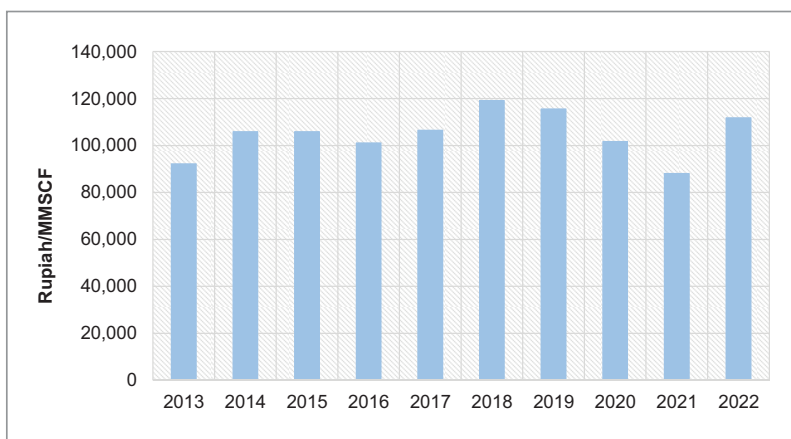


Source: HEESI 2022, processed by Sec Gen NEC

Figure 2.27 Oil Fuel Price 2013-2022

2.5.3 Natural Gas Price

Gas prices are based on an agreement between the seller or distributor and the buyer (consumer). However, since 2015 the government has issued regulations regarding industrial gas prices due to high gas prices in the industrial sector. In 2016, the gas price policy of 6 US dollars per MMBTU was issued as stated in Presidential Regulation Number 40 of 2016 concerning Determination of Gas Price. Under this Presidential Regulation, Minister of Energy and Mineral Resources Regulation Number 8 of 2020 concerning Procedure for Determining Certain Gas User and Price in the Industrial Sector is then issued. The technical regulations are outlined in Minister of Energy and Mineral Resources Decision Number 89 K/10/MEM/2020 concerning Certain Gas User and Price in the Industrial Sector. This regulation states the seven industrial sectors which receive gas at a special price of 6 US dollars per MMBTU, namely the Fertilizer, Petrochemical, Oleochemical, Steel, Ceramics, Glass and Rubber Glove Industries. Based on these regulations, this price scheme is valid from 2020 to 2024. The development of gas prices over the last ten years is shown in Figure 2.28.

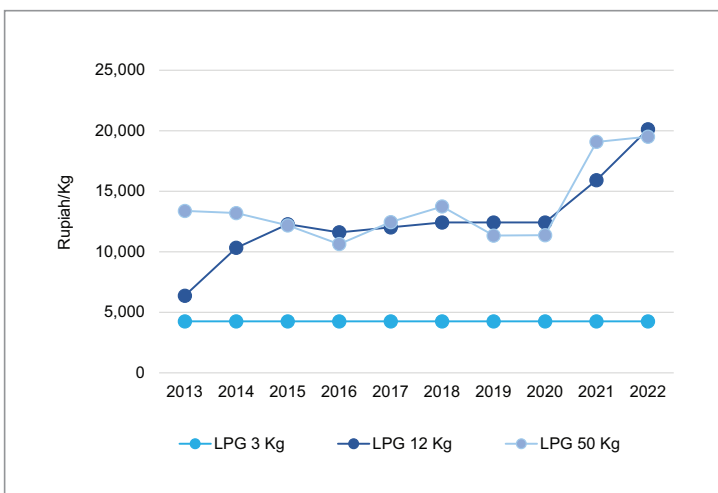


Source: PT.PLN Statistics, processed by Sec Gen of NEC

Figure 2.28 Gas Price 2013-2022

2.5.4 LPG Price

Most of LPG sold in Indonesia is subsidized LPG, namely the 3 kg cylinder LPG for low-income or poor family. However, in reality, it is used for higher income family or commercial sector, resulting in off targeted subsidies. The price of subsidized LPG is far below the price of normal LPG. However, in reality, around 85% of households actually use subsidized LPG. The comparison of subsidized and non-subsidized LPG prices can be seen in Figure 2.29.

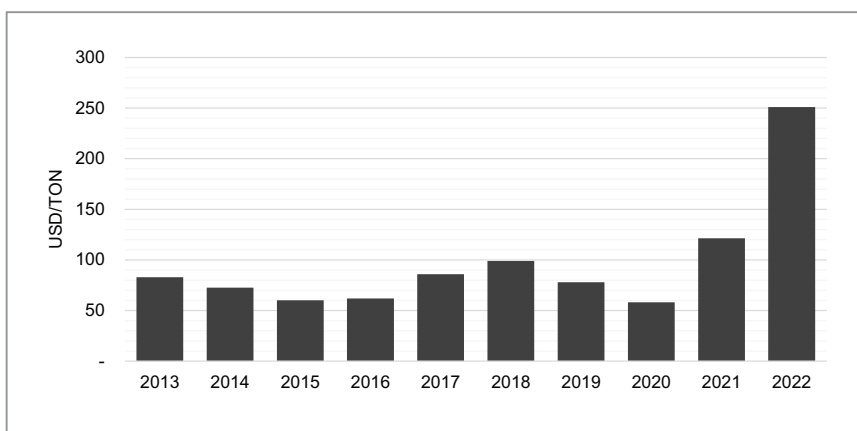


Source: HEESI 2022, processed by Sec Gen of NEC

Figure 2.29 LPG Price 2013-2022

2.5.5 Coal Price

The Directorate General of Mineral and Coal issues monthly Coal Reference Price as a benchmark price for domestic coal. The average coal reference price in 2022 reaches \$251 USD. The coal reference price in 2022 is the highest average benchmark coal price in the last 10 years (the highest Coal Reference Price is \$330.97 USD in October 2022). The development of the average coal reference price from 2012 to 2021 can be seen in Figure 2.30.



Source: DG Mineral and Coal (Coal Reference Price)

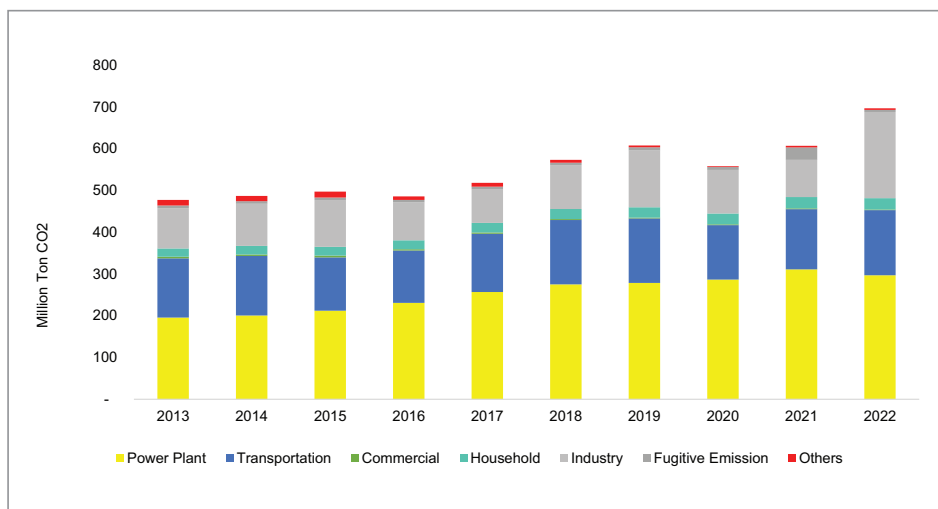
Figure 2.30 Coal Price 2013-2022

2.6 CO₂ EMISSION

Total CO₂ emissions by sector in 2022 are around 696.75 million tonnes of CO₂. It grows around 14.8% compared to emissions in 2021. The calculation of emissions using a sectoral approach is based on two main activities, namely fuel combustion and fugitive emissions. Based on fuel combustion activities in 2022, the largest CO₂ emissions come from power generation of around 297 million tons of CO₂ (42.6%). Furthermore, the use of fuel in industry contributes emissions of 206.4 million tonnes of CO₂ (29.6%), followed by activities in the transportation sector with the contribution of 155.6 million tonnes of CO₂ (22.3%) and the rest is by the household and commercial sectors with the contribution of 27.1 million tonnes of CO₂ (3.9%), 3.9 million tonnes of CO₂ (0.6%) and 1.8 million tonnes of CO₂ (0.3%) respectively.

Fugitive emissions are CO₂ emissions that are accidentally released during fuel production and supply. Fugitive emissions in 2022 are 4.9 million tons of CO₂ (0.7%), triggered by mining activities such as petroleum mining, natural gas mining and mind mouth coal.

The emission trend in 2013-2022 experiences an average growth of around 4.3% per year with the highest growth coming from the industrial sector at around 8.8% per year, followed by the electricity generation sector at around 4.8%. The trend of emissions in the last 10 years can be seen in Figure 2.31 below.



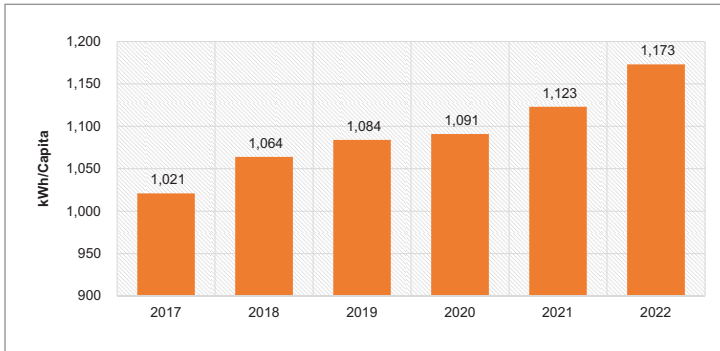
Source: Pusdatin MEMR, 2022

Figure 2.31 CO₂ Emission per Sector 2013-2022

2.7 FINAL ENERGY INDICATOR

2.7.1 Electricity Consumption per Capita

Electricity consumption per capita over the last 5 years grows by 2.8%. The trend of electricity consumption per capita for 2017-2022 based on data from Pusdatin, MEMR can be seen in Figure 2.32.

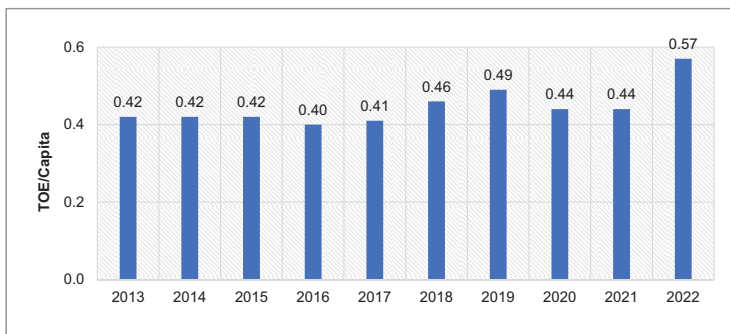


Source: Pusdatin MEMR, processed by Sec Gen NEC

Figure 2.32 Electricity Consumption per Capita 2017-2022

2.7.2 Final Energy per Capita

Final energy per capita is the average amount of energy consumed by each person. This figure is an indicator that can be used to measure the level of community welfare. In general, the higher the energy consumption per capita, the higher the people's standard of living. The trend of national final energy consumption per capita in the last 10 years can be seen in Figure 2.33.



Source: HEESI, 2022

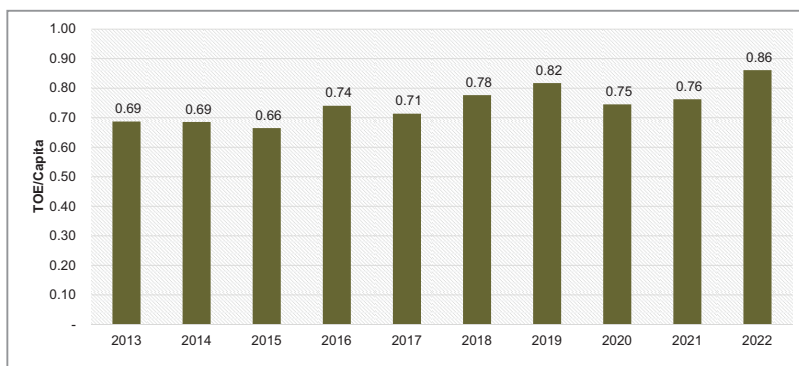
Figure 2.33 Final Energy Consumption per Capita 2013 – 2022

Final energy consumption per capita national in 2022 is 0.57 TOE/capita/year. It grows at the average of 3.3% per year since 2013.

2.7.3 Primary Energy per Capita

Indonesia primary energy per capita in 2013 - 2022 experiences a growth of around 2.5% per year. In 2022, it reaches 0.86 TOE/capita.

The utilization of primary energy is expected to continue to increase until it reaches the target in RUEN, namely 1.4 TOE/capita in 2025, and 3.2 TOE/capita in 2050. The trend of primary energy per capita can be seen in Figure 2.34.



Source: HEESI, 2022

Figure 2.34 Primary Energy Consumption per Capita 2013 – 2022

2.7.4 Emission per Capita

Emission per capita is the average amount of CO₂ emissions resulting from energy use by each person. Annual emissions growth from 2013 to 2022 is around 2.7% per year. The trend of emission per capita in 2013 - 2022 is shown in Figure 2.35.

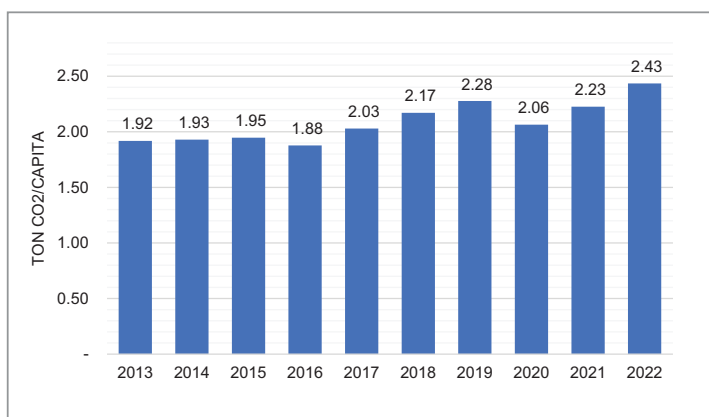
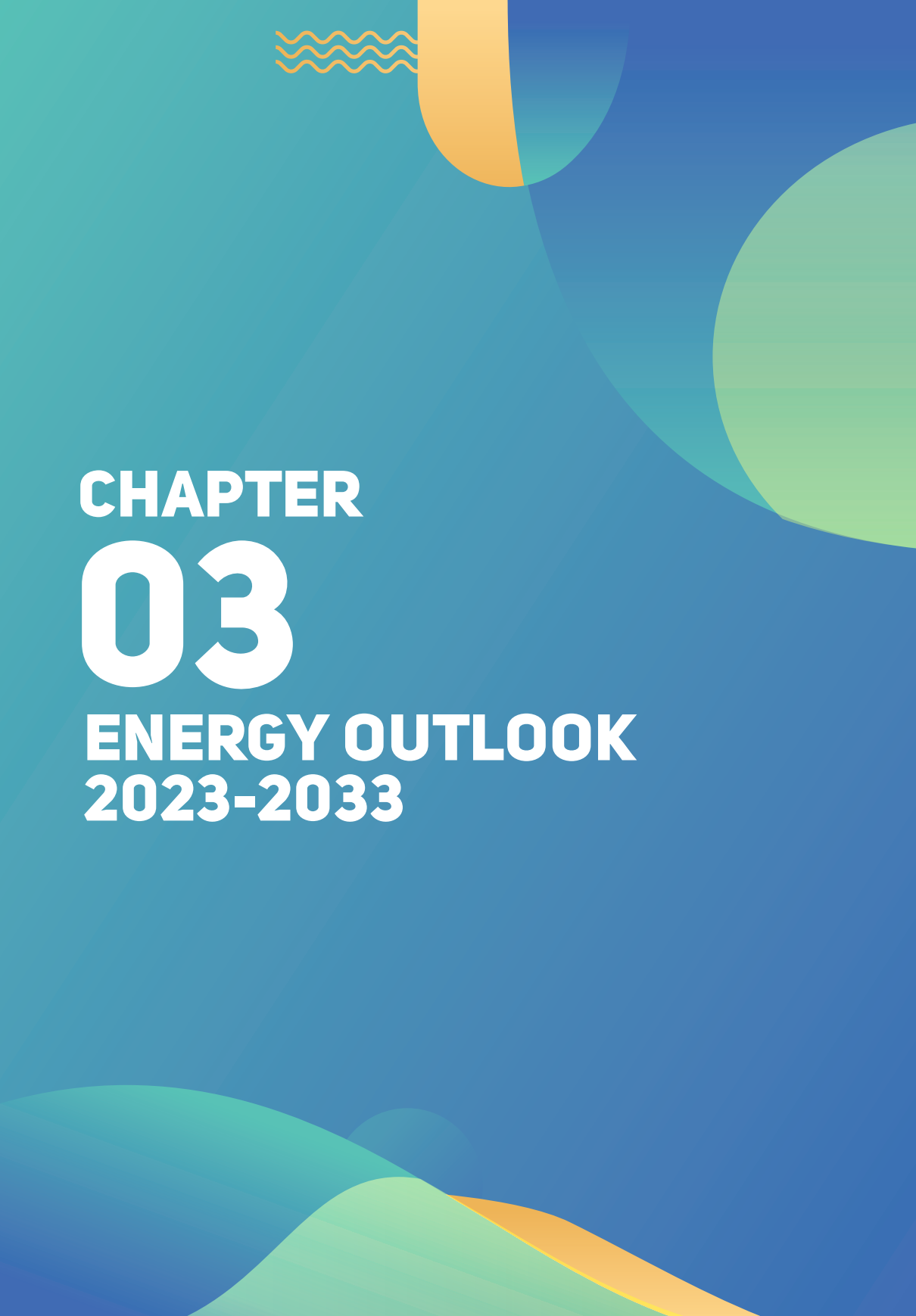


Figure 2.35 Emission per Capita 2013 – 2022



CHAPTER

03

**ENERGY OUTLOOK
2023-2033**

03

ENERGY OUTLOOK 2023-2033

3.1 FINAL ENERGY CONSUMPTION

3.1.1 Final Energy Consumption by Sector

Energy demand in each scenario is influenced by economic growth, population growth, and policies made during the projection period. In the next ten years, final energy demand in Hymne scenario is projected to grow 4.6% in average. Meanwhile, final energy demand in Mars scenario is projected to grow 3.5% in average or lower than Hymne scenario along with the energy efficiency in all sectors. The energy demand in each scenario in 2025, 2030 and 2033 is shown in Figure 3.1 below.

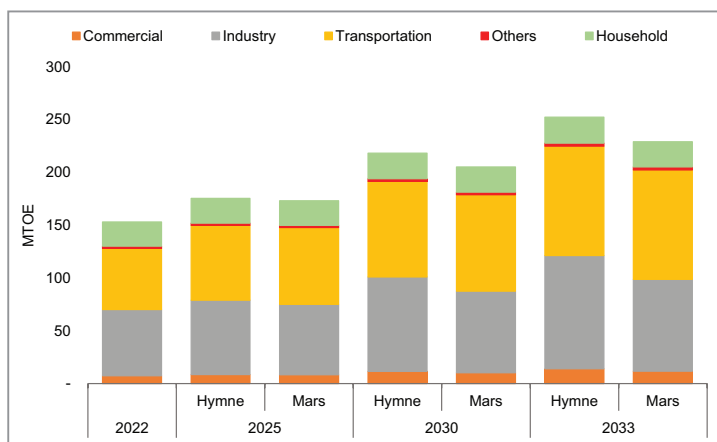


Figure 3.1 Final Energy Demand by Sector

3.1.2 Final Energy Consumption by Energy Type

Oil dominates final energy consumption in 2022 until the next ten year. Thus, in 2033 oil consumption will reach 113 million TOE in Hymne and 96 million TOE in Mars. However, the average oil fuel consumption growth in the two scenarios is below the average growth of final energy consumption namely 4% in Hymne and 2.5% in Mars. This condition is influenced by the oil fuel to power substitution, Biodiesel B40 utilization, E5 ethanol, and 0.045 million TOE hydrogen in transportation starting in 2031.

The biggest energy consumption growth based on its type is NRE (biodiesel and bioethanol) which is growing 7.4%/year in Hymne and 7.5%/year in Mars with 40% FAME (Fatty Acid Methyl Ester) mixture in diesel in 2025 and 10% ethanol in gasoline in 2025. In the next ten years, gas consumption will grow 3.2% in Hymne and 2% in Mars, while coal consumption will grow 5.4% in Hymne and 3% in Mars.

The trend of final energy consumption by energy type is shown in Figure 3.2 below.

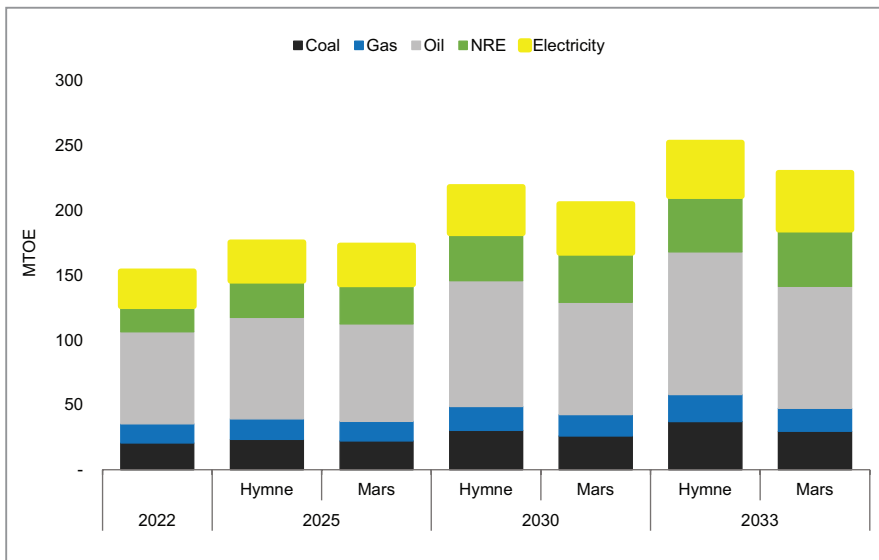


Figure 3.2 Final Energy Demand by Energy Type

3.1.3 Final Energy Consumption by Region

In 2023, energy consumption by region is dominated by Java-Bali, but the average energy consumption growth in Java-Bali is the smallest compared to other regions. Meanwhile, the biggest energy consumption growth is in Sulawesi, namely 6.9% in Hymne and 5.8% in Mars. One of them is influenced by the growing mineral processing and smelter industry especially in South East Sulawesi, South Sulawesi, Central Sulawesi, and West Sulawesi. Final energy consumption by region is as shown in Figure 3.3 below.

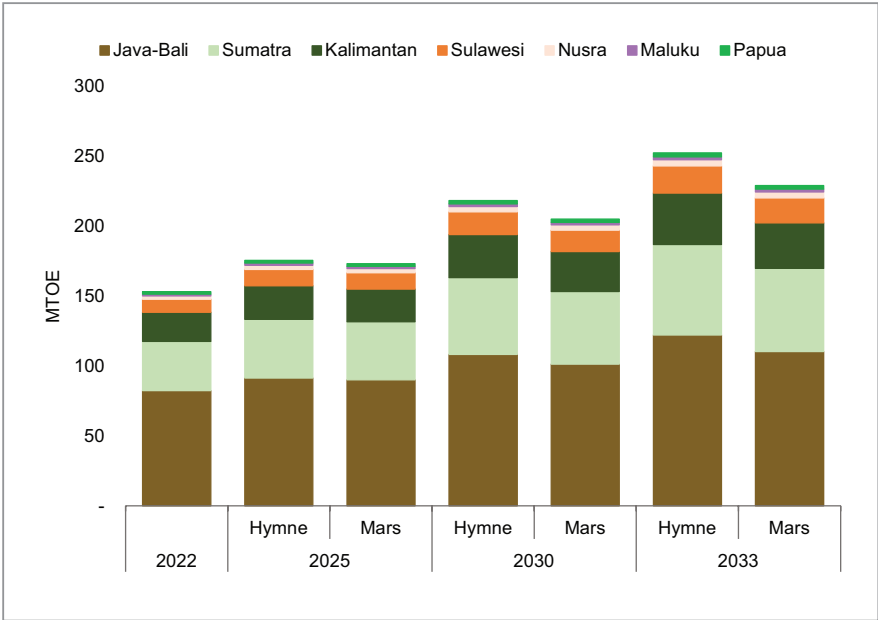


Figure 3.3 Final Energy Consumption by Region

3.2 ELECTRICITY

3.2.1 National Electricity Consumption

Total electricity demand (including electricity consumption in several smelter industries) in 2033 is projected to increase from 313 TWh in 2022 to 479 TWh (Hymne) and 488 TWh (Mars) as shown in Figure 3.4. In 2033, the biggest electricity consumer in both scenarios is the industrial sector, amounting 42% in Hymne and 44% in Mars.

The highest growth of electricity consumption in the next 10 years is the transportation sector, reaching 47% in Hymne and 60% in Mars. Thus, electricity consumption in

transportation will increase from 0.14 TWh in 2022 into 9.5 TWh in Hymne and 24 TWh in Mars in 2033. This condition is supported by the issuance of Government Regulation No 55 of 2019 on the Acceleration of Battery Based Electric Vehicle Program for public transportation. This regulation is followed by other supporting regulations such as Presidential Instruction No. 7 of 2022 on Battery Based Electric Vehicle as Operational or Civil Vehicle for Central Government and Regional Government, Minister of Industry Regulation No. 6 of 2023 on IDR 7 million subsidy for the purchase of two wheeled electric vehicle with the quota of 200,000 units in 2023 and 600,000 units in 2024.

Until Januari 2022, Traffic Unit of Indonesian Police data shows there have been 36 thousand units of electric vehicle, consisting of 30 thousand units of motorcycle and the rest are cars. With the government’s policy that supports the development of EV, it is projected that in 2033 there will be 0.9 million in Hymne and 1.9 million in Mars for cars and 1.9 million in Hymne and 9.8 million in Mars for motorcycle. The growing electricity consumption by sector during the projection year can be seen in Figure 3.4.

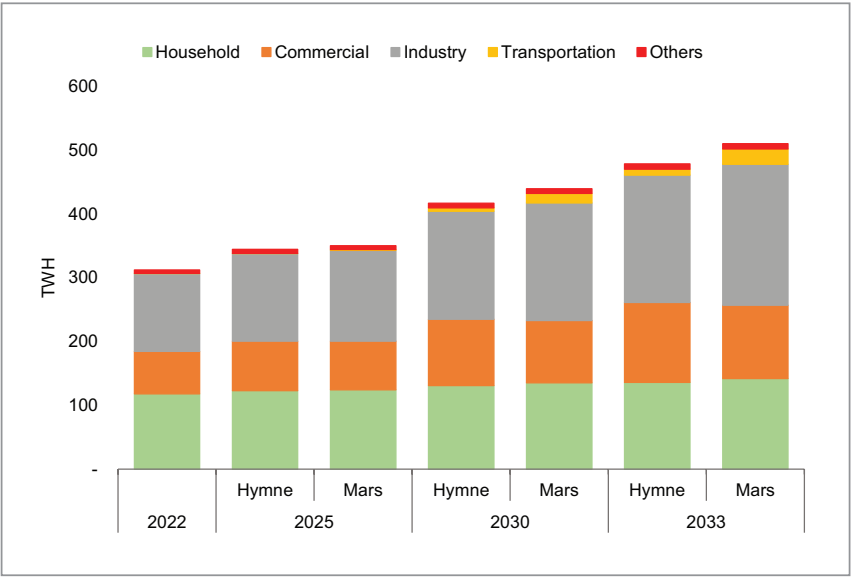


Figure 3.4 National Electricity Demand by Sector

3.2.2 Electricity Consumption by Region

Based on the regions, electricity demand until 2033 is still centered in Java-Bali since the development in Java-Bali in various sector are more advanced compared to other regions. In 2022, Papua in which the area is six times bigger that Java Island consumes

only 0.7% of the total national electricity due to the small number of population and the low development in this eastern most region of Indonesia. In the next 10 years, electricity demand in Papua will increase about 4.8% in Hymne and 8.8% in Mars. However, the share remains at only 0.7% from the total national electricity demand. The detail description of electricity consumption by region can be seen in Figure 3.5 below.

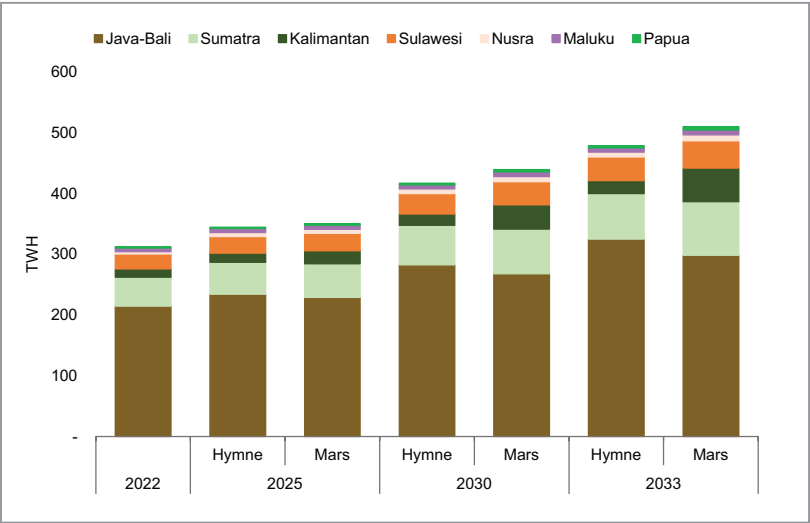


Figure 3.5 Electricity Consumption by Region

3.2.3 National Electricity Production

Total electricity production is projected to increase from 351 TWh in 2022 to 538 TWh in Hymne and 549 TWh in Mars in 2033 by taking into account 10% lost in transmission and distribution.

Electricity production in the next 10 years is still dominated by fossil energy especially coal. In 2022, electricity production from coal fired power plant reaches 71% from the total production, but the share is projected to decline into to 63% in Hymne and 38% in Mars. The declining coal share in Mars scenario is influenced by co-firing biomass program in several Coal Fired Power Plants. The capacity of coal fired power plant which implement co-firing is 14 GW in 2022 and 17 GW in 2033.

The co-firing in coal fired power plant will reach 15% starting in 2030 in Hymne scenario and 15% starting in 2030 in Mars scenario that it would give impact on the declining electricity from coal.

On the contrary, electricity production from NRE power will be growing significantly in the next ten years with 9% in Hymne and 15% in Mars. In Mars scenario, the power plant is developed to optimize NRE power plant especially solar power plant which has abundant resources in Indonesia and biomass power plant (including co-firing) as well as other bioenergy power plant such biogas and waste. In addition, the future development of power plant is also aimed to reduce the role of oil fuel in power generation replacing with NRE or gas power plant. This would give impact on the declining production of electricity from diesel power plant up to 1% in 2033. The utilization of diesel power plant still exists, but it is only prioritized for disadvantages, frontier and outermost areas and other energy inaccessible areas. The electricity production per energy type for both scenarios can be seen in Figure 3.6 below.

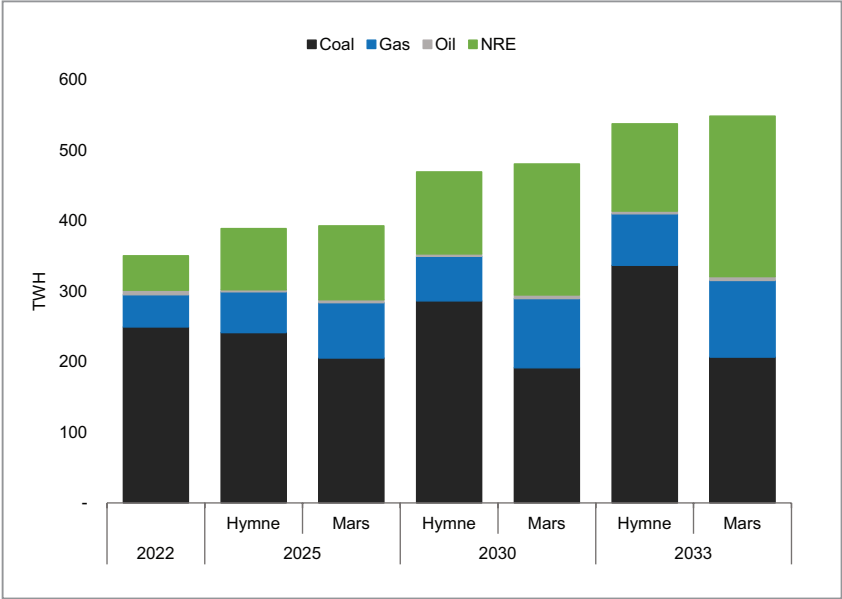


Figure 3.6 Electricity Production by Energy Type

Electricity production from NRE power plant will increase from 49 TWh in 2022 into 122 TWh in Hymne and 224 TWh in Mars. Similar to current condition, electricity production from NRE in 2033 for Hymne scenario will be dominated by Hydro Power Plant (38%), Geothermal Power Plant (33%), and Bioenergy Power Plant which include Waste Power Plant, Biomass Power Plant and Biogas Power Plant (21%). Meanwhile, electricity production in Mars scenario will be dominated by Solar Power Plant and Bioenergy Power Plant with the share of 28% and 32% respectively.

The biggest growth in NRE power plant is from Solar Power Plant with 61%. Its production increases from 0.4 TWh in 2022 into 68 TWh in 2033. This condition is supported by the continuous decline of Solar Power Plant installation cost based on data by IRENA (International Renewable Energy). The data says that Solar Power Plant cost is continuously declining from USD 0.444/kWh in 2010 into USD 0.049/kWh in 2022.

In 2032, Nuclear Power Plant will be introduced in Mars scenario with electricity production of 701 GWh. Until today, there has been no national commitment to build Nuclear Power Plant. However, in order to meet clean energy demand, nuclear needs to be considered to be included in the grid. Several site locations for Nuclear Power Plant have been prepared such as in Bangka Belitung Province and West Kalimantan. The illustration of electricity production from NRE power plant can be seen in Figure 3.7 below.

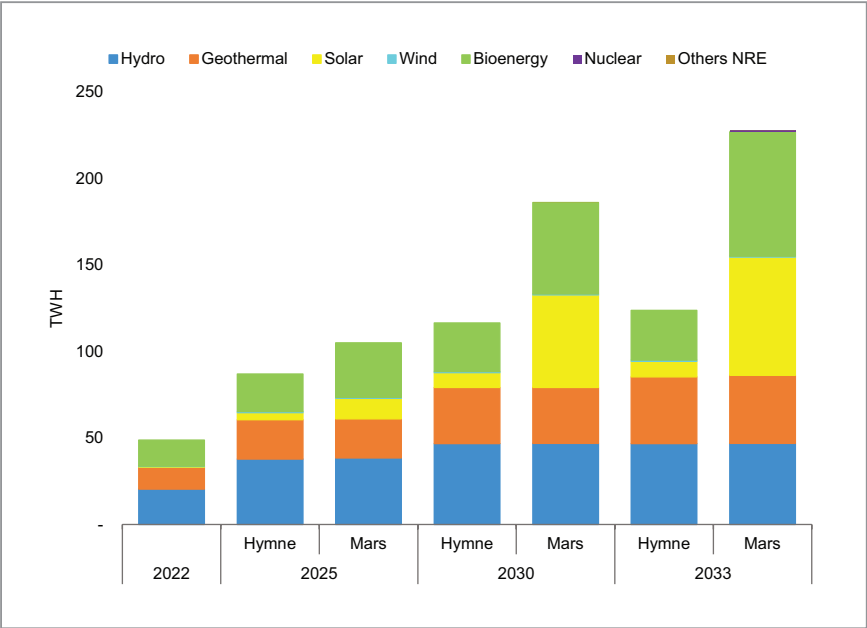


Figure 3.7 Electricity Production from NRE Power Plant

3.2.4 Electricity Production by Region

Similar to projection of electricity consumption per region, the trend of electricity production per region is also dominated by Java-Bali with 68% in Hymne and 61% in Mars. Meanwhile, the smallest share is in Papua with 0.7% in Hymne and 1% in Mars. The detail of electricity production by region is shown in Figure 3.8 below.

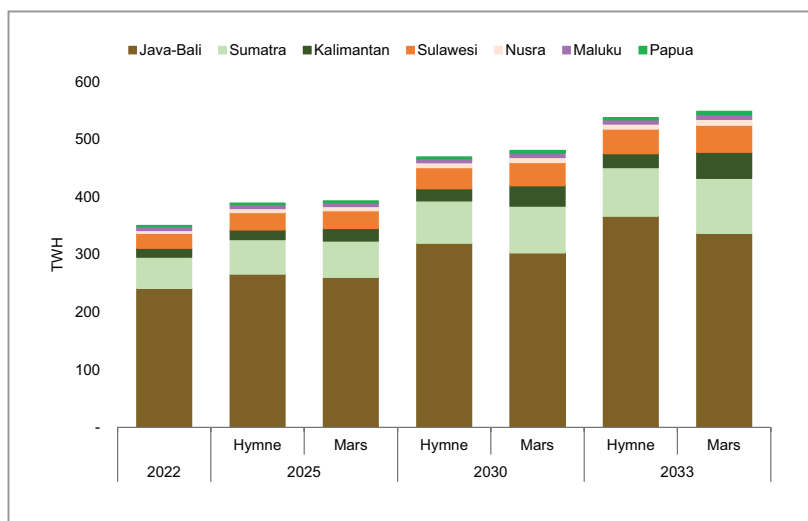


Figure 3.8 Electricity Production by Region

In 2022, Sumatra has the biggest share of NRE based-electricity production with 46% while Maluku has the smallest share of NRE based-electricity production with 0.02%. In the next 10 years, electricity production from NRE power plant is projected to increase with the highest growth in Maluku with 60% for Hymne scenario and 67% for Mars scenario. This is influenced by the condition in Maluku in which currently the electricity production is still dominated by fossil power plant especially diesel power plant. The electricity production from NRE power plant in Hymne and Mars scenario can be seen in Figure 3.9 below.

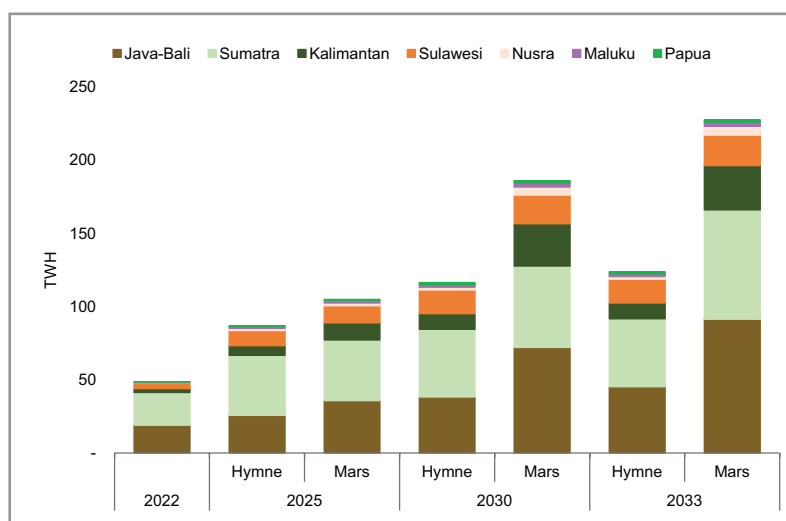


Figure 3.9 Electricity Production from NRE Power Plant by Region

3.2.5 National Power Plant Capacity

The power plant capacity will increase at the average of 4% in Hymne and 6% in Mars into 135 GW in Hymne and 188 GW in Mars in 2033. All types of power plant experiences positive growth, except diesel power plant which shows negative growth of -1.5% in Hymne scenario and -19.6% in Mars scenario. This is in line with the de-dieselizeation to reduce 5,200 diesel power plants that are currently operating.

The capacity of NRE power plant experiences the highest growth of 7% in Hymne and 14% in Mars. The capacity increases from 13 GW in 2022 into 26 GW in Hymne and 53 GW in Mars. The details of power plant capacity by scenario can be seen in Figure 3.10 below.

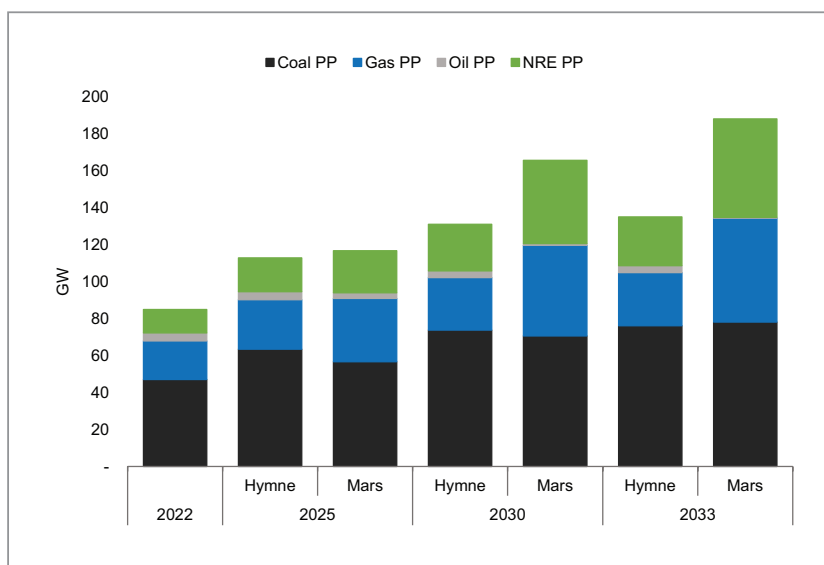


Figure 3.10 Power Plant Capacity by Scenario

The biggest growth in the capacity of NRE power plant for both scenarios are from solar power plant with 30% in Hymne and 51% in Mars. Thus, the capacity of solar power plant will increase from 0.3 GW in 2022 into 5 GW in Hymne and 26 GW in Mars in 2033.

In Hymne scenario, power plant with the biggest capacity is hydro power plant with 43% share of total capacity, but in Mars scenario solar power plant will dominate NRE power plant with 49%. The trend of power plant capacity in Hymne and Mars scenario can be seen in Figure 3.11 below.

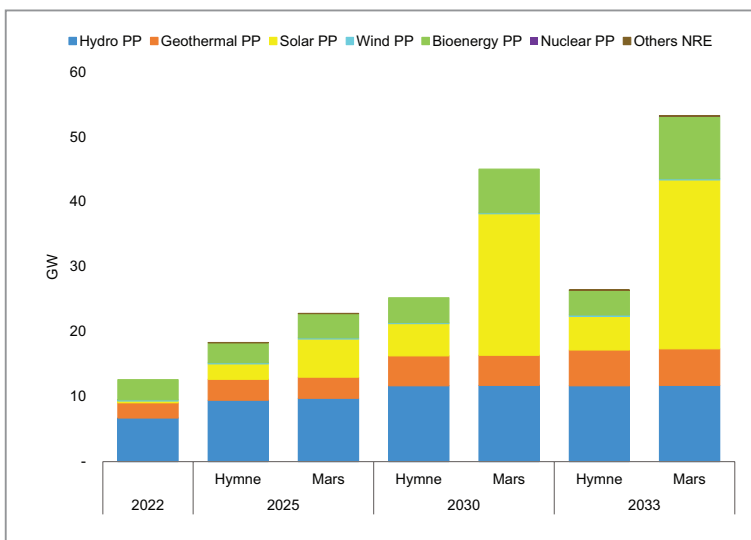


Figure 3.11 NRE Power Plant Capacity by Scenario

3.2.6 Electricity Power Plant Capacity by Region

In 2022 to 2033, 58% of power plant capacity is in Java-Bali region. Mars scenario assumes optimization in NRE power plant that it would impact the high growth in power plant capacity in Kalimantan with 17% along with New Capital City (Ibu Kota Nusantara) development program and the growing industry in this region. The details of power plant capacity by region can be seen in Figure 3.12 below.

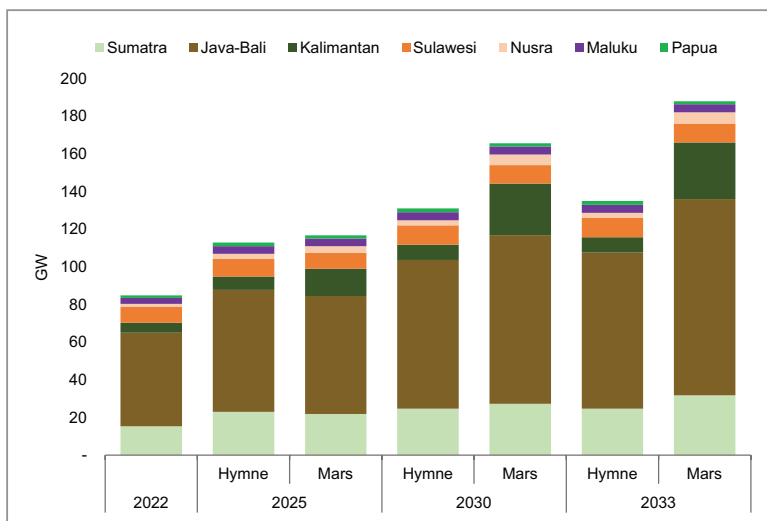


Figure 3.12 Power Plant Capacity by Region

The region with the biggest share in NRE power plant capacity in 2022 is Sumatra with 44%, followed by Java-Bali with 36%. However, in the next 10 years, it is projected that the biggest NRE power capacity will be found in Java-Bali with 38% in Hymne and 45% in Mars as the impact of solar rooftop development program which is very potential to be developed in Java.

The highest growth of power plant capacity is in Maluku region with 60%. Thus, power plant capacity in Maluku increases from 3.9 MW in 2022 into 751 MW in 2033. The details of NRE power plant capacity by region is presented in Figure 3.13 below.

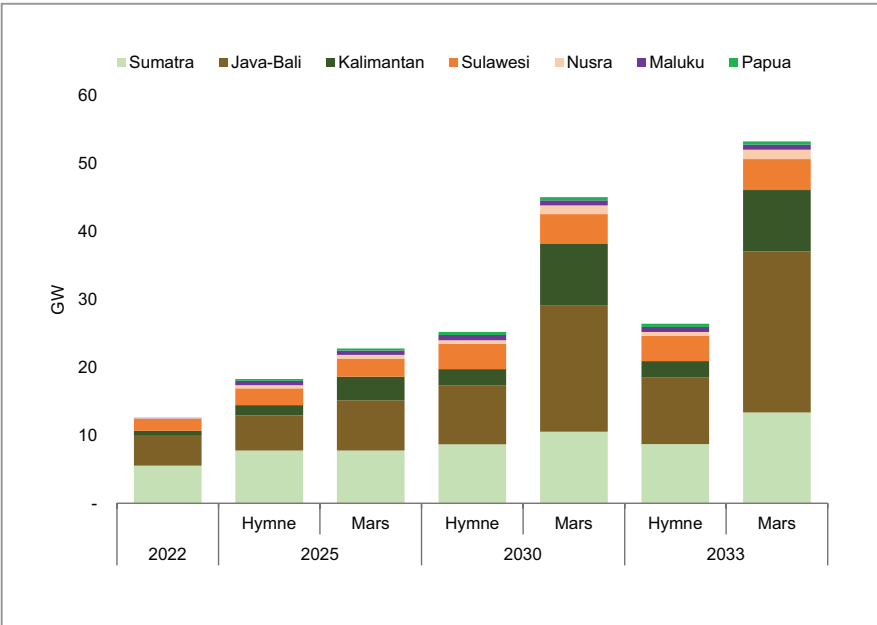


Figure 3.13 NRE Power Plant Capacity by Region

3.3 PRIMARY ENERGY SUPPLY

3.3.1 Primary Energy Supply by Energy Type

Total primary energy supply in 2033 will increase around 4.2% and 3.6% into 379 million TOE in Hymne and 353 million TOE in Mars. In Hymne, coal primary energy supply grows approximately 3.4% per year but in Mars it grows negatively of -0.4% per year due to the declining coal consumption in power plant in Mars scenario along with the deployment of co-firing program in several coal fired power plants. On the other hand,

primary energy supply from NRE shows the highest growth of 8% in Hymne and 12% in Mars that is influenced by the increasing NRE power plant capacity especially solar power plant and biofuel utilization. The projection of primary energy supply by energy type for both scenarios is shown in Figure 3.14 below.

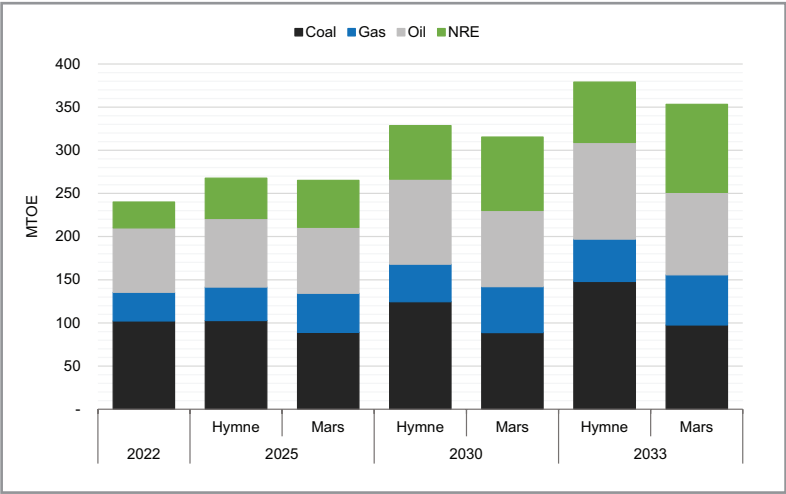


Figure 3.14 Primary Energy Supply

In 2025, NRE supply in Hymne scenario reaches 37.2 million TOE or 17.5% from the total primary energy supply. Meanwhile, in Mars scenario, NRE supply is about 45 million TOE or 20.5% from total primary energy supply. The details of primary energy mix 2025 for both scenarios are presented in Figure 3.15 below.

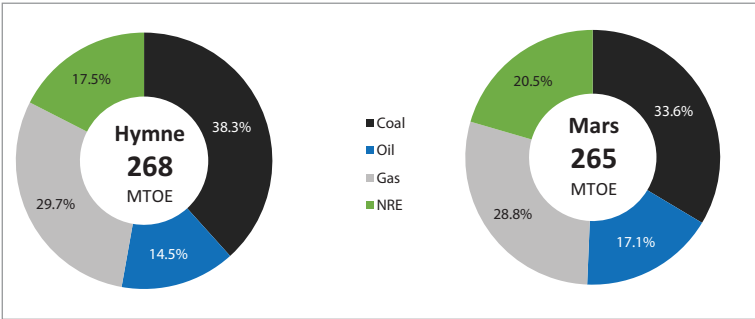


Figure 3.15 Primary Energy Mix 2025

In 2033, NRE supply in Hymne reaches 70 million TOE or around 18.5% from the total primary energy supply. Meanwhile in Mars scenario, NRE supply is around 102 million TOE or 28.9% from the total primary energy supply. The primary energy mix 2033 in both scenarios is presented in Figure 3.16 below.

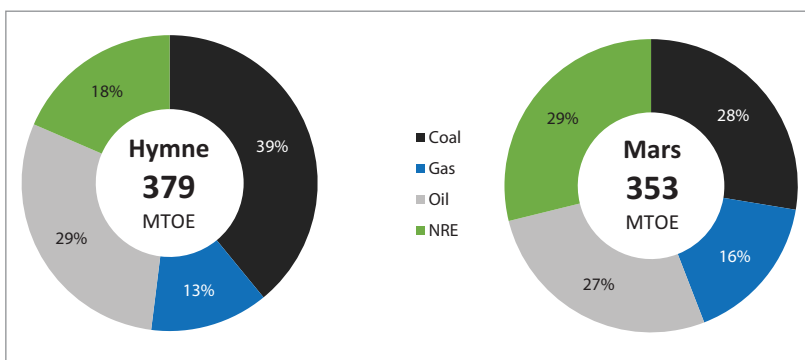


Figure 3.16 Primary Energy Mix 2033

Primary energy supply is divided into fossil energy and non-fossil energy. Fossil primary energy consists of coal, gas, and oil. Meanwhile, non-fossil energy consists of various renewable energy and new energy (NRE) such as hydro, solar, wind, geothermal, bioenergy, nuclear, hydrogen, and other renewable energy sources.

3.3.1.1 Fossil Primary Energy

The growth of fossil primary energy supply (coal, gas and oil) is 3.5% in Hymne and only 1.7% in Mars which is particularly influenced by the declining coal supply especially in power plant and the declining oil utilization in transportation due to oil fuel to EV substitution and the increasing use of biofuel. The projection of fossil primary energy supply per energy type in both scenarios is shown in Figure 3.17 below.

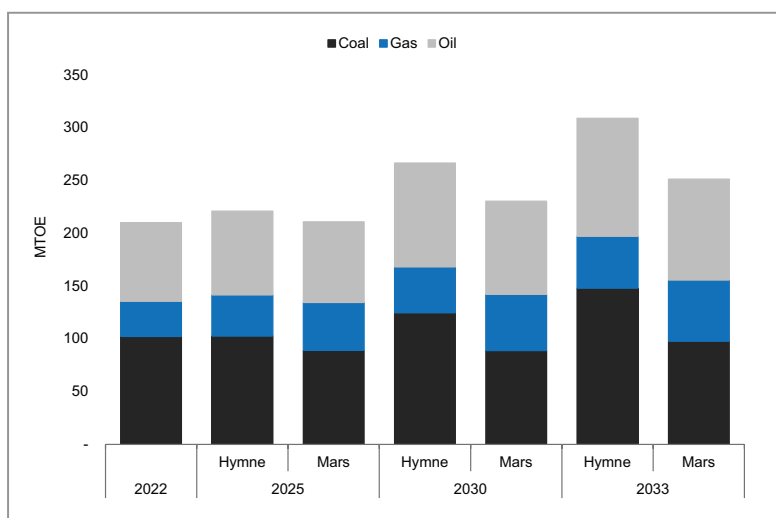


Figure 3.17 Fossil Primary Energy Supply

3.3.1.2 Non-Fossil Primary Energy

The average growth of non-fossil primary energy supply is 8% in Hymne and 12% in Mars. In 2033, the share of renewable energy still becomes the main option in Hymne, but in Mars scenario there has been the utilization of new energy amounting to 273 thousand TOE. This new energy includes nuclear energy and hydrogen. Nuclear energy will start to be utilized in 2032, while hydrogen will start to be utilized in 2031 in transportation sector. The projection of non-fossil primary energy mix 2033 in both scenarios is presented in Figure 3.18 below.

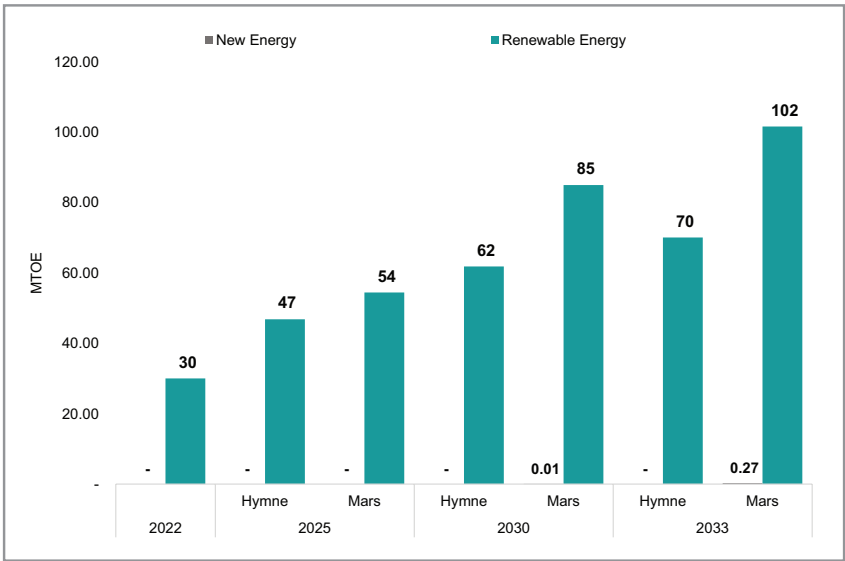


Figure 3.18 Non-Fossil Energy Supply

3.3.2 Renewable Energy and New Energy Supply

NRE supply share in 2022 is still dominated by bioenergy with 47%, followed by biofuel with 32% and hydro with 13%. Bioenergy consists of biogas, biomass, and waste energy that are used in power generation and direct use, while FAME is used as raw material of biofuel in transportation.

In 2033 in both scenarios, the biggest share of NRE supply is derived from bioenergy with 37% in Hymne and 33% in Mars as well as from biofuel with 36% in Hymne and 28% in Mars. In Mars, solar energy grows as the fastest compared to other energy sources. Its share increases from 0.4% in 2022 into 23% in 2033. Primary energy mix projection in 2033 in both scenarios is shown in Figure 3.19 below.

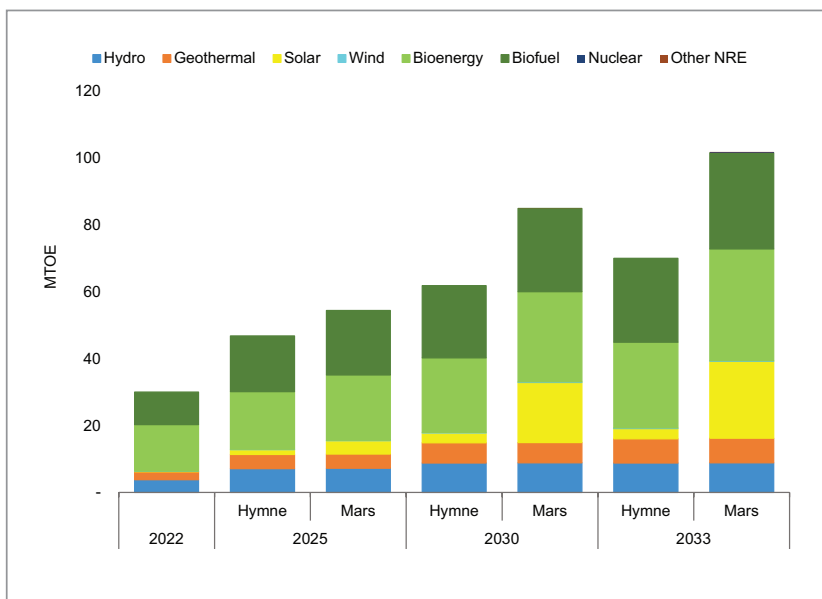


Figure 3.19 New and Renewable Energy Supply

3.3.3 Primary Energy Supply by Region

In both scenarios, the biggest primary energy supply in 2033 is projected to be from Java-Bali, followed by Sumatra. The details of primary energy supply share per region 2022-2033 is shown in Figure 3.20 below.

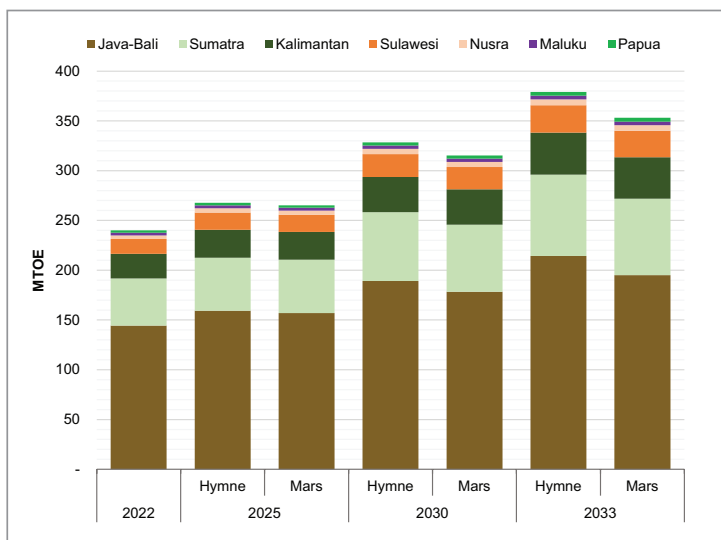


Figure 3.20 Primary Energy Supply by Region 2033

3.4 CO₂ EMISSION

3.4.1 CO₂ Emission by Sector

During the projection period, power plant is still the biggest emitter due to the dominant use of fossil energy especially coal which is amounting to 47% in Hymne and 44% in Mars in 2033. The second biggest emitter is transportation due to the dominant use of oil fuel in vehicles. The share of emission in transportation in 2033 reaches 25% in Hymne and 29% in Mars as shown in Figure 3.21 below.

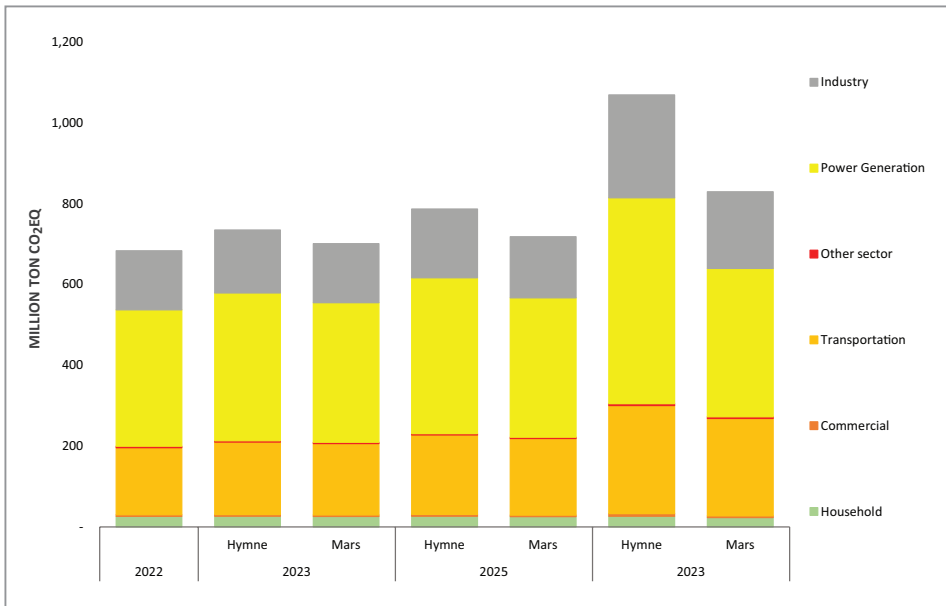


Figure 3.21 CO₂ Emission by Sector

3.4.2 CO₂ Emission by Region

In 2033, Java-Bali is still the biggest emitter compared to the other regions in both scenarios, amounting to 64% from the total national CO₂ emission as shown in Figure 3.22 below.

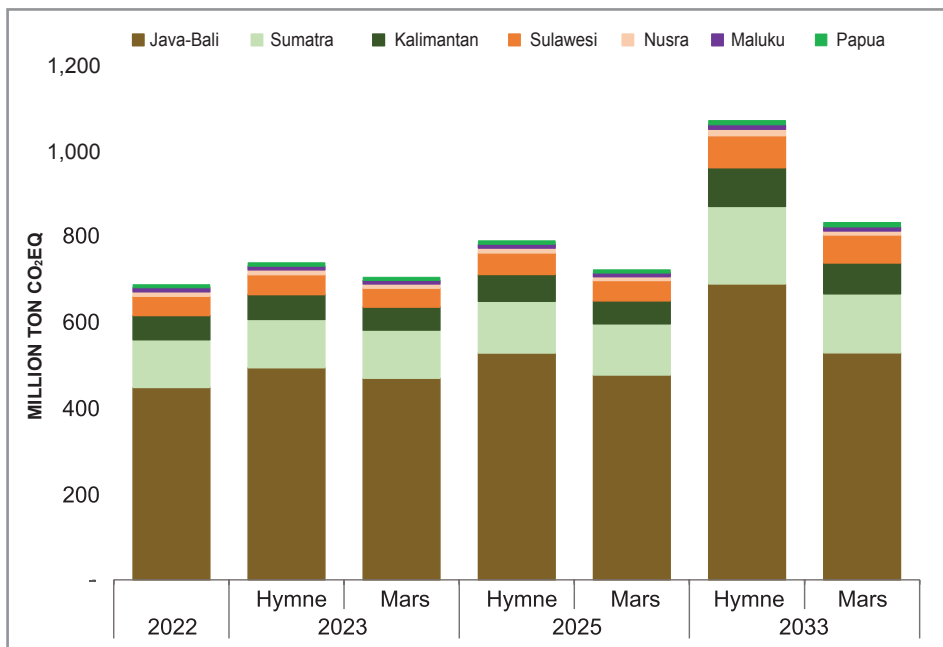


Figure 3.22 CO₂ Emission per Region

3.5 ENERGY INDICATOR

3.5.1 Electricity Consumption per Capita

In 2025, electricity demand per capita in both scenarios (Hymne and Mars) are 1,255 kWh/capita and 1,268 kWh/capita or below the target of electricity per capita in KEN of 2,500 kWh/capita in 2025. Meanwhile in 2033, electricity demand per capita will be around 1,662 kWh/capita in Mars scenario and only 1,630 kWh/capita in Hymne scenario. The low electricity per capita is influenced by the electricity consumption basic data for captive power in which the data is not yet well organized such as in mining and other sectors. The projection of electricity per capita for each scenario is shown in Figure 3.23 below.

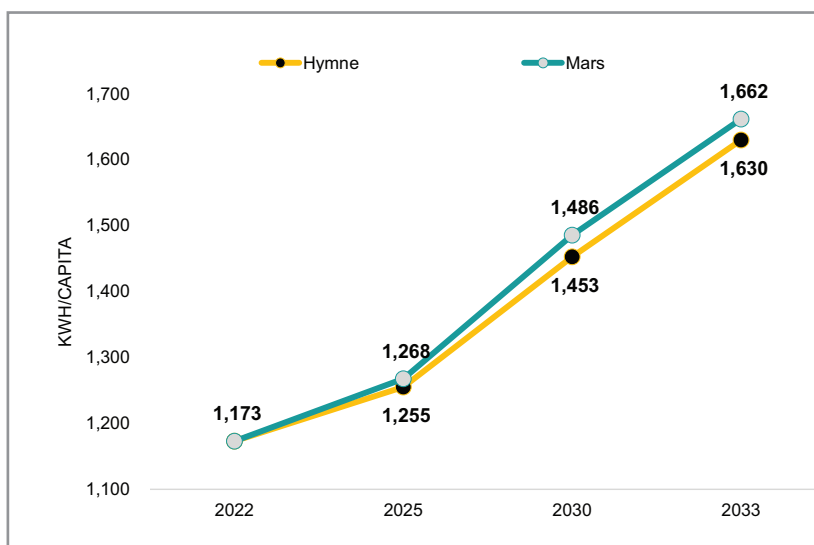


Figure 3.23 Electricity Consumption per Capita

3.5.2 Final Energy per Capita

In 2022, final energy demand per capita is 0.54 TOE/capita. However, in 2033, it is projected to increase into 0.81 TOE/capita in Hymne and 0.72% TOE/capita in Mars. This is influenced by the lower growth of final energy demand in Mars than in Hymne due to energy conversion in Mars. Final energy per capita for both scenario is presented in Figure 3.24 below.

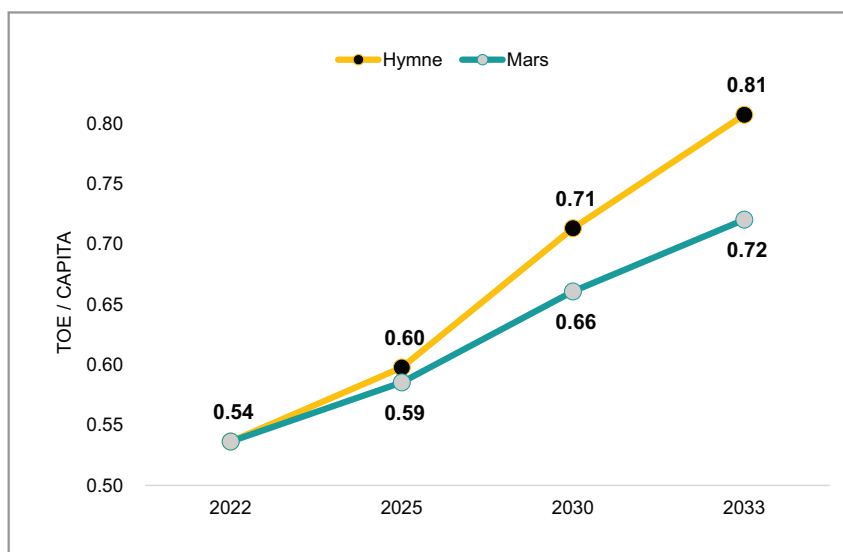


Figure 3.24 Final Energy per Capita

3.5.3 Primary Energy per Capita

In 2033, primary energy supply per capita will reach 1.20 TOE/capita in Hymne and 1.12 TOE/capita in Mars. This projection is far below the target in KEN of 1.4 TOE/capita in 2025. The projection of primary energy supply per capita in each scenario is presented in Figure 3.25 below.

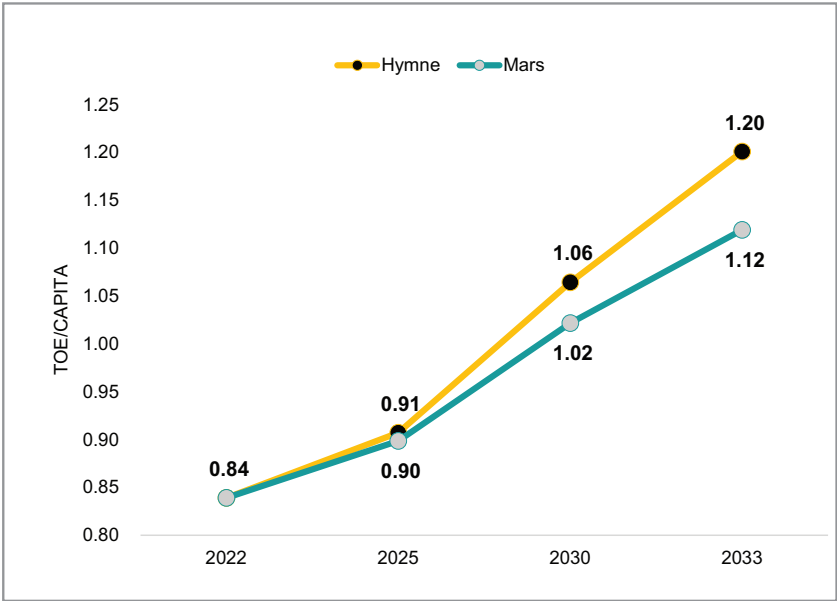


Figure 3.25 Primary Energy per Capita

3.5.4 Emission per Capita

Total emission in 2033 is projected to increase into 1,080 million ton CO₂eq in Hymne and 840 million ton CO₂eq in Mars. The emission in both scenarios is still lower than emission target set in energy sector NDC of 1,355 million ton CO₂eq in 2030. The growth of GHG emission in both scenarios per sector is presented in Figure 3.26 below.

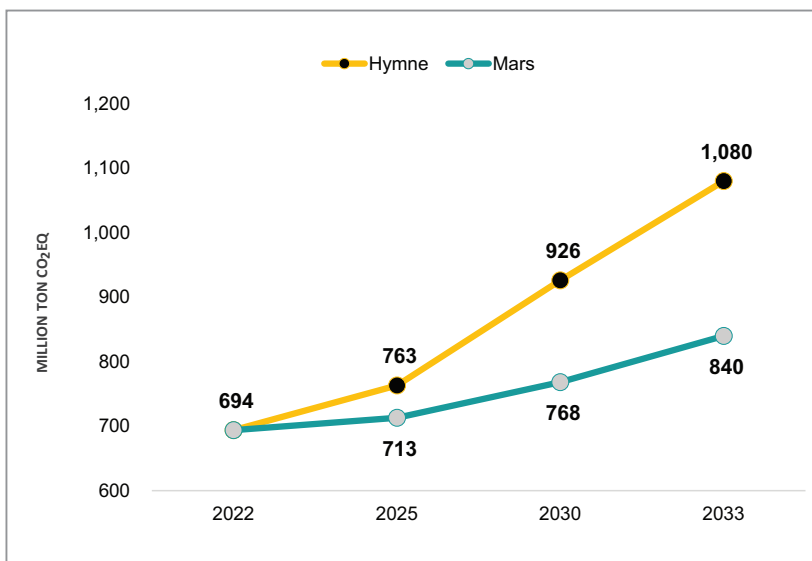


Figure 3.26 Emission per Scenario

In 2033, emission per capita is projected to increase into 3.42 million ton CO₂eq/capita in Hymne and 2.66 million ton CO₂eq/capita in Mars. The lower emission per capita in Mars is due to the increasing NRE use. The projection of national emission per capita is presented in Figure 3.27 below.

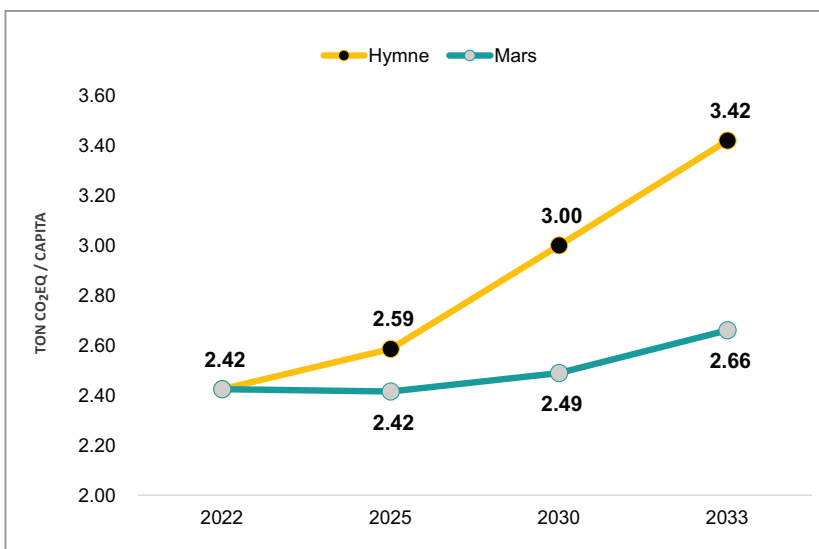


Figure 3.27 Emission per Capita

3.6 PROGRESS OF ACHIEVING PRIMARY ENERGY MIX TARGET (COMPARISON WITH RUEN)

KEN sets a number of national targets. One of them is national primary energy mix by 2025 and 2050. In 2025, the target to meet the share of at least 23% NRE, 5% oil, 30% coal and 22% gas. In details, RUEN carries out annual primary energy mix projection to meet KEN target. The comparison between primary energy mix projection in 2025 in Hymne and Mars scenario and RUEN is presented in Figure 3.28 below.

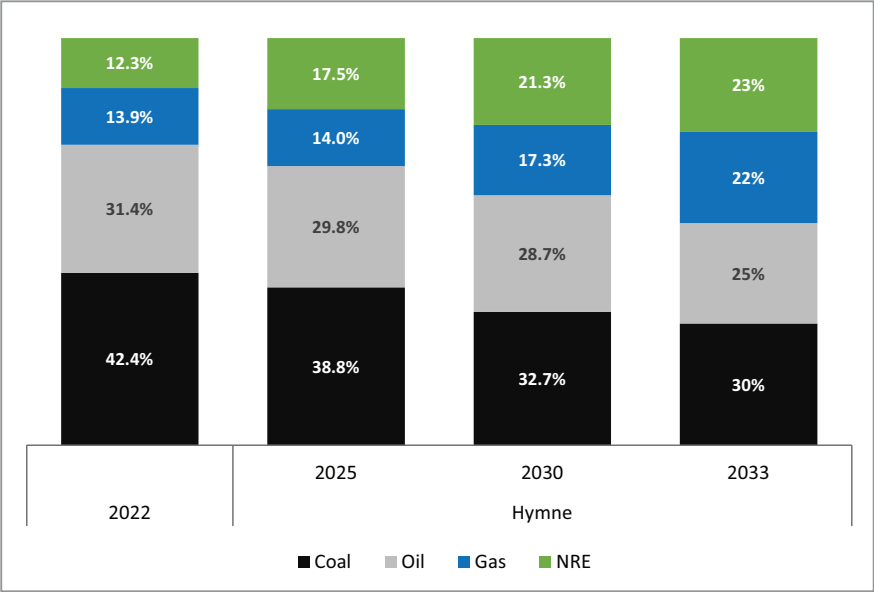


Figure 3.28 Comparison of Primary Energy Mix Projection 2025

The result of modeling, both in Hymne and Mars, shows that only coal is projected to meet the KEN target in 2025. Meanwhile, NRE is far below the target. NRE mix in 2025 is projected to reach 17.5% in Hymne and 20.5% in Mars. This is quite reasonable since NRE mix in 2022 will only reach 12.3%. In addition, RUEN predicts the need to reach the target of 16% by 2022 to meet the KEN target. On the other hand, NRE share is projected to continue increasing at least until 2033 with 18.5% in Hymne and 28.9% in Mars. In 2030, NRE share in Mars will exceed NRE share in RUEN target as shown in Figure 3.29 below.

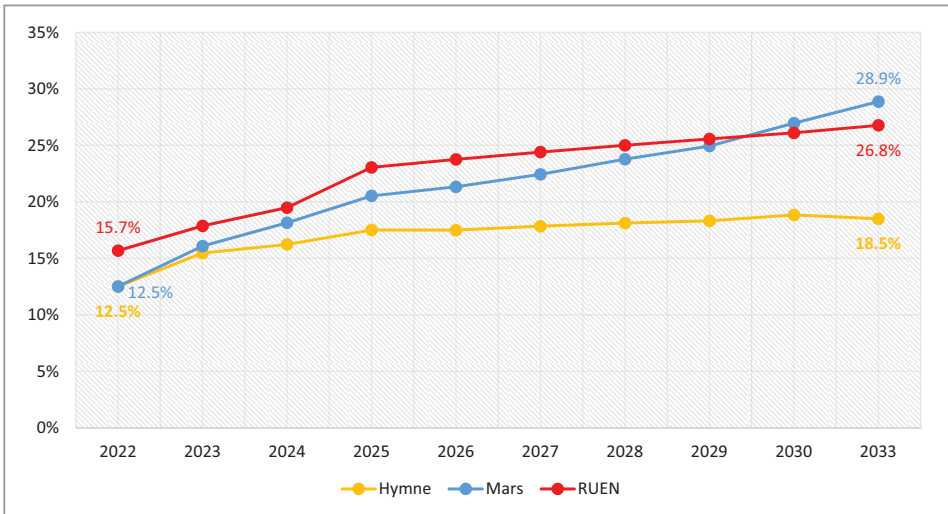
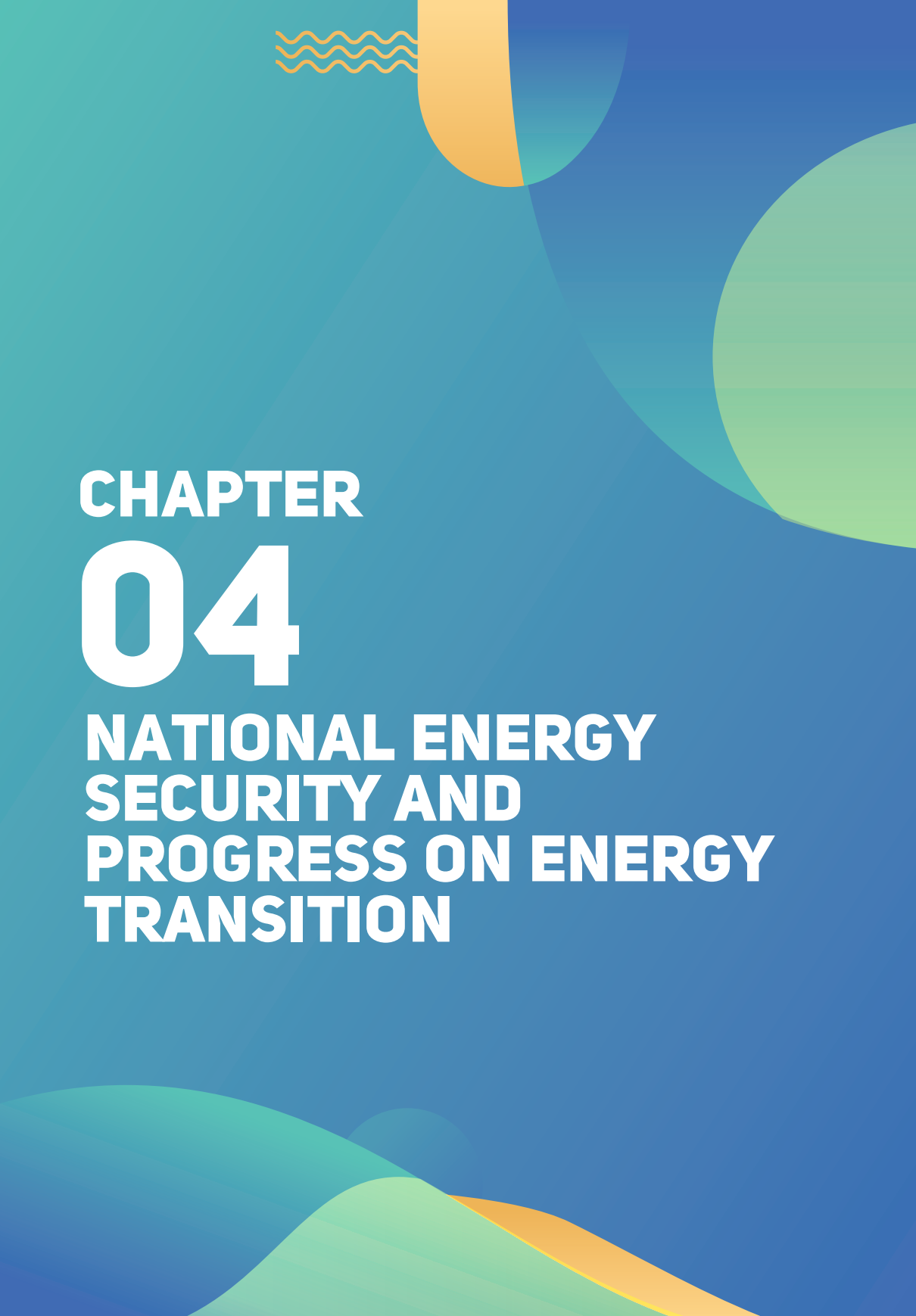


Figure 3.29 Comparison of NRE Share In Energy Mix Until 2033



CHAPTER **04**

NATIONAL ENERGY SECURITY AND PROGRESS ON ENERGY TRANSITION



04

NATIONAL ENERGY SECURITY AND PROGRESS ON ENERGY TRANSITION

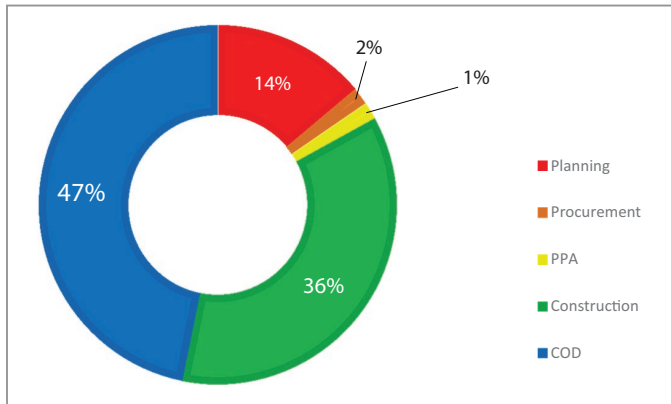
4.1 ENERGY SECURITY

4.1.1 Condition of National Electricity Supply

In 2022, there will be an additional power plant capacity of around 5.34 GW. Compared to the previous year, it increases 12%. There are also additions to transmission, substations, distribution networks, distribution substations and electric vehicle charging stations.

In 2022 there will be additions in transmission network, substations and electric vehicle charging stations that exceed the target. Transmission increases of around 3,591.76 kms, or 133% from the target of 2,709.81 kms. Substations increases of around 6,010 MVA, or 178% from the target of 3,379.5 MVA. Electric vehicle charging stations increases of around 1,415 units, or 204% from the target of 693 units. However, the addition to distribution networks and distribution substations is still below the target. The distribution network increases of around 11,537.73 kms or only around 51% of the target of 22,796 kms. Meanwhile, distribution substations increase of around 1,098 MVA, or only around 56% from the target of 1,952 MVA.

On the other hand, the development of the 35,00 MW power plant project shows that there are already 16,596 MW power plants that have in COD. Meanwhile, power plants with contracts/PPA but not yet under construction are currently in the process of fulfilling funding requirements. The development of the 35,000 WM project can be seen in Figure 4.1 below.



Source: Directorate General of Electricity MEMR

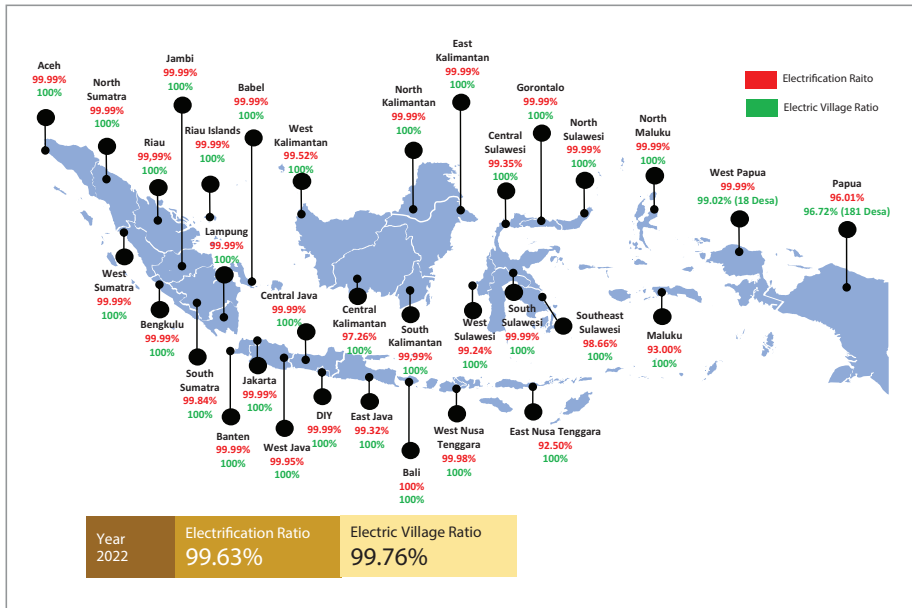
Figure 4.1 Development of 35,000 MW Power Plant

4.1.2 Electricity for Rural Areas and Energy Supply for Disadvantages, Frontier and Outermost Areas

As an archipelagic country that has a large territory and a large population, providing energy access for the community, especially in rural areas as well as in Disadvantaged, Frontier and Outermost areas, is a challenge for the Indonesian government. Various methods and strategies have been developed to provide energy in rural areas and outermost areas in an effort to create equitable development. One of the problems in providing energy in the outermost area is that the distance between villages. The difficulty in access and mobility to outermost areas has an impact on increasing investment costs for developing energy infrastructure, both networks and generators.

Based on data from the Ministry of Energy and Mineral Resources 2022, the national electrification ratio is 99.63%, meaning that there are still around 0.37% or around 260 thousand households that do not yet enjoy electricity. Meanwhile, the ratio of villages with electricity in 2022 will be around 99.76%. It means that there are still around 199 unelectrified villages.

The electrified villages are the number of villages with PLN, Non-PLN and LTSHE (Energy Saving Solar Light) electricity. Data on villages with non-PLN electricity are villages that are electrified by the Regional Government, private sector or non-governmental organization which has been consolidated and verified with the Office of Energy and Mineral Resources of the Provincial Government. The distribution of electrified villages in 2022 can be seen in Figure 4.2 below.

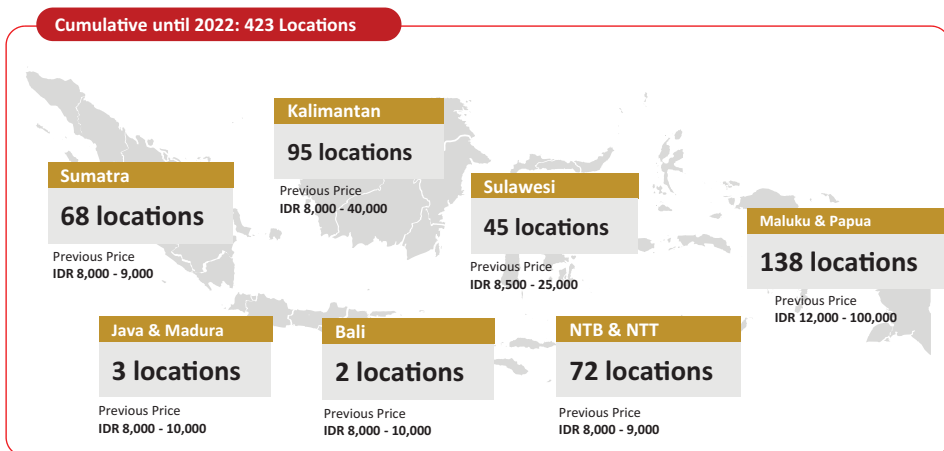


Source: Working Performance 2022 and 2023 Program, MEMR

Figure 4.2 Distribution of Electrified Villages 2022

Besides the electricity program, the government has also established a One-Price Oil Fuel program since 2017 in order to create just energy throughout Indonesia. The background of the policy is the high gap in oil fuel prices in several regions, especially in eastern Indonesia. The frontier and outermost areas are the government's focus in implementing One-Price Oil Fuel program. It is hoped that oil fuel at affordable price can be accessed by people in rural and outermost areas.

To support this plan, the Ministry of Energy and Mineral Resources has stipulated Minister of Energy and Mineral Resources Regulation Number 36 of 2016 concerning the Acceleration of One-Price for Certain Types of Oil Fuel and Assigned Special Types of Oil Fuel Nationally to support this policy. This Ministerial Regulation mandates oil fuel distribution business entities to establish distributors in certain locations, especially locations with no distributors of certain types of oil fuel and special types of oil fuel on assignment. Thus, public can buy oil fuel at the retail selling price set by the government. This One-Price Oil Fuel Program applies to RON 90 gasoline at Rp. 10,000 per litre, and Diesel Oil at Rp. 6,800/litre. Until 2022, there have been 423 points of One-Price Oil Fuel (the total addition in 2022 is 92 distribution points of One-Price Fuel). The Ministry of Energy and Mineral Resources has determined the addition of 71 One-Price Oil Fuel locations until 2023. The distribution of One-Price Oil Fuel points until 2022 can be seen in Figure 4.3 below.



Source: Working Performance 2022 and 2023 Program, MEMR

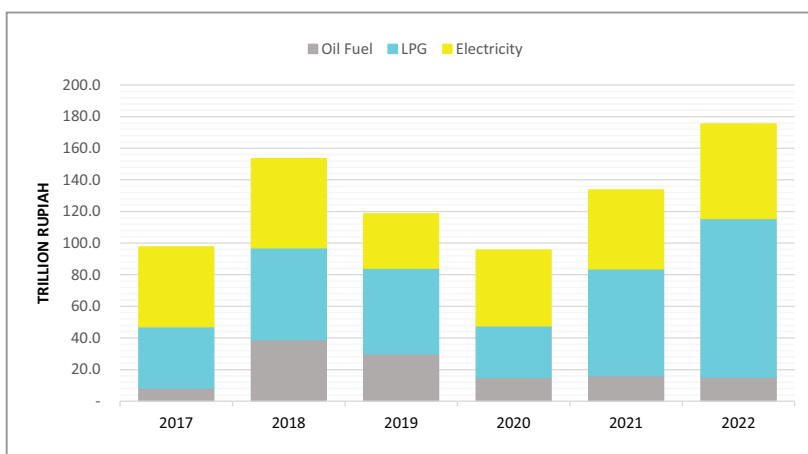
Figure 4.3 Distribution of One-Price Oil Fuel Until 2022

4.1.3 Energy Subsidy

The realization of oil fuel subsidies has fluctuated in the last 5 years. The highest fluctuation occurs in 2018 amounting to 38.9 trillion Rupiah due to an increase in diesel oil subsidies. However, in 2020 the subsidy decreases to 14.9 trillion Rupiah due to the Covid-19 pandemic. In 2021, it increases again to 16.2 trillion Rupiah, and then decreases again to 15.2 trillion Rupiah in 2022.

The realization of LPG subsidies in 2016 is 24.9 trillion Rupiah. It continues to increase until 2018 to 58.1 trillion Rupiah. This increasing LPG load is another form of commitment to convert kerosene to clean energy. LPG subsidies in 2019 decreases of around 6.9% due to the decline in gas prices in the world market. It continues to decline in 2020 to 32.8 trillion Rupiah due to the Covid-19 pandemic. However, in 2021 LPG subsidies increase again to 67.6 trillion Rupiah and in 2022 it increases to around 100.4 trillion Rupiah.

The electricity subsidy also fluctuates. In 2022 electricity subsidy reaches 59.8 trillion Rupiah. The trend of the realization of energy subsidies in 2017-2022 can be seen in Figure 4.4 below.



Source: Pusdatin, MEMR

Figure 4.4 Development of Energy Subsidy 2017-2022

4.2 POLICY ON ENERGY SUBSIDY

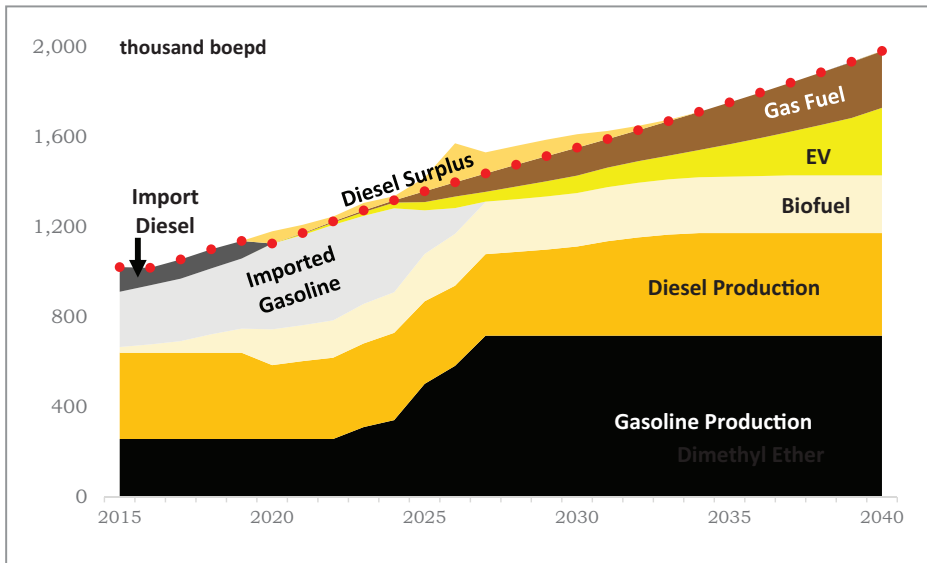
4.2.1 Status of Electric Vehicle Development

To encourage the development of electric cars in 2019, the government issues Presidential Regulation Number 55 of 2019 concerning the Acceleration of the Battery-Based Electric Vehicle (KBLBB) Program for Road Transportation. This regulation is followed by several derivative regulations, including Presidential Instruction Number 7 of 2022 concerning the Use of Battery Based Electric Vehicles as Operational Service Vehicles and/or Individual Service Vehicles for Central Government and Regional Government Agencies. In addition, the Government of Indonesia has also issued policies and incentives to support the growth of the electric vehicle market including tax incentives, exemption from import duties for electric vehicle components, and reduced registration fees for electric vehicles. These steps aim to increase the penetration of electric vehicles as an effort to reduce the dependence on oil fuel imports.

Until November 2022, the Ministry of Energy and Mineral Resources recorded 33,800 units of electric vehicles have been operating in Indonesia. This number is dominated by 25,782 units of electric motorcycle. This transition is also a concern for online transportation companies such as GOJEK and GRAB in their efforts to reduce their carbon footprint. These two transportation companies are starting to provide services using electric motorcycles and cars by providing more than 8,000 units by the end of 2022. In addition, taxi and Transjakarta bus companies are also starting to operate electric-based vehicles.

Despite of the quite significant development of electric vehicles in Indonesia, there are still several challenges, such as high prices, limited electric charging infrastructure and concerns regarding limited mileage or travel distance. However, with government support and public awareness, it is hoped that the development of electric vehicles in Indonesia will continue to increase.

One of the programs in the National Energy Grand Strategy (GSEN) is reducing oil fuel imports through increasing the use of gas fueled vehicles, biofuel and battery based electric vehicle. The target is to increase the number of 2 million units of electric cars and 12 million units of electric motorcycle by 2030. The battery based electric vehicle program is expected to stop oil fuel imports before 2030, as shown in Figure 4.5 below.

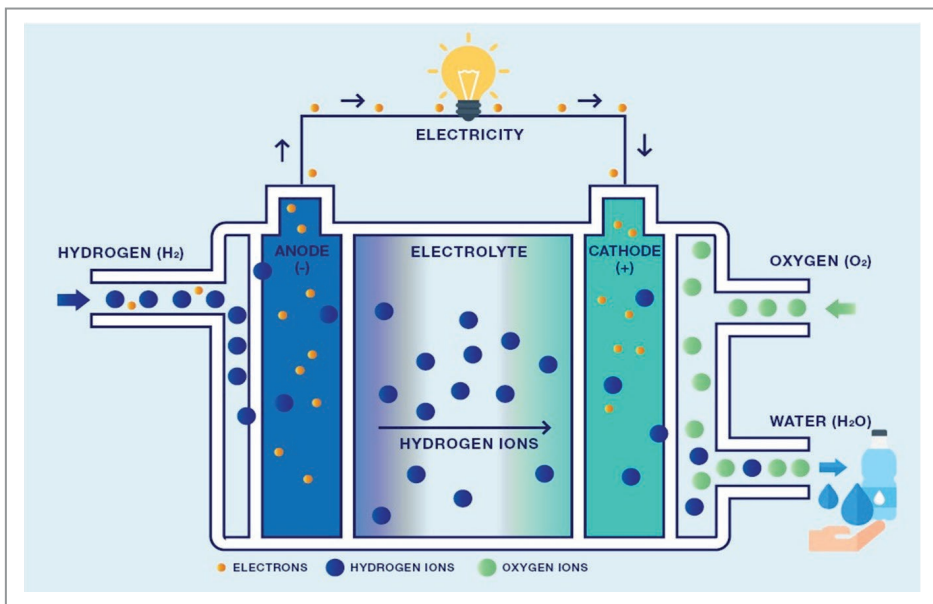


Source: GSEN, 2020

Figure 4.5 GSEN Program to Stop Oil Fuel Import

4.2.2 Status of Hydrogen Development (Including Hydrogen Vehicle)

Hydrogen fuel is a zero-emission fuel that is used as fuel for generating electricity together with oxygen using a unit called a hydrogen fuel cell. Similar to a battery, a fuel cell has an anode and cathode poles where hydrogen (H_2) and oxygen (O_2) flow to the two different poles as shown in Figure 4.6 below.



Source: MEMR, 2021

Figure 4.6 Hydrogen Fuel Cell

Hydrogen can be produced through these following processes:

a. *Gas Steam Reforming*

In this process, the methane contained in natural gas is extracted and reacted with steam to produce hydrogen. About 95% of the hydrogen on the market is produced by this process. This process can produce gray hydrogen and blue hydrogen if the carbon produced is absorbed through CCS technology.

b. *NRE Power Plant Electrolysis*

The electrolysis process uses a device called an electrolyzer. The main concept in this process is the separation of hydrogen and oxygen molecules from water using a reaction caused by an electric current. If this process is carried out at an NRE power plant, green hydrogen will be obtained.

c. *Coal Gasification*

In the gasification process, coal or biomass material is reacted with oxygen and steam to produce synthesis gas. Then, the hydrogen molecules are separated from the synthesis gas using a separation system. The coal gasification process will produce brown hydrogen or blue hydrogen if the carbon produced is absorbed through CCS technology.

d. Biological Process

Microbes such as bacteria and microalgae can produce hydrogen by biological reactions using sunlight or organic materials. This technology is still at the research and development stage.

In Indonesia, hydrogen development is still at the research and pilot project stage and there are no commercial projects yet. Research on hydrogen as a fuel has been carried out by the Agency for the Study and Application of Technology (BPPT), University of Indonesia (UI), Bandung Institute of Technology (ITB) and Lemigas since 2007. Furthermore, in 2012, ITB created a fuel cell car concept and Korea International Cooperation Agency (KOICA) provided a grant to the Indonesian government for the construction of a fuel cell pilot project with a capacity of 300 kilowatts (kW) in Ancol, Jakarta. Then in 2014, the Indonesian Association for Fuel Cell and Hydrogen Energy (INAFHE) was founded to accelerate hydrogen development in Indonesia. In 2017, PT Telkomsel developed fuel cells at the Base Transceiver Station (BTS) as backup power. After that, in 2018, BPPT and Toshiba ESS signed a cooperation agreement regarding the development of the H2One Autonomous Hydrogen Energy System for Off Grid Systems. In 2019, the Ministry of Energy and Mineral Resources, PT Kereta Api Indonesia (KAI) and Allstrom began collaborating in developing hydrogen fuel trains. PT HDF Energi took the initiative to develop green hydrogen from hybrid Solar PP and Wind PP on Sumba Island with a capacity of 7-8 MW during the day and 1-2 MW at night from hydrogen storage. Apart from that, there is also collaboration between PT Pertamina and GIZ in developing a green hydrogen pilot project from geothermal energy. In 2023 PT PLN (Persero) through its subholding PLN Nusantara Power inaugurated the first Green Hydrogen Plant (GHP) in Indonesia which is located in the Muara Karang Steam Gas Power Plant (PLTGU) area, Pluit, Jakarta. This GHP is 100 percent sourced from solar energy and is capable of producing 51 tons of hydrogen per year. Of the total production, 43 tons can be used for 147 cars that can cover a distance of 100 km every day.

4.2.3 Status of Electric Stove

The Indonesian government has encouraged the use of electric stoves as part of energy transition efforts to reduce dependence on fossil fuels and reduce greenhouse gas emissions. The government has also tested the use of electric stoves in several areas and distributed free electric stoves to the public. This trial aims to measure the effectiveness and public acceptance on the use of electric stoves.

The government plans to remove subsidies for LPG in the long term and replace them with the use of electric stoves. This is aimed to reduce the burden of energy subsidies and encourage the use of clean energy. Meanwhile, one effort to reduce LPG imports is to use electric stoves. In GSEN, several programs to reduce LPG imports include: Development of the City Gas Network, Increasing the Production Capacity of New Oil Refineries, Development of Dimethyl Ether (DME) and the use of electric stoves.

In contrast to the LPG supply from imports, the electricity supply in Java-Bali based on 2021 RUPTL data shows that the installed capacity of power plants on Java-Bali Island is 41 GW, while the peak load is 29 GW. It means that there is a reserve margin above 30% in Java-Bali region. One of the efforts to increase the growth of electricity consumption is by increasing the use of electric stoves in Java-Bali. The estimated energy consumption for using an electric stove with a power of 2,000 Watt based on a study by the Research and Development Agency, Ministry of Energy and Mineral Resources is around 82 kWh/month. Meanwhile, the number of household customers in Java-Bali with 1,300 VA and above is around 10.3 million.

If it is assumed that there is additional use of electric stoves for 1 million household customers per year, there will be an increase in electricity consumption of around 82 GWh or 414 MW. If all household customers with the power of 1,300 VA and above use electric stoves, there will be an increase in electricity consumption of 841 GWh or an increase in load of 4.2 GW. However, in 2022 the proposed program to convert 3 kg LPG to electric stoves is postponed as it has not received approval from the House of Representatives.

4.2.4 Status of CCS/CCUS Development

The energy transition is carried out by transforming the use of fossil fuels with clean technology, accelerating the use of new and renewable energy and increasing energy conservation activities. The energy transformation towards Net Zero Emissions is expected to realize energy independence, energy security, sustainable development, climate resilience and low carbon conditions

The use of CCS (Carbon Capture and Storage) or CCUS (Carbon Capture, Utilization, and Storage) technology has several significant benefits in the efforts to reduce greenhouse gas emissions and overcome climate change since CCS/CCUS can capture and store carbon dioxide (CO₂) produced from various sources, such as coal-fired power plants, chemical factories and other industries. By reducing the amount of CO₂ released into

the atmosphere, this technology can help reduce greenhouse gas emissions that cause global warming. The use of CCS/CCUS can also be a bridge in the transition from fossil-based energy to clean energy. By reducing CO₂ emissions from sources that are difficult to replace in the near future, this technology can provide additional time to develop and implement more sustainable renewable energy sources

The Government of Indonesia has developed Ultra Super Critical (USC) technology in Coal fired Power Plant starting with the construction of Java-7 located in Serang (Banten) in 2017. In 2021 there are several USC being built, including Java 9 & 10, Central Java (Batang), and Java 4 (Tanjung Jati B) power plant.

In 2020, the Government of Indonesia carried out a feasibility study for the CCUS Pilot Project in Gundih Oil and Gas field, Central Java. The total potential of CO₂ reduction in this pilot project is projected to be 2.92 million tons over 10 years. PT Pertamina also has a roadmap for implementing CCUS in 7 locations related to natural oil and gas production, namely in Subang, Cilamaya, Jatibarang, Merbau, Jambaran Tiung Biru, Natuna and Matindok. Based on data from the Directorate General of Oil and Gas, there are 16 oil field locations that will implement CCS and it will be implemented no later than 2030.

4.2.5 Status of Nuclear Development and NEPIO

The report of Nuclear Energy and Energy Security in the Energy Transition released by the IEA in June 2022 states that nuclear plays an important role in securing the global energy transition path towards net zero emissions. Nuclear energy has made a major contribution to slowing the rise in global CO₂ emissions since the 1970s. Globally, between 1971 and 2020, around 66 gigatons (Gt) of CO₂ were reduced. Without the contribution of nuclear energy, total emissions from power plants would have been almost 20% higher and total energy with those emissions is 6% higher over the period.

Nuclear energy, with a capacity of 413 gigawatts (GW) operating in 32 countries, contributes to both by reducing 1.5 Gt of global emissions and 180 billion cubic meters (bcm) of global gas demand per year.

The role of nuclear as a low-carbon technology is widely accepted by many countries, philanthropists and environmental NGOs, who recognize the role of nuclear energy in meeting net zero emissions commitments. In addition, recent developments show that GHG emissions from nuclear energy are comparable to wind energy and lower than solar energy.

Nuclear Power Plants have various advantages compared to other power plants. Nuclear power plants produce very large and reliable power supply. The nuclear power plant can operate for two years non-stop without changing fuel. Nuclear power plants have varied types of technology. It provides stable and large-scale power supplies, ranging from 10 MWe to 1,600 MWe per unit. Nuclear power plants rely more on technological developments (technology bases) than on energy sources (resource bases). Thus, they do not depend on the location of the nuclear fuel source. Nuclear power plant has the excellence in terms of environmental externalities. Nuclear power plants cause very small environmental impacts. It has zero local emissions (SO_x, NO_x, and fly ash), and minimum global CO_x emissions. With a small environmental impact, nuclear power plants also have the lowest mortality percentage compared to other types of energy generation. However, several countries are reluctant to build nuclear power plants due to security and safety risks.

The presence of nuclear minerals in Indonesia is spread across 26 locations in Sumatra, Kalimantan, Sulawesi, Maluku and Papua and estimates of uranium and thorium resources have been carried out at 9 deposit locations. In 2021 the total uranium resources are 89,483 tons of U₃O₈ (the heat generation value for this amount of uranium is equivalent to the heat generation by 1.85 trillion tons of coal), while the total thorium resources are 143,234 tons. The details of radioactive mineral resources in Indonesia can be seen in Table 4.1.

Table 4.1 Indonesia Radioactive Mineral Resources 2021

No.	Potential Sector	Resource								
		Measured		Indicated	Concessed		Hypothetic		Speculative	
		U ₃ O ₈	Th	U ₃ O ₈	U ₃ O ₈	Th	U ₃ O ₈	Th	U ₃ O ₈	Th
1	Aloban, Sibolga - North Sumatra				490*					
2	Singkep - Riau						1.298*	433*		
3	Bangka Belitung	2.840*	4.729*				1.224*	10.361*	25.715*	111.298*
4	Ketapang - West Kalimantan						736*	4.767*		
5	Kalan, Melawai - West Kalimantan	2.394*		5.903*	2.914*		5.058*			
6	Mentawa dan Darab, Seruyan - Central Kalimantan				623*		9.669*			
7	Katingan - Central Kalimantan						572*	2.261*		
8	Kawat, Mahakam Hulu - East Kalimantan						17.861*			
9	Mamuju - West Sulawesi				769*	3.424*	3.023*	3.138*	8.393*	2.823*
Total		5.234*	4.729**	5.903*	4.796*	3.424**	39.441*	20.960**	34.108*	114.121**
		Measured U	Measured Th	Indicated U	Con-cessed U	Concessed Th	Hipothet-ic U	Hipothet-ic Th	Speculative U	Speculative Th
Total U ₃ O ₈		89.483*								
Total Th		143.234**								

Note: (*) Unit Ton U₃O₈

(**) Unit Ton Th

Source: ORTN BRIN, 2021

For countries that do not yet have a nuclear power plant and intend to develop one, it is strongly recommended by the International Atomic Energy Agency (IAEA) to establish a Nuclear Energy Program Implementing Organization (NEPIO) as part of the government's commitment to develop nuclear energy. Therefore, in 2021, the Minister of Energy and Mineral Resources Decision Number 250.K/HK.02/MEM/202 concerning the Preparation Team for the Establishment of a Nuclear Energy Program Implementing Organization has been issued. Currently, Nuclear Energy Program Implementation Committee is being proposed to the Ministry of Administrative and Bureaucratic Reform to obtain the draft of Presidential Decree on Nuclear Energy Program Implementation Committee.

4.2.6 Status of Implementation on Carbon Economic Value

A carbon tax, also known as a carbon emissions tax or carbon pollution tax, is a policy instrument implemented by governments to reduce greenhouse gas emissions and encourage reductions in carbon pollution. Carbon taxes are usually imposed on companies or individuals that produce carbon emissions, based on the amount of carbon dioxide or carbon equivalent produced by their activities.

Carbon taxes aim to provide economic incentives for companies and individuals to reduce their carbon emissions. By providing a surcharge on carbon emissions, carbon taxes encourage the use of clean energy, energy efficiency, and low-carbon technological innovation.

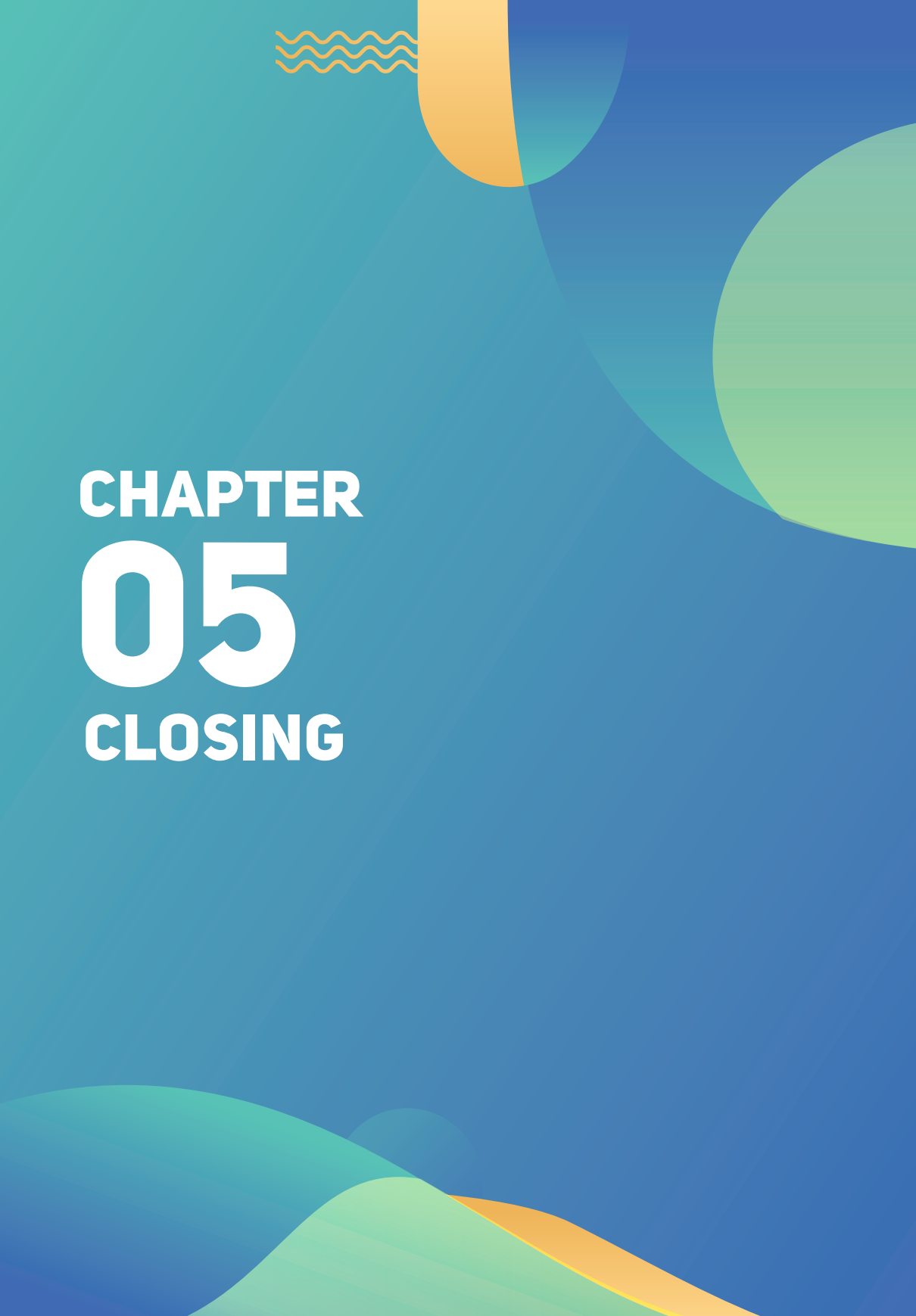
The implementation of carbon tax in Indonesia is regulated in Law Number 7 of 2021 concerning Harmonization of Tax Regulations. This law provides the legal basis for the government to impose a carbon tax as well as regulates tariffs and implementation mechanisms.

The carbon tax in Indonesia uses a cap and tax scheme, which regulates the permitted emission cap for each industry. Companies that exceed the emission cap will be subject to carbon tax according to the specified rates. A carbon market trial was conducted in 2021 for power plants. In this trial, the upper CAP value for GHG emissions has been determined based on the weighted average GHG emission intensity value in 2019 for the 3 (three) groups of coal fired power plant participating in the carbon trading trial and taking into account excess emission quota allocations. Meanwhile, capacity distribution is determined based on Minister of Energy and Mineral Resources Regulation Number 9 of 2020 concerning PT PLN (Persero) Electricity Power Supply Efficiency. CAP values based on power plant capacity classification are shown in Table 4.2 below.

Table 4.2 Classification on Capacity and Value of GHG Emission CAP

Type of Power Plant	Installed Capacity (MW)	Cap Value (ton CO ₂ /MWh)
Coal PP	$X > 400$	0.918
Coal PP	$100 \leq X \leq 400$	1.013
Mine Mouth PP	$100 \leq X \leq 400$	1.094

The implementation of the carbon tax in Indonesia was previously planned to start in 2023. However, Minister of Finance announced that the implementation of the carbon tax was postponed until 2025 since there were still a need to carry out technical preparations and information dissemination.



CHAPTER **05** **CLOSING**

05

CLOSING

5.1. SUMMARY

1. Final energy consumption in the next 10 years is projected to increase to 259 million TOE (Hymne scenario) and 232 million TOE (Mars scenario) with the growth of 4.6% and 3.5% respectively. The efficiency in Mars scenario means that final energy consumption will be more economical compared to the Hymne scenario which is projected using BaU assumptions.
2. In 2022, the largest electricity consumption is the household sector. However, in 2033 in Mars scenario, where the assumptions are towards optimizing the achievement of NZE, the largest demand for electricity will shift to the industrial sector (41%).
3. Based on region, the largest final energy consumption for the next 10 years is still dominated by Java-Bali region, as well as electricity consumption with the share of 67% (Hymne scenario) and 61% (Mars scenario).
4. The NRE mix in primary energy supply for both scenarios is still below the RUEN target, namely 17,5% in Hymne scenario and 20,5% in Mars scenario since the economic growth in RUEN is being projected to be higher than current conditions.

5.2. RECOMMENDATION

1. To achieve the 2060 NZE target, Indonesia must achieve high industrial growth especially outside Java to become a developed country.

2. It is necessary to make massive use of NRE in the power generation sector, especially coal co-firing, and solar PV, as well as to increase the use of biodiesel mix and bioethanol to exceed the NRE mix target in RUEN by 2030 and be in line with the NZE target in 2060.
3. Massive use of electric vehicles and electric stoves, especially in Java-Bali region, needs to be done immediately to absorb the excess power of PLN power plants.



ATTACHMENT

ATTACHMENT 1

OUTLOOK SUMMARY

No	Energy Indicator	Unit	National Projection of Hymne Scenarios										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	165	173	181	188	196	205	214	224	235	247	259
2	Final Energy Demand for Electricity	TWh	323	334	345	357	370	385	400	417	436	456	479
3	Primary Energy Supply	Mill. TOE	247	258	268	279	290	302	315	328	344	360	379
4	Power Plant Capacity	GW	98	105	113	118	122	124	127	131	133	135	135
5	Electricity Production	TWh	363	376	389	403	418	433	451	470	491	513	538
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	720	745	763	797	827	857	893	926	975	1021	1080

No	Energy Indicator	Unit	National Projection of Mars Scenarios										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	163	170	177	183	189	195	201	208	216	223	232
2	Final Energy Demand for Electricity	TWh	325	337	348	362	376	392	409	427	446	466	488
3	Primary Energy Supply	Mill. TOE	246	256	265	275	284	294	305	315	327	339	353
4	Power Plant Capacity	GW	98	108	117	124	133	143	152	166	175	188	188
5	Electricity Production	TWh	366	380	393	408	425	442	461	481	502	525	549
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	704	711	713	729	742	752	766	768	784	798	840

No	Energy Indicator	Unit	Sumatra Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	38	40	43	45	47	50	53	56	59	62	66
2	Final Energy Demand for Electricity	TWh	49	51	53	55	57	59	62	65	68	71	75
3	Primary Energy Supply	Mill. TOE	49	51	53	56	60	63	66	69	73	77	82
4	Power Plant Capacity	GW	20	21	23	23	22	23	23	25	25	25	25
5	Electricity Production	TWh	56	58	60	62	65	67	70	73	77	81	85
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	108	114	111	122	133	139	147	148	159	170	182

No	Energy Indicator	Unit	Sumatra Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	38	40	42	44	46	48	50	52	55	57	60
2	Final Energy Demand for Electricity	TWh	51	53	56	59	62	65	68	72	76	80	84
3	Primary Energy Supply	Mill. TOE	49	52	53	56	60	62	65	67	70	73	77
4	Power Plant Capacity	GW	19	20	22	22	23	24	25	27	29	32	32
5	Electricity Production	TWh	58	61	63	66	70	73	77	81	86	91	96
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	109	113	110	120	129	133	139	132	130	129	139

No	Energy Indicator	Unit	Java-Bali Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	89	93	96	99	102	106	110	114	118	123	128
2	Final Energy Demand for Electricity	TWh	221	227	234	242	251	261	271	283	296	310	325
3	Primary Energy Supply	Mill. TOE	149	154	159	164	170	176	182	189	197	205	214
4	Power Plant Capacity	GW	56	60	65	70	74	75	77	79	81	83	83
5	Electricity Production	TWh	249	257	266	275	285	295	307	320	334	350	367
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	482	499	516	533	546	566	588	611	638	662	696

No	Energy Indicator	Unit	Java-Bali Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	89	92	95	97	99	101	104	106	109	112	115
2	Final Energy Demand for Electricity	TWh	219	224	229	236	243	251	259	268	277	288	299
3	Primary Energy Supply	Mill. TOE	148	152	157	161	165	169	174	178	184	189	195
4	Power Plant Capacity	GW	56	59	63	68	73	79	84	89	95	104	104
5	Electricity Production	TWh	247	254	260	268	276	284	293	303	313	325	337
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	470	474	478	482	484	488	494	499	507	514	534

No	Energy Indicator	Unit	Kalimantan Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	22	23	24	25	26	28	29	31	33	35	37
2	Final Energy Demand for Electricity	TWh	14	14	15	16	16	17	18	19	20	20	21
3	Primary Energy Supply	Mill. TOE	26	27	28	29	31	32	34	35	37	40	42
4	Power Plant Capacity	GW	6	7	7	7	7	8	8	8	8	8	8
5	Electricity Production	TWh	16	16	17	18	18	19	20	21	22	23	24
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	56	58	60	63	65	68	70	74	79	85	91

No	Energy Indicator	Unit	Kalimantan Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	21	22	23	24	25	26	26	28	29	30	31
2	Final Energy Demand for Electricity	TWh	16	17	19	21	23	26	28	31	34	37	40
3	Primary Energy Supply	Mill. TOE	26	27	28	29	31	32	34	36	37	39	42
4	Power Plant Capacity	GW	9	12	15	16	18	20	22	27	29	30	30
5	Electricity Production	TWh	18	20	22	24	26	29	32	35	38	41	45
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	55	54	54	54	54	55	55	57	61	66	72

No	Energy Indicator	Unit	Sulawesi Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	10	11	12	12	13	14	15	16	17	18	19
2	Final Energy Demand for Electricity	TWh	26	26	27	29	30	31	32	34	35	37	39
3	Primary Energy Supply	Mill. TOE	15	16	17	18	19	20	22	23	24	26	27
4	Power Plant Capacity	GW	9	9	9	10	10	10	10	10	10	10	10
5	Electricity Production	TWh	28	29	30	31	33	34	35	37	39	41	43
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	45	47	49	51	53	55	58	61	65	70	75

No	Energy Indicator	Unit	Sulawesi Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	10	10	11	12	13	13	14	15	16	17	17
2	Final Energy Demand for Electricity	TWh	26	27	28	30	31	33	35	37	39	41	43
3	Primary Energy Supply	Mill. TOE	15	16	17	18	19	20	21	22	24	25	27
4	Power Plant Capacity	GW	8	8	9	9	9	9	10	10	10	10	10
5	Electricity Production	TWh	28	29	31	33	34	36	38	40	43	45	47
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	44	45	46	48	49	51	53	55	58	62	66

No	Energy Indicator	Unit	Nusra Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	3	3	3	3	3	3	4	4	4	4	4
2	Final Energy Demand for Electricity	TWh	5	7	7	7	7	7	7	7	8	8	8
3	Primary Energy Supply	Mill. TOE	4	4	4	5	5	5	5	5	6	6	6
4	Power Plant Capacity	GW	2	2	3	3	3	3	3	3	3	3	3
5	Electricity Production	TWh	5	7	7	7	7	8	8	8	8	9	9
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	11	11	11	11	12	12	13	13	14	15	15

No	Energy Indicator	Unit	Nusra Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	2	3	3	3	3	3	3	4	4	4	4
2	Final Energy Demand for Electricity	TWh	5	7	7	7	7	7	8	8	8	9	9
3	Primary Energy Supply	Mill. TOE	4	4	4	4	5	5	5	5	5	5	6
4	Power Plant Capacity	GW	2	3	3	4	4	5	5	6	6	6	6
5	Electricity Production	TWh	7	7	8	8	8	8	9	9	10	10	0
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	10	9	9	9	9	8	8	9	9	9	0

No	Energy Indicator	Unit	Maluku Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	1	1	2	2	2	2	2	2	2	2	2
2	Final Energy Demand for Electricity	TWh	6	6	6	6	7	7	7	7	7	7	7
3	Primary Energy Supply	Mill. TOE	3	3	3	3	3	3	3	3	3	3	4
4	Power Plant Capacity	GW	3	4	4	4	4	4	4	4	4	4	4
5	Electricity Production	TWh	6	7	7	7	7	7	7	7	7	7	7
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	10	9	9	9	9	9	9	10	10	10	10

No	Energy Indicator	Unit	Maluku Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	1	1	1	1	2	2	2	2	2	2	2
2	Final Energy Demand for Electricity	TWh	6	6	7	7	7	7	7	7	8	8	8
3	Primary Energy Supply	Mill. TOE	3	3	3	3	3	3	3	3	3	4	4
4	Power Plant Capacity	GW	3	4	4	4	4	4	4	4	4	4	4
5	Electricity Production	TWh	7	7	7	7	7	7	7	8	8	8	8
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	10	8	8	8	8	9	9	8	9	9	9

No	Energy Indicator	Unit	Papua Regional Projection Hymne Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	2	2	2	2	2	2	2	2	3	3	3
2	Final Energy Demand for Electricity	TWh	2	2	2	2	2	3	3	3	3	3	3
3	Primary Energy Supply	Mill. TOE	2	2	3	3	3	3	3	3	3	4	4
4	Power Plant Capacity	GW	1	2	2	2	2	2	2	2	2	2	2
5	Electricity Production	TWh	2	2	3	3	3	3	3	3	3	4	4
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	8	8	8	8	8	9	8	8	9	9	10

No	Energy Indicator	Unit	Papua Regional Projection Mars Scenario										
			2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Final Energy Demand	Mill. TOE	2	2	2	2	2	2	2	2	2	2	2
2	Final Energy Demand for Electricity	TWh	2	2	3	3	3	3	4	4	4	5	5
3	Primary Energy Supply	Mill. TOE	2	2	3	3	3	3	3	3	3	4	4
4	Power Plant Capacity	GW	1	2	2	2	2	2	2	2	2	2	2
5	Electricity Production	TWh	3	3	3	3	4	4	4	5	5	5	6
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	7	8	7	8	8	9	8	9	9	10	10

ATTACHMENT 2

CONVERSION FACTOR

Energy	Original Unit	Multiplier Factor to BOE (Barrel Oil Equivalent)
Coal		
Anthracite	Ton	4.9893
Coal Import	Ton	4.2766
Coal Kalimantan	Ton	4.2
Coal Ombilin	Ton	4.8452
Coal Tanjung Enim	Ton	3.7778
Lignite	Ton	3.0649
Riau Peat	Ton	2.5452
Briket	Ton	3.5638
Biomassa		
Charcoal	Ton	4.9713
Firewood	Ton	2.2979
Natural Gas	MSCF	0.1796
Gas Products		
City Gas	Thousand Kcal	0.0007
CNG	Thousand Kcal	0.0007
LNG	Ton	8.0532
LNG	MMBTU	0.1796
LPG	Ton	8.5246

Attachment 2 Continued

Energy	Original Unit	Multiplier Factor to BOE (Barrel Oil Equivalent)
Oil		
Condensate	Barrel	0.9545
Crude Oil	Barrel	1
Oil Fuel		
Aviation Gasoil (Avgas)	Kilo Liter	5.553
Aviation Turbine Gas (Avtur)	Kilo Liter	5.8907
Super TT gasoline	Kilo Liter	5.8275
Premix gasoline	Kilo Liter	5.8275
Premium gasoline	Kilo Liter	5.8275
Kerosene	Kilo Liter	5.9274
Diesel oil	Kilo Liter	6.4871
MDF	Kilo Liter	6.6078
FO	Kilo Liter	6.9612
Oil Products		
Other oil products	Barrel	1.02
Refinery Fuel		
Refinery Fuel Gas (RFG)	Barrel	1.6728
Refinery Fuel Oil (RFO)	Barrel	1.1236
Feed Stock	Barrel	1.0423
Power Plant	MWh	0.613

Source: HEESI, 2022 or Energy Balance 1990-1994,
Department of Mining and Energy

ATTACHMENT 3

EMISSION FACTOR

No.	Energy Type	Net Energy Content	% Carbon Content	CO ₂ (Kg CO ₂ / TJ)
1	Natural Gas	0	73.4	57,600
2	Gasoline	5.8	84.6	73,290
3	Avtur	5.9	85	71,348
4	Kerosene	5.9	85	72,430
5	Diesel Oil	6.5	86.5	76,000
6	LPG	5.6	82	59,500
7	Coal	4	74.6	100,000
8	Briquettes	3.5	74.6	100,000
9	Avgas	5.6	84.6	71,348
10	FO	7	86.5	77,900
11	DME	5.6	82	57,600

Source: Pusdatin



ATTACHMENT 4

GLOSSARY

Avgas

A type of aviation fuel used in piston engine aircraft.

Avtur (Aviation Turbine Fuel)

A type of aviation fuel designed for gas turbine engines-aircraft.

Coal

Sedimentary rocks that can burn, are formed from organic deposits, mainly plant remains and are formed through the process of coalescence.

Biomass

Renewable energy from organic materials such as animals and plants.

Biofuel

Any fuel, whether solid, liquid or gas, that is produced from organic materials which may be derived directly from plants or indirectly from industrial, commercial, domestic or agricultural waste.

B30

The result of mixing diesel oil with 30% vegetable oil derived from palm oil or Crude Palm Oil (CPO).

NRE (New Renewable Energy)

Energy that comes from natural processes that is produced continuously and sustainably without having to wait for millions of years like fossil energy.

Carbon Emission

The remaining results of fuel combustion in internal combustion engines, external combustion engines, or jet engines that are released through the engine exhaust system.

Final Energy

Energy that can be directly consumed by end users.

Primary Energy

The energy that is first consumed in an area, both which comes directly or indirectly from nature.

Gas

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the form of a gas phase obtained from the oil and gas mining process.

Industry

A field or economic activity related to the processing/manufacture of raw materials or finished goods in factories using skills and labor and the use of tools in the field of processing agricultural products and their distribution as the main activity.

KEN (National Energy Policy)

Government Regulation of The Republic of Indonesia Number 79 of 2014, a national energy management policy based on just, sustainable, and environmentally friendly principles to create national energy independence and security.

Commercial

An activity carried out with the aim of gaining profit, generally in the form of activities related to commerce.

LEAP (Low Emission Analysis Platform)

An energy planning simulation model that is capable of analyzing energy from demand to supply in an integrated manner.

Oil

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the liquid or solid phase.

Power Plant

A collection of equipment and machines used to generate electricity through the process of transforming energy from various energy sources.

Region

Regional division based on geographic proximity used in energy modeling (LEAP).

Household

Consists of one or several people who live in the same residence which include one family or group of people.

Other Sectors

Several activities that correlate with supporting commercial businesses consist of agricultural, construction and mining activities.

Scenario

Conditions or circumstances that may occur based on the assumptions input into energy modeling.

Transportation

One part of the goods and passenger distribution network that has developed very dynamically and plays a role in supporting political, economic, socio-cultural as well as defense and security development.

NZE (Net Zero Emission)

Conditions where the amount of carbon emissions released into the atmosphere does not exceed the amount of emissions that the earth can absorb.

RUEN (National Energy Plan)

The policy by the central government regarding the national energy management plan as the elaboration and implementation of the National Energy Policy (KEN) which is cross-sectoral in nature to achieve the National Energy Policy (KEN) targets.

GDP (Gross Domestic Product)

The value of all goods and services produced by a country in a certain period.

ATTACHMENT 5

ABBREVIATIONS

APBN	: Anggaran Pendapatan dan Belanja Negara (State Budget)
APK DEN	: Anggota Pemangku Kepentingan Dewan Energi Nasional (Members of National Energy Council from Stakeholders)
BaU	: Bussines as Usual
Bappenas	: Badan Perencanaan Pembangunan Nasional (National Development Plan Agency)
BBM	: <i>Bahan Bakar Minyak</i> (Fuel)
BBN	: <i>Bahan Bakar Nabati</i> (Biofuel)
BOE	: Barrel Oil Equivalent
BPP	: Biaya Pokok Penyediaan (Basic Cost of Supply)
BPS	: Badan Pusat Statistik (Statistics Indonesia)
BPH	: Badan Pengatur Hilir (Downstream Task Force)
BPPT	: Badan Pengkajian dan Penerapan Teknologi (Technology Assessment and Application Agency)
BTS	: Base Transceiver Station
CCS	: Carbon Capture and Storage
CCUS	: Carbon Capture, Utilization and Storage
CO ₂	: Carbon Dioxide
CO ₂ eq	: Carbon Dioxide equivalent
Covid-19	: Corona Virus Disease 2019
DEN	: Dewan Energi Nasional (National Energy Council)
Ditjen Migas	: Direktorat Jenderal Minyak dan Gas Bumi (Directorate General of Oil and Gas)
DME	: Dimethyl Ether
EV	: Electric Vehicle

EBT	: <i>Energi Baru Terbarukan</i> (New Renewable Energy)
EBTKE	: <i>Energi Baru Terbarukan dan Konservasi Energi</i> (New Renewable Energy and Energy Conservation)
ETEB	: <i>Energi Baru dan Energi Terbarukan</i> (New Energy and Renewable Energy)
GIZ	: Deutsche Gesellschaft für Internationale Zusammenarbeit
GRK	: <i>Gas Rumah Kaca</i> (Green House Gas)
GSEN	: <i>Grand Strategi Energi Nasional</i> (National Energy Grand Strategy)
Gt	: Giga tonne
GW	: Giga Watt
Gwe	: Giga Watt electric
GWh	: Giga Watt hour
HBA	: <i>Harga Batubara Acuan</i> (Coal Reference Price)
HEESI	: Handbook Of Energy & Economic Statistics Of Indonesia
H ₂	: <i>Hidrogen</i> (Hydrogen)
IAEA	: International Atomic Energy Agency
IEA	: International Energy Agency
IKN	: <i>Ibu Kota Nusantara</i> (Capital City)
INAFHE	: Indonesian Association for Fuel Cell and Hydrogen Energy
ITB	: Institut Teknologi Bandung
Jargas	: <i>Jaringan Gas</i> (City gas)
JBKP	: <i>Jenis BBM Khusus Penugasan</i> (Certain Fuel for Assignment)
JBT	: <i>Jenis BBM Tertentu</i> (Certain Fuel)
KAI	: Kereta Api Indonesia (Indonesia railway)
KBLBB	: <i>Kendaraan Bermotor Listrik Berbasis Baterai</i> (Battery based electric vehicles)
KEN	: <i>Kebijakan Energi Nasional</i> (National Energy Policy)
KESDM	: Kementerian Energi dan Sumber Daya Mineral (Ministry of Energy and Mineral Resources)
KOICA	: Korea International Cooperation Agency
kW	: kilo Watt
kWh	: kilo Watt hour
LEAP	: Low Emissions Analysis Platform
LPEM	: Lembaga Penyelidikan Ekonomi dan Masyarakat (Institute for Research on Economics and Society)

LPG	: Liquefied Petroleum Gas
LNG	: Liquefied Natural Gas
LSM	: Lembaga Swadaya Masyarakat (NGO)
LTSHE	: <i>Lampu Tenaga Surya Hemat Energi</i> (Energy Saving Solar Lights)
MBSD	: Million Barel Steam Per Day
MMBTU	: Million British Thermal Unit
MT	: Metrik Ton
MTPA	: Metric Ton per Annum
MMLTPA	: Million Metric Ton per Annum
MVA	: Mega Volt Ampere
MW	: Mega Watt
MWh	: Mega Watt hour
NEPIO	: Nuclear Energy Program Implementing Organization
NDC	: Nationally Determined Contributions
NTB	: <i>Nusa Tenggara Barat</i> (West Nusa Tenggara)
NTT	: <i>Nusa Tenggara Timur</i> (East Nusa Tenggara)
Nusra	: <i>Nusa Tenggara</i>
NZE	: Net Zero Emission
OEI	: Outlook <i>Energi Indonesia</i> (Indonesia Energy Outlook)
O ₂	: <i>Oksigen</i> (Oxygen)
PSO	: Public Service Obligation
PLN	: Perusahaan Listrik Negara (Electricity State Own Enterprise)
Perpres	: <i>Peraturan Presiden</i> (Presidential Regulation)
PLTA	: <i>Pembangkit Listrik Tenaga Air</i> (Hydro Power Plant)
PLTB	: <i>Pembangkit Listrik Tenaga Bayu</i> (Wind Power Plant)
PLTD	: <i>Pembangkit Listrik Tenaga Diesel</i> (Diesel Power Plant)
PLTG	: <i>Pembangkit Listrik Tenaga Gas</i> (Gas Power Plant)
PLTM	: <i>Pembangkit Listrik Tenaga Minihidro</i> (Minihydro Power Plant)
PLTMH	: <i>Pembangkit Listrik Tenaga Mikrohidro</i> (Microhydro Power Plant)
PLTN	: <i>Pembangkit Listrik Tenaga Nuklir</i> (Nuclear Power Plant)
PLTP	: <i>Pembangkit Listrik Tenaga Panas Bumi</i> (Geothermal Power Plant)
PLTS	: <i>Pembangkit Listrik Tenaga Surya</i> (Solar Power Plant)
PLTU	: <i>Pembangkit Listrik Tenaga Uap</i> (Coal Fired Power Plant)

PP	: <i>Peraturan Pemerintah</i> (Government Regulation)
Pusdatin	: Pusat Data dan Teknologi Informasi (Data and Information Technology Center)
Puslitbang	: Pusat Penelitian dan Pengembangan (Research and Development Center)
PUPR	: Pekerjaan Umum dan Perumahan Rakyat (Public Works and Housing)
RUPTL	: <i>Rencana Usaha Penyediaan Tenaga Listrik</i> (Power Supply Business Plan)
RUEN	: <i>Rencana Umum Energi Nasional</i> (National Energy Plan)
RON	: Research Octane Number
Setjen/SJ DEN	: Sekretariat Jenderal Dewan Energi Nasional (Secretariat General of the National Energy Council)
SPBE	: <i>Stasiun Pengisian Bulk Elpiji</i> (LPG Filling Station)
SPBU	: <i>Stasiun Pengisian Bahan Bakar Umum</i> (Fuel Filling Stations)
SPBKLU	: <i>Stasiun Penukaran Baterai Kendaraan Listrik Umum</i> (Public Electric Vehicle Battery Swap Station)
SPKLU	: <i>Stasiun Pengisian Kendaraan Listrik Umum</i> (Public Electric Vehicle Charging Station)
SR	: <i>Sambungan Rumah</i> (Household Connection)
SWH	: Solar Water Heater
TOE	: Tonne of Oil Equivalent
TSCF	: Trillion Standard Cubic Feet
TWh	: Terra Watt hour
UI	: Universitas Indonesia
USC	: Ultra Super Critical
USD	: United States Dollar
VA	: Volt Ampere
3T	: <i>Tertinggal, Terdepan dan Terluar</i> (Disadvantages, Frontier and Outermost)



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